

TANGENT STABILITY ON TORIC FANO BLOWUPS AND BUNDLES

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ABSTRACT. We study anticanonical slope stability of tangent bundles under two constructions of smooth toric Fano varieties. For codimension-two blowups of products of projective spaces indexed by a tree, all subspace stability inequalities reduce to one divisor-degree inequality at each vertex. We express the exceptional contributions as mixed intersections on the blowup centres. The criterion proves stability for every tree of maximum degree three whose nonleaf factor dimensions equal their valences, with arbitrary positive leaf dimensions; a specified degree-four neighbourhood forces instability, although other degree-four and degree-five arrangements remain stable. For locally trivial toric bundles with fibre a product of degree-six del Pezzo surfaces and base a product of projective spaces, we prove that the maximal stable Picard number in dimension $n \geq 4$ is $\lfloor 5n/4 \rfloor + 1$, allowing arbitrary integral twists. Every fixed subcubic tree with an edge yields maximizers after sufficiently long independent subdivisions. Consequently the number of torically distinct stable extremizers is unbounded as the dimension grows through multiples of four. The proofs combine relative tangent obstructions, subspace-rank formulas and integral comparisons; the bundle existence proofs also use exact rational calculations with uniform error estimates.

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1. INTRODUCTION

Let X be a smooth toric Fano variety of dimension n . We study slope stability of its tangent bundle with respect to $H = -K_X$. The tangent bundle of a nontrivial product is not stable. Connecting the factors by a toric bundle or by invariant blowups can produce stability, but it can also concentrate too much anticanonical degree in a proper tangent subsheaf. This paper gives stability criteria and obstructions for two classes of such connections, in unbounded dimension and Picard number.

For an invariant prime divisor D_ρ , write

$$\delta_\rho = \frac{nD_\rho H^{n-1}}{H^n}.$$

These positive weights sum to n . The toric tangent-bundle criterion says that stability is equivalent to

$$\sum_{u_\rho \in V} \delta_\rho < \dim V$$

for every nonzero proper subspace spanned by primitive ray generators [Kly90, HNS22]. Our first result reduces all these inequalities to one divisor-degree inequality at each vertex of a tree of blowups. We interpret the terms in that inequality as mixed intersections on the exceptional centres. Our second result determines a sharp Picard maximum for a class of toric bundles with arbitrary integral twists. The maximum is attained by many different fans: every fixed subcubic tree supplies extremizers after sufficiently long independent subdivisions.

Blowup trees and exceptional intersections. Let G be a finite simple graph and choose integers $b_v \geq 1$. Start with $\prod_v \mathbb{P}^{b_v}$ and assign distinct coordinate hyperplanes at each factor to its incident edges. For each edge $\{v, w\}$, blow up the codimension-two centre defined by its two assigned hyperplanes. These star subdivisions commute. The resulting variety $Y(G, b)$ is Fano exactly when $b_v \geq d_v := \deg_G(v)$ for every v , and then

$$\dim Y(G, b) = \sum_v b_v, \quad \rho(Y(G, b)) = |V(G)| + |E(G)|.$$

At each vertex there is an unassigned hyperplane. Its inverse image has divisor class $D_v = f_v^* \mathcal{O}_{\mathbb{P}^{b_v}}(1)$, where f_v is the original factor projection composed with the blowdown. Set $A_v = nD_v H^{n-1} / H^n$.

Theorem 1.1. *If G is a tree with at least one edge and $b_v \geq d_v$ at every vertex, then $T_{Y(G, b)}$ is anticanonically stable if and only if $A_v < 1$ for every v . It is semistable if and only if $A_v \leq 1$ for every v .*

Theorem 3.4 controls subspaces meeting arbitrarily many factors, not just individual ray lines. At a nonleaf vertex with $b_v = d_v$, the distinguished ray defines a rank-one foliation with first Chern class D_v . Its degree satisfies the identity

$$(b_v + 1)A_v = b_v + \sum_{e \ni v} J_{v, e}.$$

Each $J_{v, e}$ is a positive first moment of the exceptional facet. Section 4 expresses it instead by ordinary mixed intersections on the blowup centre, with the two normal directions ordered by the factor projection. Moreover,

$$J_{v, e} + J_{w, e} = \frac{nE_e H^{n-1}}{H^n}, \quad \sum_{e \ni v} J_{v, e} = \frac{n f_v^*(-K_{\mathbb{P}^{b_v}}) H^{n-1}}{H^n} - b_v.$$

Thus the number of incident blowups counts the contributions, but their sizes and distribution determine whether the foliation violates stability. The mixed-intersection formula applies to compatible blowup morphisms without a tree hypothesis.

Call a vertex *saturated* when $b_v = d_v$.

Theorem 1.2. *Let G be a tree with at least one edge and maximum degree at most three. If every nonleaf vertex is saturated and the leaf dimensions are arbitrary positive integers, then $T_{Y(G,b)}$ is anticanonically stable. By contrast, a saturated degree-four vertex adjacent to four saturated degree-four vertices forces instability, independently of the farther branches of a Fano tree.*

The assertions are Theorems 6.1 and 6.4. No separation or symmetry of the branching vertices is required in the stable case. A saturated tree with M edges has dimension $2M$ and Picard number $2M + 1$. The degree bound three is sharp as a guarantee for every saturated tree, but higher degree alone does not force instability. There are stable trees with two branching vertices of degree four or five and arbitrarily long connecting paths (Proposition 6.6). The neighbouring dimensions and the arrangement of branches enter the exceptional first moments.

A sharp bundle extremum and many extremizers. Let S_6 denote the blowup of $\mathbb{P}^2$ at its three torus-fixed points. Consider locally trivial toric bundles with fibre S_6^j over $\prod_{c=1}^k \mathbb{P}^{b_c}$, where $j, k, b_c \geq 1$, in the fan presentation (47). The twists are arbitrary integral vectors in the fibre lattice.

Theorem 1.3. *For every $n \geq 4$, the largest Picard number among smooth toric Fano n -folds in this bundle class with anticanonically stable tangent bundle is*

$$\left\lfloor \frac{5n}{4} \right\rfloor + 1.$$

The upper bound has three geometric sources. Fano positivity bounds the number of nonzero twists at each base factor by its dimension. A relative tangent slope obstruction requires each surface factor to meet at least two base factors. Stability also excludes a disconnected incidence graph. The resulting inequalities give the bound (Theorem 7.3). Paths of hexagon factors attain it in dimension $4j$; end attachments give the other residue classes (Corollary 9.3).

For a tree Γ , assign one hexagon factor to each edge and a base $\mathbb{P}^{\deg(v)}$ to each vertex. Twist the hexagon by opposite primitive rays at its endpoints, as in Definition 8.1, and denote the bundle by $X(\Gamma)$.

Theorem 1.4. *Every fixed finite tree of maximum degree three with at least one edge gives stable $T_{X(\Gamma)}$ after replacing each edge by an independently chosen sufficiently long path. A sufficient minimum length depends only on the number s of trivalent vertices and is $O(\log s)$ as $s \rightarrow \infty$. If Γ has j edges, then*

$$\dim X(\Gamma) = 4j, \quad \rho(X(\Gamma)) = 5j + 1.$$

For each integer $q \geq 1$, every sufficiently large j admits at least q pairwise nonisomorphic toric fans of such stable extremizers.

The subdivision assertion is Theorem 10.1. Its abundance consequence follows from recovering the graph from the fan (Lemma 8.4 and Corollary 10.2). It concerns isomorphisms of toric fans. We do not claim a classification of all extremizers. The exponential contraction of

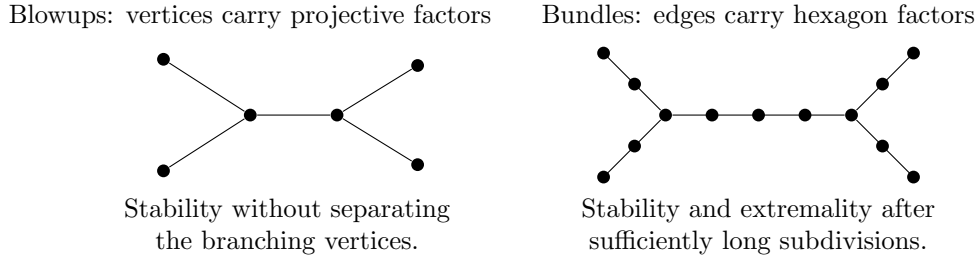


FIGURE 1. Two uses of a subcubic tree. On the left an edge specifies a codimension-two blowup; on the right it specifies a fibre factor with opposite twists. The right-hand subdivisions are schematic: the theorem requires the stated sufficient lengths.

positive slice functions explains the effect of separation, while a rank formula at a cut controls subspaces extending across several branches. Figure 1 distinguishes this separation phenomenon from the blowup theorem. Subdivision is not a general stability-preserving operation: it can destabilize the blowup examples (Corollary 5.4).

Relation to earlier work and proof structure. The subspace criterion and its interpretation as cone-volume concentration are established toric stability theory [Kly90, HNS22]. Hering–Nill–Süß determine stability regions for smooth toric surfaces and varieties of Picard rank two. Dasgupta–Dey–Khan independently treat Picard rank two and classify the smooth toric Fano fourfolds of Picard rank three [DDK20]; the Picard-rank-at-most-two fourfold case is due to Biswas–Dey–Genc–Poddar [BDGP21]. Reynolds treats all 124 smooth toric Fano fourfolds and their unstable Harder–Narasimhan filtrations [Rey23]. Devey develops finite matroid tests for equivariant reflexive sheaves [Dev22]. We use saturated subsheaves, as required by the erratum [DDK21]. The single-edge blowup with dimensions $(1, 3)$ is already stable by [DDK20, Proposition 5.6.1(2)].

Our results address families with no bound on the dimension or number of factors. For blowups, the new steps are the tree reduction, the exceptional mixed-intersection identity, and uniform estimates that separate stable from destabilizing arrangements. The estimates use Efron’s monotonicity theorem through [SW18, Theorems 3.2–3.3]. For bundles, the relative tangent obstruction uses a divergence identity related to [HL14, Lemma 2.2 and Eq. (4.4)]. Unlike a centred unweighted fibre polytope, its density weighted by base-slice volumes may have nonzero first moment. This distinction gives the obstruction. The polarization is the fixed divisor $-K_X$; results for polarizations near a pulled-back or contracted class [NT24, CNT25] do not give the needed inequalities. The preceding paper [Wue26] constructs stable root-twisted bundles with $\rho = n + 2$; allowing the number of surface factors to grow produces the larger extremum and branching families here.

The bundle existence proofs combine exact finite rational inequalities with the Birkhoff–Hopf contraction theorem [EN95] and induction. The finite calculations are premises of proofs covering all lengths, not numerical extrapolations. Their matrices, inequalities and error estimates are printed in the appendices and reproduced by the ancillary code. The blowup-tree arguments use uniform identities, integral comparisons and induction, with explicit polynomial arithmetic. Appendix F also records completed classification checks, including the unique stable eightfold of Picard number eleven.

After the common stability criterion, Sections 3–6 develop the blowup results. Sections 7–10 treat the bundle obstruction, sharpness and abundance. Complete path, attachment and three-arm calculations, followed by secondary families and classification data, are in the appendices. Both Picard maxima concern specified constructions. The unrestricted extremum remains open and is discussed in Section 11.

2. SLOPE STABILITY AND SYMMETRIES OF THE RAYS

Let $X = X_\Sigma$ be a smooth toric Fano variety of dimension n , with lattice N , dual lattice M , and primitive ray generators $u_\rho \in N$ for $\rho \in \mathcal{R} := \Sigma(1)$. Write D_ρ for the corresponding invariant prime divisor, and put

$$d_\rho = D_\rho \cdot (-K_X)^{n-1}, \quad D = (-K_X)^n = \sum_{\rho \in \mathcal{R}} d_\rho, \quad \delta_\rho = \frac{nd_\rho}{D}.$$

Thus $d_\rho > 0$, $\delta_\rho > 0$, and $\sum_\rho \delta_\rho = n$. For a torsion-free sheaf $\mathcal{E}$, its slope is $\mu(\mathcal{E}) = c_1(\mathcal{E}) \cdot (-K_X)^{n-1} / \text{rk } \mathcal{E}$. Stability, respectively semistability, means that every nonzero subsheaf of smaller rank has strictly smaller slope, respectively no larger slope. In particular, $\mu(T_X) = D/n$.

Lemma 2.1 (Toric slope criterion). *The tangent bundle T_X is stable, respectively semistable, if and only if every nonzero proper complex subspace $V \subset N_\mathbb{C}$ satisfies*

$$(1) \quad w(V) := \sum_{\rho: u_\rho \in V} \delta_\rho < \dim V, \quad \text{respectively } w(V) \leq \dim V.$$

It suffices to test subspaces spanned by ray generators.

Proof. This is the criterion of Hering–Nill–Süß [HNS22, Proposition 1.2 and Remark 2.6], obtained from Klyachko’s filtrations [Kly90]. For completeness, the decreasing filtrations for T_X on $N_\mathbb{C}$ are

$$E^\rho(i) = \begin{cases} N_\mathbb{C}, & i \leq 0, \\ \mathbb{C}u_\rho, & i = 1, \\ 0, & i \geq 2. \end{cases}$$

The saturated equivariant subsheaf $\mathcal{E}_V$ associated to V has induced filtrations $V \cap E^\rho(i)$ [DDK21, Corollary 0.0.2], rank $\dim V$, and degree $\sum_{u_\rho \in V} d_\rho$. Both slope stability and semistability can be tested on these subsheaves [HNS22, Proposition 2.3 and Remark 2.4]. If V contains rays, their span V' contains exactly the same rays and has dimension at most $\dim V$, so it has at least the same slope. If V contains no ray, then $w(V) = 0 < \dim V$. $\square$

For a rational ray-generated subspace V , the saturated subsheaf $\mathcal{E}_V$ is also the toric foliation whose general leaves are orbits of the corresponding subtorus. Its first Chern class is

$$c_1(\mathcal{E}_V) = \sum_{u_\rho \in V} D_\rho.$$

This follows from the same induced filtrations. In particular, a line containing a single ray has first Chern class D_ρ ; a line containing two opposite rays has the sum of the two divisors.

There is an equivalent convex-geometric normalization. Let

$$P = \{m \in M_\mathbb{R} : \langle m, u_\rho \rangle \geq -1 \text{ for all } \rho\}$$

be the anticanonical polytope, with facets F_ρ . With lattice volume, $H^n = n! \text{Vol}(P)$ and $D_\rho H^{n-1} = (n-1)! \text{Vol}(F_\rho)$, where $H = -K_X$. The probability measure assigning to u_ρ the mass

$$\nu_P(\rho) = \frac{\text{Vol}(\text{conv}(0, F_\rho))}{\text{Vol}(P)} = \frac{\delta_\rho}{n}$$

is the cone-volume measure, written on ray directions. The criterion is precisely $\nu_P(V) < \dim(V)/n$. This connection with subspace concentration is established in [HNS22]; our use of it will be through the explicit divisor degrees of the two constructions.

Lemma 2.2. *The divisor degrees satisfy*

$$\sum_{\rho \in \mathcal{R}} d_\rho u_\rho = 0.$$

Proof. For $m \in M$, the principal divisor of the character χ^m is $\sum_\rho \langle m, u_\rho \rangle D_\rho$. Its intersection with $(-K_X)^{n-1}$ vanishes. Pairing with all $m \in M$ gives the assertion. $\square$

2.1. The ray matroid and invariant subspaces. The *ray matroid* is the linear matroid on $\mathcal{R}$ whose rank function is

$$r(S) = \dim_{\mathbb{C}} \text{span}\{u_\rho : \rho \in S\}, \quad S \subseteq \mathcal{R}.$$

A circuit is a minimally linearly dependent set of ray generators. The components of the matroid are the equivalence classes generated by requiring all members of a circuit to belong to the same class; a ray in no circuit forms a singleton class. The matroid is *connected* if it has a single component.

Lemma 2.3. *If $C_1, \dots, C_s$ are the components of the ray matroid and $V_a = \text{span}\{u_\rho : \rho \in C_a\}$, then*

$$N_{\mathbb{C}} = V_1 \oplus \dots \oplus V_s.$$

Every direct-sum decomposition of $N_{\mathbb{C}}$ in which each ray generator lies in a summand is obtained by grouping these component subspaces. In particular, stability of T_X implies connectedness of its ray matroid.

Proof. Choose a basis $B \subset \mathcal{R}$. For $\rho \notin B$, the ray u_ρ and the basis vectors occurring with nonzero coefficient in its expansion form a circuit. All those basis vectors therefore belong to the same component as ρ . Consequently V_a is spanned by the subset $B \cap C_a$ of this basis, proving the direct sum. Conversely, a circuit cannot meet two summands of any decomposition containing every ray in a summand: projecting its dependence to either summand would give a dependence on a proper subset of the circuit. Thus each component is contained in one summand. Since the rays span $N_{\mathbb{C}}$, each summand is the sum of the component subspaces it contains.

If $s > 1$, the equalities $\sum_a w(V_a) = n = \sum_a \dim V_a$ imply $w(V_a) \geq \dim V_a$ for at least one component, contradicting Lemma 2.1. $\square$

Proposition 2.4 (Symmetry reduction). *Let A be any subgroup of the lattice automorphisms preserving Σ . Then T_X is stable if and only if its ray matroid is connected and*

$$w(V) < \dim V$$

for every nonzero proper A -invariant subspace $V \subset N_{\mathbb{C}}$ spanned by ray generators.

Proof. The group A permutes the rays and preserves their divisor degrees, so $w(aV) = w(V)$ for $a \in A$. Necessity follows from Lemmas 2.1 and 2.3. We prove sufficiency by linear algebra.

For any two subspaces $U, V \subseteq N_{\mathbb{C}}$, counting the rays in their union and intersection gives

$$(2) \quad w(U) + w(V) \leq w(U + V) + w(U \cap V).$$

Let

$$\lambda = \max_{0 \neq V \subseteq N_{\mathbb{C}}} \frac{w(V)}{\dim V}.$$

The maximum is attained among the finitely many subspaces spanned by rays, and $\lambda \geq 1$ because $w(N_{\mathbb{C}}) = n$. If U, V attain this maximum, then (2) and the bound $w(W) \leq \lambda \dim W$ imply that $U + V$ also attains it. They also imply $w(U \cap V) = \lambda \dim(U \cap V)$, including when the intersection is zero. Every maximizing subspace is spanned by rays, since $\lambda > 0$. If $\lambda > 1$, the sum of all maximizing subspaces is therefore a nonzero proper maximizing subspace. It is spanned by rays and is A -invariant, contrary to the hypothesis. Thus $\lambda = 1$, and T_X is semistable by Lemma 2.1.

Call a subspace V *tight* if $w(V) = \dim V$. Every nonzero tight subspace is spanned by rays: replacing it by the span of its rays would otherwise contradict $w(W) \leq \dim W$. The preceding argument shows that sums and intersections of tight subspaces are tight. Moreover, if U and V are tight, equality holds in (2). Since all ray weights are positive, every ray in $U + V$ then lies in U or in V .

There are finitely many nonzero tight subspaces. Let $U_1, \dots, U_q$ be the minimal ones under inclusion. Distinct U_i have zero intersection, since an intersection is tight. Their sum is in fact direct. To see this inductively, put $W = U_1 + \dots + U_{k-1}$. The intersection $U_k \cap W$ is tight, so minimality implies that it is either zero or all of U_k . In the latter case, choose a ray in U_k . Repeated use of the preceding observation about rays in a sum shows that this ray lies in some U_i with $i < k$, contradicting $U_i \cap U_k = 0$. Thus

$$S = U_1 \oplus \dots \oplus U_q$$

is tight, is spanned by rays, and is A -invariant, since A permutes the minimal nonzero tight subspaces.

If $S \neq N_{\mathbb{C}}$, it contradicts the assumed strict inequality on invariant subspaces. If $S = N_{\mathbb{C}}$, every ray belongs to one of the U_i , again by the same observation. Connectedness of the ray matroid and Lemma 2.3 force $q = 1$. Hence $U_1 = N_{\mathbb{C}}$ is minimal among nonzero tight subspaces, so there is no proper nonzero tight subspace. All the inequalities in Lemma 2.1 are therefore strict. $\square$

3. BLOWUPS OF PRODUCTS OF PROJECTIVE SPACES

We first study blowups of products of projective spaces along invariant codimension-two centres. A simple graph specifies the two factors meeting each centre. The main result of this section is Theorem 3.4: on a tree, stability against all tangent subsheaves is equivalent to one divisor-degree inequality at each vertex. The proof combines a rank formula for the rays with a divergence identity for the anticanonical polytope. Section 4 interprets the positive terms in that identity as exceptional intersection numbers; Sections 5–6 then estimate them to obtain uniform stability and local instability theorems.

3.1. The fan and its primitive collections. Let $G = (V, E)$ be a finite nonempty simple graph, and write $k = |V|$, $m = |E|$, and $d_v = \deg_G(v)$. Assign a positive integer b_v to each vertex. In

$$N = \bigoplus_{v \in V} \mathbb{Z}^{b_v},$$

let R_v be the $b_v + 1$ primitive ray generators of the standard fan of $\mathbb{P}^{b_v}$ in its summand. Thus $\sum_{r \in R_v} r = 0$. Suppose $d_v \leq b_v + 1$, and assign the edges incident to v injectively to rays $r_{v,e} \in R_v$. For every $e = \{v, w\}$, subdivide the original two-cone $\langle r_{v,e}, r_{w,e} \rangle$ by the ray

$$(3) \quad s_e = r_{v,e} + r_{w,e}.$$

Denote the resulting fan by $\Sigma(G, b)$ and its variety by $Y(G, b)$. The injectivity condition makes these pairs of original rays disjoint. All choices of the assignments are related by permutations of the rays within the projective-space factors and hence give isomorphic fans.

Proposition 3.1. *The fan $\Sigma(G, b)$ is independent of the order of subdivision and is smooth and projective. With $n = \sum_v b_v$, one has*

$$(4) \quad \dim Y(G, b) = n, \quad \rho(Y(G, b)) = k + m.$$

Its primitive collections are precisely

$$(5) \quad \mathcal{C}_e = \{r_{v,e}, r_{w,e}\}, \quad e = \{v, w\},$$

$$(6) \quad \mathcal{P}_v(J) = (R_v \setminus \{r_{v,e} : e \in J\}) \cup \{s_e : e \in J\}, \quad J \subseteq E(v),$$

where $E(v)$ is the set of edges incident to v . Their primitive relations are

$$(7) \quad r_{v,e} + r_{w,e} = s_e, \quad \sum_{r \in \mathcal{P}_v(J)} r = \sum_{e=\{v,w\} \in J} r_{w,e}.$$

Every $Y(G, b)$ is weak Fano, and

$$(8) \quad Y(G, b) \text{ is Fano} \iff b_v \geq d_v \quad (v \in V).$$

If G is connected, $Y(G, b)$ admits no torically locally trivial bundle presentation with positive-dimensional complete fibre and positive-dimensional base.

Proof. In an original maximal cone, the subdivision centres that it contains use disjoint pairs of basis vectors. The subdivisions are therefore products of subdivisions of two-dimensional orthants. They commute and remain unimodular. Each remaining original centre persists after the other subdivisions. Every step is a toric blowup along a smooth invariant centre, so the resulting fan is projective. It has $n + k + m$ rays, proving (4).

We first describe all its faces. For a subset A of the final rays, write B for its original rays and J for the edges whose exceptional rays belong to A . Then A spans a cone if and only if

- (1) B contains no endpoint pair $\mathcal{C}_e$;
- (2) the union of B with both endpoints of every $e \in J$ contains no complete set R_v .

Indeed, the smallest original cone containing s_e contains both its endpoints, whereas the original endpoint pair is removed by subdivision. This proves necessity. For sufficiency, the second condition places the expanded set in an original product cone. Within that cone, the subdivisions act on disjoint coordinate pairs, and the first condition is exactly the condition for the chosen rays to lie in one of the resulting cones.

The sets in (5) are minimal nonfaces. The expansion of $\mathcal{P}_v(J)$ contains R_v , and deleting any one member removes its unique representative of one ray of R_v . At any other vertex, that expansion contains at most one ray, because G is simple; it cannot contain the full

set of at least two rays of that factor. There is also no original endpoint pair in $\mathcal{P}_v(J)$. The face criterion proves its minimality, including when every member is an exceptional ray. Conversely, a nonface containing no $\mathcal{C}_e$ expands to a set containing some R_v . Choosing one original or exceptional representative for each ray of R_v produces a subset $\mathcal{P}_v(J)$. The list is complete.

The vector identities (7) follow from $\sum_{r \in R_v} r = 0$. On the right of the second identity there is at most one ray in each neighbouring factor. No two of these rays form another centre, since incidence rays are assigned injectively. They therefore span a cone of the final fan, and their positive sum lies in its relative interior. These are the primitive relations, of degrees 1 and $b_v + 1 - |J|$, respectively. Batyrev's primitive-collection criterion for nefness and ampleness [CLS11, Theorem 6.4.9] proves that $-K$ is nef and gives (8). The anticanonical polytope contains the origin in its interior and is bounded by completeness, so $-K$ is also big.

Finally, the cone complex of a torically locally trivial bundle is the join of its fibre and base cone complexes. Its rays partition into fibre rays and one lift of each base ray, and every primitive collection lies entirely in one part. This follows on the invariant affine charts, where the fan is the product of the fibre fan and the base cone; the identifications agree on common faces. In our fan, the primitive collection $R_v = \mathcal{P}_v(\emptyset)$ joins all original rays of one factor. The collections $\mathcal{C}_e$ join neighbouring factors, and $\mathcal{P}_v(\{e\})$ joins s_e to $b_v \geq 1$ original rays. For connected G , all rays consequently lie in one equivalence class generated by membership in a common primitive collection. They admit no partition into two nonempty parts respecting every primitive collection. This excludes every bundle presentation of the stated kind, including alternative choices of fibre and base. $\square$

The last assertion concerns locally trivial toric bundles; it does not exclude toric morphisms with special fibres or arbitrary non-equivariant fibrations. Simplicity of G and injectivity of the incidence assignments are part of the construction. For example, two parallel edges using the opposite pairs of rays of $\mathbb{P}^1 \times \mathbb{P}^1$ give the degree-six del Pezzo surface, which does not satisfy (8) with these vertex dimensions.

3.2. Divisor degrees and a divergence identity. For the remainder of this section assume $b_v \geq d_v$, and put $Y = Y(G, b)$. Use the normalized degrees of Section 2:

$$(9) \quad \delta_r = \frac{n D_r \cdot (-K_Y)^{n-1}}{(-K_Y)^n}, \quad \sum_r \delta_r = n.$$

They are all positive. Set $C_e = \delta_{s_e}$. There is a common value A_v for the degrees of the unassigned original rays at v , and

$$(10) \quad \delta_{r_{v,e}} = A_v - C_e, \quad \sum_v (b_v + 1) A_v - \sum_e C_e = n.$$

To see this, project the relation $\sum_r \delta_r r = 0$ of Lemma 2.2 onto the v -th lattice summand. The coefficient of an assigned ray is $\delta_{r_{v,e}} + C_e$, and that of an unassigned ray is its degree. The unique relation among R_v has all coefficients equal, proving the first assertion. Summing all ray degrees gives the second. In particular,

$$(11) \quad 0 < C_e < A_v \quad (e \in E(v)).$$

There are $q_v = b_v + 1 - d_v \geq 1$ unassigned rays at v . Choose one as the negative sum ray and write the others as an integral basis $r_{v,1}, \dots, r_{v,b_v}$. If $M = \text{Hom}(N, \mathbb{Z})$, the affine coordinates

$a_{v,j} = 1 + \langle x, r_{v,j} \rangle$ on $M_{\mathbb{R}}$ preserve lattice measure. We write $a_{v,e}$ for the coordinate assigned to the incidence (v, e) . The anticanonical polytope P is

$$(12) \quad a_{v,j} \geq 0, \quad \sum_{j=1}^{b_v} a_{v,j} \leq b_v + 1, \quad a_{v,e} + a_{w,e} \geq 1 \quad (e = \{v, w\}).$$

Throughout, volumes are lattice volumes without factorial normalization. If F_r is the facet corresponding to r , then

$$(13) \quad (-K_Y)^n = n! \operatorname{vol}(P), \quad \delta_r = \frac{\operatorname{vol}(F_r)}{\operatorname{vol}(P)}.$$

The second formula follows from $D_r \cdot (-K_Y)^{n-1} = (n-1)! \operatorname{vol}(F_r)$.

Lemma 3.2. *For an edge $e = \{v, w\}$, define*

$$J_{v,e} = \frac{1}{\operatorname{vol}(P)} \int_{F_{s_e}} a_{v,e} d \operatorname{vol}.$$

Then

$$(14) \quad (b_v + 1)A_v = b_v + \sum_{e \in E(v)} J_{v,e}, \quad J_{v,e} + J_{w,e} = C_e,$$

and $0 < J_{v,e} < C_e < A_v$. In particular, $A_v < 1$ at every vertex of degree one, for every positive value of b_v .

Proof. Apply the divergence theorem to the vector field equal to a_v in the v -th coordinate block and zero elsewhere. Its divergence is b_v . The coordinate facets have zero flux. The facet $\sum_j a_{v,j} = b_v + 1$ has flux $(b_v + 1)$ times its lattice volume, and the exceptional facet F_{s_e} has flux $-\int_{F_{s_e}} a_{v,e} d \operatorname{vol}$. All remaining facets have zero flux. For a primitive integral normal, its Euclidean length in hypersurface measure cancels the reciprocal factor in the unit-normal flux. Division by $\operatorname{vol}(P)$ proves the first identity.

On F_{s_e} one has $a_{v,e} + a_{w,e} = 1$. Its relative interior has both coordinates strictly between zero and one, since intersection with either coordinate facet has smaller dimension. This proves the second identity and the strict bounds on $J_{v,e}$; the remaining bound is (11). At a leaf, the first identity gives $(b_v + 1)A_v = b_v + J_{v,e} < b_v + A_v$, hence $A_v < 1$. $\square$

3.3. A cographic rank formula. Construct a multigraph H with vertices o , the vertices v of G , and one vertex z_e for each edge of G . Its edges are indexed by the rays of $\Sigma(G, b)$ as follows:

- an assigned ray $r_{v,e}$ corresponds to an edge $z_e \rightarrow v$;
- each unassigned original ray at v corresponds to an edge $o \rightarrow v$;
- the exceptional ray s_e corresponds to an edge $o \rightarrow z_e$.

The orientations specify an incidence matrix and do not affect graph connectivity. Thus H has $k + m + 1$ vertices and $n + k + m$ edges. It is connected, even if G is disconnected: each z_e joins the root, each nonisolated v joins some z_e , and an isolated v has unassigned rays joining it to the root.

Proposition 3.3. *For a subset S of the rays, let $H \setminus S$ denote the graph obtained by deleting its corresponding edges and retaining all vertices. Then*

$$(15) \quad r(S) = |S| + 1 - c(H \setminus S),$$

where c counts connected components, including isolated vertices. Consequently the ray matroid is the cographic matroid of H : its bases are exactly the complements of spanning trees of H .

Proof. Let B be the signed incidence matrix of H , with positive entries at the heads of oriented edges, and let $R : \mathbb{Q}^{E(H)} \rightarrow N_{\mathbb{Q}}$ send each edge basis vector to its ray. The row at v maps to $\sum_{r \in R_v} r = 0$, and the row at z_e maps to $s_e - r_{v,e} - r_{w,e} = 0$. Thus the row space of B lies in $\ker R$. Both spaces have dimension $k + m$: the incidence matrix has that rank because H is connected, and the original rays span $N_{\mathbb{Q}}$. Hence $\ker R = \text{im } B^{\top}$.

A relation supported on S is therefore $B^{\top}f$, where f is constant on every component of $H \setminus S$. Modulo the global constants, these functions have dimension $c(H \setminus S) - 1$. Subtracting this relation dimension from $|S|$ proves (15). Rational and complex ranks agree. A subset of n rays is a basis precisely when its complement is connected; that complement has $k + m$ edges on $k + m + 1$ vertices, so it is a spanning tree. $\square$

3.4. The stability criterion for trees.

Theorem 3.4. *Suppose G is a tree with at least one edge and $b_v \geq d_v$ for all v . Then $T_{Y(G,b)}$ is anticanonically stable if and only if $A_v < 1$ for every vertex, and is semistable if and only if $A_v \leq 1$ for every vertex. Thus it is strictly semistable when $\max_v A_v = 1$ and unstable when $\max_v A_v > 1$.*

Proof. Let K be the tree obtained by subdividing every edge of G at z_e , so H is obtained from K by adjoining the root and its edges. Give each edge of H its normalized ray degree. There are q_v root edges of weight A_v at v , one root edge of weight C_e at z_e , and the edge vz_e has weight $A_v - C_e$.

For a ray subset S , set $E' = E(H) \setminus S$, write $\delta(E')$ for its total weight, and put

$$\beta(E') = |E'| - |V(H)| + c(E').$$

This is the cycle rank, with all vertices retained. By Proposition 3.3, $r(S) = n - \beta(E')$. Since the total weight is n , the slope inequality becomes

$$(16) \quad \delta(E') > \beta(E') \quad \text{when } 0 < r(S) < n,$$

with a weak inequality for semistability.

For a nonempty connected vertex set Q of K , let $q(Q)$ count all root edges at its vertices, including multiplicity, and let $W(Q)$ be the sum of their weights and the weights of all K -edges internal to Q . Define

$$D(Q) = W(Q) - (q(Q) - 1).$$

For $Q = K$ this is zero, because the cycle rank and total weight of H both equal n . For $Q = \{z_e\}$ it is $C_e > 0$.

Otherwise the original vertices in Q form a nonempty connected subtree I of G . All exceptional vertices on internal edges of I belong to Q . Let ∂I be the edges with exactly one endpoint in I , write $v(e)$ for that endpoint, and set $t_e = 1$ if $z_e \in Q$, and $t_e = 0$ otherwise. Counting the original-ray contributions and then using (14) gives

$$\begin{aligned} W(Q) &= \sum_{v \in I} (b_v + 1)A_v - \sum_{e \in \partial I} (1 - t_e)A_{v(e)} - \sum_{e \text{ internal to } I} C_e \\ &= \sum_{v \in I} b_v + \sum_{e \in \partial I} J_{v(e),e} - \sum_{e \in \partial I} (1 - t_e)A_{v(e)}. \end{aligned}$$

There are $|I| - 1$ internal edges, and $\sum_{v \in I} d_v = 2(|I| - 1) + |\partial I|$. Consequently

$$q(Q) - 1 = \sum_{v \in I} b_v - |\partial I| + \sum_{e \in \partial I} t_e,$$

and hence

$$(17) \quad D(Q) = \sum_{e \in \partial I} [J_{v(e),e} + (1 - t_e)(1 - A_{v(e)})].$$

Assume now $A_v \leq 1$ for every v . If I is proper, its boundary is nonempty, so this sum is strictly positive. If I contains every original vertex, connectivity forces $Q = K$. Together with the exceptional singleton case, this proves $D(Q) > 0$ for every nonempty proper connected Q .

The selected non-root edges of E' form a forest in K . Each component has a connected vertex set Q and contains every K -edge internal to that set, since K is a tree. If it has h selected root edges, counted with multiplicity, it contributes $\max(h - 1, 0)$ to $\beta(E')$. This remains true when several components join the common root: each component with a root edge adds one connection and its remaining root edges add cycles. For $h = 0$, its contribution to weight minus cycle rank is nonnegative, and positive if it contains an edge. For $h \geq 1$, completing the missing root edges gives the exact contribution

$$(18) \quad D(Q) + \sum_{r \text{ a missing root edge at } Q} (1 - \delta_r).$$

All root degrees are at most one; indeed $C_e < A_v \leq 1$. Thus every component contribution is nonnegative, proving the weak inequalities in (16).

If every $A_v < 1$, all root degrees are strictly below one. Contributions from proper components with root edges are already positive. A full component $Q = K$ is also positive if a root edge is missing, by (18); if none is missing, $E' = E(H)$ and $S = \emptyset$. Components without root edges contribute positively whenever they contain an edge. The only remaining graph with no positive contribution is the empty E' , which has $r(S) = n$. Both exceptions lie outside the tested ranks. This proves every strict inequality.

Conversely, an unassigned ray of degree A_v belongs to a rank-one subspace of weight at least A_v . This is a proper subspace because $n \geq 2$. Stability therefore requires $A_v < 1$, and semistability requires $A_v \leq 1$. These necessity statements, the inequalities just proved, and Lemma 2.1 give all three alternatives. $\square$

The reduction to vertex inequalities uses two different features of the construction. Proposition 3.3 describes all tangent subspace ranks for any simple graph. For trees, the boundary formula (17) expresses the remaining inequalities as positive sums of vertex and exceptional contributions. In particular, the scalar criterion is not obtained by disregarding higher-rank subsheaves. The tree hypothesis is used in this passage from local degrees to all subspace inequalities.

Corollary 3.5. *For every $p, q \geq 1$, the blowup of $\mathbb{P}^p \times \mathbb{P}^q$ along the invariant codimension-two subvariety corresponding to a cone generated by one ray from each factor has anticanonically stable tangent bundle.*

Proof. Both vertices of its graph have degree one. Lemma 3.2 gives both strict inequalities required by Theorem 3.4. $\square$

For $(p, q) = (1, 3)$, this recovers [DDK20, Proposition 5.6.1(2)].

3.5. Stable paths and the largest Picard number.

Theorem 3.6. *Let G be a path with $L \geq 1$ edges, endpoint dimensions $p, q \geq 1$, and dimension two at every internal vertex. Then $T_{Y(G,b)}$ is anticanonically stable, and*

$$(19) \quad \dim Y(G, b) = 2L + p + q - 2, \quad \rho(Y(G, b)) = 2L + 1.$$

At each internal vertex its unassigned normalized degree satisfies $A_v \leq 24/25$.

Proof. Fix the incidence coordinate a on one side of an edge and integrate the coordinates of the branch on the other side. Denote its volume by $\phi(a)$. We shall use the following properties:

$$(20) \quad \phi(a) > 0, \quad \phi(a) = c := \phi(1) \ (a \geq 1), \quad \phi(a) \geq ac \ (0 \leq a \leq 1).$$

A leaf of dimension b gives

$$\phi_b(a) = \frac{(b + \min(a, 1))^b}{b!}.$$

Indeed, translating its assigned coordinate by the lower bound $\max(0, 1 - a)$ leaves a simplex of that capacity. Bernoulli's inequality gives

$$\frac{\phi_b(a)}{\phi_b(1)} = \left(1 - \frac{1-a}{b+1}\right)^b \geq 1 - \frac{b}{b+1}(1-a) \geq a,$$

so (20) holds for every endpoint dimension.

At a degree-two vertex, integration of the coordinate assigned to the parent transforms an incoming function by

$$(21) \quad g(a) = \int_0^{2+\min(a,1)} (2 + \min(a, 1) - t)\phi(t) dt.$$

For a positive input constant beyond one, set $I_0 = \int_0^1 \phi(t) dt$ and $I_1 = \int_0^1 t\phi(t) dt$. On $0 \leq a \leq 1$, the output is $g(a) = g_0 + g_1 a + g_2 a^2$, where

$$g_0 = 2I_0 - I_1 + c/2, \quad g_1 = I_0 + c, \quad g_2 = c/2.$$

All coefficients are positive, and $g_0 - g_2 = 2I_0 - I_1 > 0$. Hence

$$g(a) - ag(1) = (1-a)(g_0 - ag_2) \geq 0.$$

The output is constant beyond one. Induction along each branch proves (20) for every incoming function, including branches whose first input is not quadratic.

At an internal vertex let ϕ and ψ be the two incoming functions. The branches have disjoint coordinate blocks, so their product is the integrand over $a, b \geq 0$, $a + b \leq 3$. For either input use subscripts to distinguish c , I_0 and I_1 . Splitting this simplex at $a = 1$ and $b = 1$ gives its full volume V and its unassigned facet volume F as

$$(22) \quad V = I_{0,\phi}I_{0,\psi} + c_\psi(2I_{0,\phi} - I_{1,\phi}) + c_\phi(2I_{0,\psi} - I_{1,\psi}) + \frac{1}{2}c_\phi c_\psi,$$

$$(23) \quad F = c_\phi c_\psi + c_\psi I_{0,\phi} + c_\phi I_{0,\psi}.$$

The first term of V is the unit square, the next two are the regions with just one coordinate at least one, and the last is the remaining triangle. Eliminating one coordinate from $a + b = 3$ gives the lattice measure in the formula for F .

Define

$$i_\phi = I_{0,\phi}/c_\phi, \quad h_\phi = (I_{0,\phi} - I_{1,\phi})/c_\phi,$$

and similarly for ψ . From (20), $i_\phi \geq 1/2$ and $h_\phi \geq 1/6$. Equations (22)–(23) become

$$\frac{F}{c_\phi c_\psi} = 1 + i_\phi + i_\psi, \quad \frac{V - F}{c_\phi c_\psi} = i_\phi i_\psi + h_\phi + h_\psi - \frac{1}{2}.$$

Write $i_\phi = 1/2 + x$, $i_\psi = 1/2 + y$, $h_\phi = 1/6 + u$ and $h_\psi = 1/6 + v$, with $x, y, u, v \geq 0$. Then

$$\frac{24V - 25F}{c_\phi c_\psi} = 24xy + 11(x + y) + 24(u + v) \geq 0.$$

Thus (13) gives $A_v = F/V \leq 24/25 < 1$ at every internal vertex. Lemma 3.2 gives $A_v < 1$ at both endpoints. Theorem 3.4 proves stability, also for $L = 1$, when there is no internal vertex. The dimension and Picard number follow from Proposition 3.1. $\square$

Corollary 3.7. *For every $n \geq 1$, the largest Picard number among the smooth Fano n -folds $Y(G, b)$ with stable tangent bundle is*

$$(24) \quad 2 \left\lfloor \frac{n}{2} \right\rfloor + 1.$$

Here G ranges over the finite nonempty simple graphs with injective incidence-ray assignments in the construction above.

Proof. If G is disconnected, its fan is the product of the fans for its connected components. The tangent bundle is a direct sum of at least two positive-rank bundles; one summand has slope at least the slope of the whole bundle. Stability therefore forces connectedness. Consequently $k \leq m + 1$, whereas the Fano condition gives

$$n = \sum_v b_v \geq \sum_v d_v = 2m.$$

It follows that $\rho = k + m \leq 2m + 1 \leq 2\lfloor n/2 \rfloor + 1$.

For $n = 2L \geq 2$, take the length- L path with endpoint dimensions $(1, 1)$. For $n = 2L + 1 \geq 3$, take endpoint dimensions $(1, 2)$ instead. Theorem 3.6 gives stability, and (19) gives equality in (24). For $n = 1$, the single-vertex graph with dimension one gives $\mathbb{P}^1$, whose rank-one tangent bundle is stable and whose Picard number is one. $\square$

This maximum concerns the specified blowups of products of projective spaces. It is separate from the bundle-class maximum in Section 7 and does not give an upper bound for arbitrary smooth toric Fano varieties.

4. EXCEPTIONAL INTERSECTIONS AND TANGENT FOLIATIONS

The quantities $J_{v,e}$ in Lemma 3.2 measure more than the number of blowups incident to a factor. They are first moments of exceptional facets, and they can be recovered from ordinary intersection numbers on the blowup centres. We establish this interpretation without assuming that the remaining factors are indexed by a tree. It identifies the tangent subsheaf whose degree is being estimated and gives a criterion in terms of a specified collection of exceptional contractions.

4.1. The radial foliation. Let B be a smooth projective toric variety and let $b \geq 2$. Choose a torus-fixed point $p \in \mathbb{P}^b$, distinct invariant hyperplanes $\mathcal{H}_1, \dots, \mathcal{H}_m$ through p , where $0 \leq m \leq b$, and distinct invariant prime divisors $D_1, \dots, D_m$ on B . Let

$$\pi : X \longrightarrow \mathbb{P}^b \times B$$

be the successive blowup of the strict transforms of $\mathcal{H}_i \times D_i$. Assume that X is Fano, and put

$$(25) \quad n = \dim X, \quad H = -K_X, \quad f = \text{pr}_1 \circ \pi, \quad L = f^* \mathcal{O}_{\mathbb{P}^b}(1).$$

The pairs of rays defining the centres are disjoint. In every original smooth cone their subdivisions act on disjoint pairs of basis vectors. They therefore commute, all the centres remain smooth, and each exceptional divisor E_i can be contracted while retaining the other blowups. The centres themselves need not be disjoint.

Projection from p induces a rational map

$$X \dashrightarrow \mathbb{P}^{b-1} \times B.$$

Let $\mathcal{F}$ be the saturation in T_X of the tangent directions to its general fibres. Thus $\mathcal{F}$ is the birational transform of the foliation by lines through p , with the B -coordinate fixed.

Proposition 4.1. *The sheaf $\mathcal{F}$ is a rank-one toric foliation and $\mathcal{F} \simeq \mathcal{O}_X(L)$. Its normalized slope is*

$$(26) \quad A := \frac{\mu_H(\mathcal{F})}{\mu_H(T_X)} = \frac{nLH^{n-1}}{H^n} = \frac{d}{dt} \log \text{vol}(H + tL) \Big|_{t=0}.$$

In particular, $A \geq 1$ excludes stability of T_X , and $A > 1$ gives a strict rank-one destabilizer.

Proof. Choose basis rays $r_1, \dots, r_b$ for the hyperplanes through p and put $r_0 = -\sum_j r_j$. On the dense torus, the radial foliation has tangent line $\mathbb{C}r_0$. Since $b \geq 2$, no other ray of the projective-space factor lies on this line. Rays of B lie in the other lattice summand, and the new exceptional rays have nonzero components in both summands. Thus r_0 is the only ray on the line. The induced tangent filtrations from Lemma 2.1 give $c_1(\mathcal{F}) = D_{r_0}$. The invariant hyperplane opposite p contains none of the blowup centres, so its strict transform has class $D_{r_0} = L$.

A saturated rank-one subsheaf of a locally free sheaf on a smooth variety is reflexive: the quotient is torsion-free, and the depth lemma gives depth at least two in codimension at least two. It is therefore a line bundle. This proves the sheaf identification; it does not assert that the foliation is nonsingular everywhere. The first equality in (26) now follows from $\mu_H(T_X) = H^n/n$. Near the ample class H , algebraic volume is the top intersection number. Differentiating $(H + tL)^n$ proves the last equality and the slope conclusions. $\square$

The restriction $b \geq 2$ is essential. For a $\mathbb{P}^1$ factor the two original rays lie on the same line, and both contribute to its tangent subsheaf. This is why the leaf estimates in Lemma 3.2 are kept separate from the present foliation statement.

4.2. Mixed intersections on the exceptional centres. Contract only E_i , obtaining $\pi_i : X \rightarrow Y_i$ with smooth centre Z_i of dimension $d = n - 2$. The centre is the transverse intersection of the transforms of $\mathcal{H}_i \times B$ and $\mathbb{P}^b \times D_i$. Their normal line bundles on Z_i , in this order, will be denoted $N_{i,A}$ and $N_{i,B}$. Thus

$$N_{Z_i/Y_i} = N_{i,A} \oplus N_{i,B}, \quad q_i : E_i = \mathbb{P}_{\text{lines}}(N_{i,A} \oplus N_{i,B}) \longrightarrow Z_i.$$

Let $S_{i,0}$ be the section corresponding to the normal direction $N_{i,B}$, and let $S_{i,1}$ correspond to $N_{i,A}$. The first is the intersection with the strict transform of $\mathcal{H}_i \times B$. Identify the sections with Z_i and set

$$(27) \quad M_{i,0} = H|_{S_{i,0}} = -K_{Z_i} + N_{i,A}, \quad M_{i,1} = H|_{S_{i,1}} = -K_{Z_i} + N_{i,B}.$$

These are ample classes. The displayed equalities use additive notation for line bundles. Indeed, if $\xi = c_1(\mathcal{O}_{\mathbb{P}_{\text{lines}}(N)}(1))$, then

$$E_i|_{E_i} = -\xi, \quad H|_{E_i} = q_i^*(-K_{Y_i}|_{Z_i}) + \xi,$$

while ξ restricts to $-N_{i,B}$ on $S_{i,0}$ and to $-N_{i,A}$ on $S_{i,1}$. Adjunction gives (27).

For the next proposition all products of the $M_{i,j}$ are top intersection numbers on Z_i . Define

$$(28) \quad Q_i = \sum_{k=0}^{n-2} (k+1) M_{i,0}^{n-2-k} M_{i,1}^k, \quad C_i = \frac{n E_i H^{n-1}}{H^n}.$$

The order of the two normal directions is part of these data.

Proposition 4.2. *Let P be the anticanonical polytope of X and F_i its facet corresponding to E_i . On F_i , let t be the affine coordinate which is zero and one on the faces corresponding to $S_{i,0}$ and $S_{i,1}$, respectively. Then*

$$(29) \quad J_i := \frac{1}{\text{vol}(P)} \int_{F_i} t \, d\text{vol} = \frac{Q_i}{H^n},$$

$$(30) \quad C_i = \frac{n}{H^n} \sum_{k=0}^{n-2} M_{i,0}^{n-2-k} M_{i,1}^k.$$

Interchanging the sections gives a positive contribution J_i^{opp} with $J_i + J_i^{\text{opp}} = C_i$. If $M_{i,1} - M_{i,0}$ is nef, then $J_i \geq C_i/2$; if $M_{i,0} - M_{i,1}$ is nef, the inequality reverses. Numerically equal endpoint classes give $J_i = C_i/2$.

Proof. Let a_i and c_i be the affine support coordinates of the two original rays defining the i -th blowup. The exceptional equation is $a_i + c_i = 1$, with $a_i, c_i \geq 0$, so $t = a_i$ has the stated endpoints. The original two rays extend to an integral basis. On the exceptional facet, an integral tangent vector can therefore increase a_i by one and decrease c_i by one. Thus its lattice measure disintegrates as dt times the lattice measure on the centre, without an index factor.

The slice at t is, up to translation, the moment polytope of

$$M_{i,t} = (1-t)M_{i,0} + tM_{i,1}.$$

One way to see this is to restrict the supporting inequalities of $H|_{E_i}$ along the two sections: their constants interpolate affinely, giving precisely the supporting inequalities of $M_{i,t}$. Both endpoints are ample, so the normal fan remains the fan of Z_i along the entire segment. The slice volume is consequently $M_{i,t}^{n-2}/(n-2)!$. Since $\text{vol}(P) = H^n/n!$,

$$(31) \quad J_i = \frac{n(n-1)}{H^n} \int_0^1 t M_{i,t}^{n-2} dt.$$

For $d = n-2$, the coefficient of $M_{i,0}^{d-k} M_{i,1}^k$ in the integral is

$$\binom{d}{k} \int_0^1 t^{k+1} (1-t)^{d-k} dt = \frac{k+1}{(d+1)(d+2)}.$$

This proves (29). Integration without t gives (30), also the projective-bundle formula for $(H|_{E_i})^{n-1}$. Replacing t by $1-t$ replaces the coefficient $k+1$ by $n-1-k$, so the two contributions sum to C_i . Their positivity follows from ampleness.

Finally, write $I_k = M_{i,0}^{d-k} M_{i,1}^k$. If $M_{i,1} - M_{i,0}$ is nef, then

$$I_{k+1} - I_k = M_{i,0}^{d-k-1} M_{i,1}^k (M_{i,1} - M_{i,0}) \geq 0.$$

Pairing k with $d - k$ in $\sum_k (k - d/2)I_k$ proves $J_i - C_i/2 \geq 0$. The reverse nef condition and numerical equality are treated identically. $\square$

Theorem 4.3. *For the blowup and foliation in (25),*

$$(32) \quad (b+1)A = b + \sum_i J_i, \quad A - 1 = \frac{\sum_i Q_i - H^n}{(b+1)H^n}.$$

Consequently $\sum_i Q_i \geq H^n$ excludes stability of T_X , and strict inequality makes $\mathcal{F}$ a strict destabilizer. Equivalently, the total contribution is

$$(33) \quad \begin{aligned} \sum_i J_i &= n \frac{f^*(-K_{\mathbb{P}^b})H^{n-1}}{H^n} - b \\ &= (n-b) - n \frac{(-K_{X/\mathbb{P}^b})H^{n-1}}{H^n}. \end{aligned}$$

Here $-K_{X/\mathbb{P}^b} = H - f^*(-K_{\mathbb{P}^b})$ denotes the relative anticanonical divisor class.

Proof. In central affine coordinates the anticanonical polytope has inequalities

$$a_j \geq 0, \quad \sum_{j=1}^b a_j \leq b+1, \quad a_i + c_i \geq 1 \quad (1 \leq i \leq m),$$

together with the inequalities from the rays of B . Apply divergence to $\sum_j a_j \partial/\partial a_j$, whose divergence is b . The coordinate facets and the facets from B have zero flux. The opposite central facet contributes $(b+1)$ times its lattice volume, and F_i contributes $-\int_{F_i} a_i$. As in Lemma 3.2, the conversion between Euclidean and lattice facet measures cancels the normal-length factor. Division by $\text{vol}(P)$ gives the first identity in (32). Proposition 4.2 gives the second, and Proposition 4.1 gives the slope conclusions. Substituting $f^*(-K_{\mathbb{P}^b}) = (b+1)L$ proves (33). $\square$

Thus an exceptional divisor contributes its normalized degree C_i multiplied by the mean of t in its slice-volume distribution. The two section polarizations determine how this degree is divided. For example, the geometric hypotheses $M_{i,1} - M_{i,0}$ nef for every i and $\sum_i C_i \geq 2$ imply failure of stability; a strict weighted inequality implies instability. We do not impose these nef hypotheses in the tree theorems below. The identity, which does not require them, retains the effect of all other blowups through Z_i and $M_{i,0}, M_{i,1}$. It also distinguishes an individual foliation test from a test of every subsheaf: $\sum_i Q_i < H^n$ controls $\mathcal{F}$ alone.

4.3. Geometric slice profiles. There is a second useful description when $m = b$ and $B = \prod_{i=1}^b B_i$, with D_i pulled back from a smooth invariant prime divisor on B_i . Put $d_i = \dim B_i$ and $h_i = -K_{B_i}$. Assume that h_i and $h_i - D_i$ are ample, as well as $H = -K_X$. Define the positive continuous functions

$$(34) \quad p_i(a) = \begin{cases} (h_i - (1-a)D_i)^{d_i}/d_i!, & 0 \leq a \leq 1, \\ h_i^{d_i}/d_i!, & a \geq 1. \end{cases}$$

These functions express the volume of a divisor segment on the base.

Proposition 4.4. *In this product-base setting, let*

$$Z(t) = \int_{\substack{a_i \geq 0 \\ \sum_i a_i \leq t}} \prod_i p_i(a_i) da_1 \cdots da_b.$$

Then

$$(35) \quad A = \frac{Z'(b+1)}{Z(b+1)}, \quad \frac{p'_i(a)}{p_i(a)} = \frac{d_i D_i (h_i - (1-a)D_i)^{d_i-1}}{(h_i - (1-a)D_i)^{d_i}} \quad (0 < a < 1).$$

If q_i are positive continuous log-concave functions, constant beyond one, and every p_i/q_i is nonincreasing, then A is at most the corresponding ratio $Z'_q(b+1)/Z_q(b+1)$. Nondecreasing ratios give the reverse inequality.

Proof. At fixed $a_i \in [0, 1]$, the exceptional constraint moves the facet of D_i inward by $1 - a_i$. The remaining polytope is therefore the moment polytope of $h_i - (1 - a_i)D_i$. The ample endpoint assumption keeps the whole segment in the same ample cone and gives exactly (34). For $a_i \geq 1$ the new constraint is redundant. Independence of the base factors gives the product integrand. Hence $Z(b+1)$ is the full polytope volume, and $Z'(b+1)$ is the lattice volume of the opposite central facet. Their ratio is A . Differentiating the intersection polynomial gives the second formula in (35). The comparison is Lemma 6.2, proved below from Efron's conditional-sum theorem; it requires log-concavity of the reference profiles only. $\square$

For a vertex v of a blowup tree, deleting v separates the remaining construction into factors B_i . The divisor at the former parent incidence becomes an unassigned divisor D_i on its component. Its class is the pullback of a hyperplane at that component's root. The classes $h_i - D_i$ are ample: after deleting the parent edge, every primitive relation of root type has degree at least two. Subtracting D_i lowers each such degree by one, while edge and non-root primitive relations are unchanged. Thus the incoming functions in the tree calculations are precisely the intersection polynomials (34).

The proof of Theorem 6.1 bounds their logarithmic derivatives above by $3/(2+a)$ when every nonleaf dimension equals its valence and the maximum degree is three. The degree-four obstruction instead supplies four profiles bounded below in ratio order by (46). Proposition 4.4 allows either comparison to be tested on other polarized base pairs, without assigning a construction graph to those factors. Outside the tree criterion, an upper bound concerns only the radial foliation; a lower bound exceeding one still proves instability of T_X .

Neither this description nor (33) asserts that a large Picard number forces the required contraction data or intersection inequality. That is a separate structural problem. Also, the edges of the bundle constructions in Section 8 represent twists, not the exceptional divisors used here; their subdivision theorems do not assert monotonicity of these blowup weights.

5. BRANCHING AND NEIGHBOURING FACTOR DIMENSIONS

We retain the mixed-blowup construction $Y(G, b)$ of Section 3, with distinct original rays assigned to the incidences at each vertex. Write $d_v = \deg_G(v)$. Throughout this section assume $b_v \geq d_v$ and $b_v \geq 1$ at every vertex, so that $Y(G, b)$ is Fano. A vertex v is *saturated* if $b_v = d_v$; equivalently, exactly one of its original rays is unassigned. The degree of its unassigned divisor depends on the geometry beyond its incident centres. We first determine stability for saturated stars, including the stable degree-four and degree-five examples. We then show that replacing the rest of the geometry by independent leaves of the same neighbouring dimensions can only decrease this degree. This comparison allows cycles and does not require saturation at the other vertices.

5.1. The exact transition for stars. Let Y_m be the mixed blowup associated with a star with $m \geq 1$ edges, central dimension m , and dimension one at each leaf. In the product

lattice, choose bases $e_1, \dots, e_m$ for the central factor and u_i for leaf i . Put

$$e_0 = -\sum_{i=1}^m e_i, \quad s_i = e_i + u_i.$$

Thus Y_m is obtained from $\mathbb{P}^m \times (\mathbb{P}^1)^m$ by subdividing the m original cones $\langle e_i, u_i \rangle$. Let A_m denote the normalized divisor degree of e_0 , with the normalization used in Section 3.

Theorem 5.1. *For every $m \geq 1$, the variety Y_m is smooth Fano, has dimension $2m$ and Picard number $2m + 1$, and admits no torus-equivariantly locally trivial bundle presentation with positive-dimensional complete fibre and positive-dimensional base. Its tangent bundle is anticanonically stable exactly when $1 \leq m \leq 5$. For every $m \geq 6$, the line $\mathbb{C}e_0$ defines a strictly destabilizing saturated subsheaf.*

The proof, including the finite sign calculation and the uniform tail estimate, is given in Appendix A.

For the two stable stars with higher valence, the central degrees and total exceptional contributions are

$$A_4 = \frac{44080}{47119}, \quad \sum_i J_{0,i} = \frac{31924}{47119} < 1; \quad A_5 = \frac{100780}{103289}, \quad \sum_i J_{0,i} = \frac{88235}{103289} < 1.$$

These values follow from the table and (14). They show that neither four nor five incident blowups alone forces the radial foliation of Proposition 4.1 to destabilize. The neighbouring slice profiles, rather than the number of contributions, determine whether their sum reaches one.

The same line is the first Harder–Narasimhan term for $m \geq 6$ (Corollary A.1).

5.2. Comparison at a saturated vertex. The star calculation gives a necessary condition for stability in the entire simple-graph construction. In the next statement, the graph need not be connected and may contain cycles.

Theorem 5.2. *Let G be a finite simple graph with positive dimensions $b_w \geq d_w$ at every vertex, and use injective incidence assignments in the construction of $Y(G, b)$. Suppose v is saturated and $b_v = d_v = m \geq 1$. Let A_v be the normalized divisor degree of its unique unassigned original ray. Enumerate the neighbors as $w_1, \dots, w_m$ and put $c_i = b_{w_i}$. Define*

$$\phi_c(a) = \frac{(c + \min(a, 1))^c}{c!}, \quad Z_c(t) = \int_{\substack{a_i \geq 0 \\ \sum_i a_i \leq t}} \prod_{i=1}^m \phi_{c_i}(a_i) da.$$

Then

$$A_v \geq \frac{Z'_c(m+1)}{Z_c(m+1)} \geq A_m.$$

The first ratio is nondecreasing in each neighboring dimension c_i . The dimensions at vertices other than v may exceed their degrees.

Proof. At every vertex w , choose an unassigned original ray as the negative sum ray, and use the other b_w rays as a lattice basis. This is possible because $b_w \geq d_w$. In affine coordinates $a_{w,j} = 1 + \langle x, r_{w,j} \rangle$, the anticanonical polytope is

$$(36) \quad a_{w,j} \geq 0, \quad \sum_{j=1}^{b_w} a_{w,j} \leq b_w + 1, \quad a_{w,e} + a_{z,e} \geq 1 \quad (e = \{w, z\}).$$

Here $a_{w,e}$ is the coordinate assigned to the incidence (w, e) . The coordinate change preserves lattice volume.

Write $a = (a_1, \dots, a_m)$ for the coordinates at v . Before imposing the simplex inequality at v , integrate over all other vertices and denote the resulting volume by $W(a)$, for every $a_i \geq 0$. It is bounded by the product of the other simplex volumes and is coordinatewise nondecreasing, since increasing a_i only relaxes one inequality. It is constant in each coordinate once that coordinate is at least one.

Moreover, $W(0) > 0$. At every other vertex choose its assigned coordinates equal to $1 + \varepsilon$ and its remaining coordinates equal to ε . For sufficiently small $\varepsilon > 0$, the sum at w is $d_w + b_w \varepsilon < b_w + 1$, and all mixed inequalities are strict. An open set of such choices has positive volume. Monotonicity gives $W(a) > 0$ throughout the nonnegative orthant. The function W is continuous: its domain is contained in a fixed compact product of simplexes, and under a convergent change of a the defining indicators converge away from finitely many affine hyperplanes of measure zero.

We first prove that

$$(37) \quad R(a) = \frac{W(a)}{\prod_{i=1}^m \phi_{c_i}(a_i)}$$

is coordinatewise nondecreasing. Fix every central coordinate except a_i , and let w be the other endpoint of its edge. Put $b = b_w$. At w , let x be the coordinate assigned to this edge and let $z = (z_1, \dots, z_{b-1})$ be its remaining coordinates. Injectivity of the incidence assignments ensures that no other mixed inequality uses x . Simplicity ensures that w has no second edge to v .

Temporarily omit the simplex inequality at w and integrate over every vertex other than v, w . The resulting function $\Psi(z)$ is nonnegative and coordinatewise nondecreasing on the nonnegative orthant: increasing a component of z only enlarges the feasible set. It may depend on the other fixed central coordinates, but it is independent of x and a_i . Edges among the other vertices remain inside this integral, so no independence of neighbouring components is required.

For $0 \leq a_i \leq 1$, the interval for x is $[1 - a_i, b + 1 - \sum_j z_j]$. With $s = b + a_i$, integration over this interval and then the substitution $z = sy$ give

$$(38) \quad \begin{aligned} W(a) &= \int_{\substack{z_j \geq 0 \\ \sum_j z_j \leq s}} (s - \sum_j z_j) \Psi(z) dz \\ &= s^b \int_{\substack{y_j \geq 0 \\ \sum_j y_j \leq 1}} (1 - \sum_j y_j) \Psi(sy) dy. \end{aligned}$$

When $b = 1$, the last integral is zero-dimensional and the same identity holds. The integral factor is nondecreasing in s . Thus, with the other coordinates fixed and $0 \leq a \leq a' \leq 1$,

$$\frac{W(a')}{W(a)} \geq \left(\frac{b + a'}{b + a} \right)^b = \frac{\phi_b(a')}{\phi_b(a)}.$$

This proves monotonicity of R below one in each coordinate. Above one both W and ϕ_{c_i} are constant, and continuity gives the assertion across one. The function R is also bounded, since each ϕ_{c_i} has a positive lower bound.

To compare the simplex integrals, introduce independent random variables $X_1, \dots, X_m$ with respective densities

$$f_i(x) = C_i e^{-x} \phi_{c_i}(x) \mathbf{1}_{\{x \geq 0\}},$$

where $C_i > 0$ is the normalizing constant. Each density is log-concave. On $(0, 1)$ its logarithmic derivative is $-1 + c_i/(c_i + x)$, which decreases; at one this derivative jumps downward to -1 and stays constant thereafter. Its support is the interval $[0, \infty)$.

Efron's conditional-sum theorem states that, for independent log-concave random variables, the conditional expectation of a bounded coordinatewise nondecreasing function is nondecreasing in their sum; the version for general log-concave densities and any finite number of variables is given in [SW18, Theorems 3.2–3.3]. Extend R to $\mathbb{R}^m$ by replacing each argument by its nonnegative part. The extension is bounded and coordinatewise nondecreasing. The theorem therefore implies that

$$h(s) = \mathbb{E}[R(X_1, \dots, X_m) \mid X_1 + \dots + X_m = s]$$

has a nondecreasing version. When $m = 1$, the same assertion follows directly from $h(s) = R(s)$.

On the slice $\sum_i a_i = s$, the product exponential factor equals e^{-s} and cancels from the conditional expectation. Thus $h(s)$ is the ratio of the W -weighted slice volume to the $\prod_i \phi_{c_i}(a_i)$ -weighted slice volume. These integrals are continuous for $s > 0$, by rescaling the compact sum simplex to a fixed simplex and applying dominated convergence. The denominator is positive. Their ratio consequently agrees everywhere on $(0, \infty)$ with the nondecreasing version supplied by the theorem.

Set

$$F(t) = \int_{\substack{a_i \geq 0 \\ \sum_i a_i \leq t}} W(a) da, \quad q(t) = Z'_c(t) > 0.$$

Using the lattice slice measure obtained by eliminating one coordinate, we have

$$F'(t) = q(t)h(t), \quad F(t) = \int_0^t q(s)h(s) ds \leq h(t)Z_c(t).$$

All quantities are positive, so

$$\frac{F'(t)}{F(t)} \geq \frac{Z'_c(t)}{Z_c(t)}.$$

At $t = m + 1$, the denominator on the left is the full anticanonical polytope volume and its numerator is the lattice volume of the unassigned facet at v . Their ratio is A_v , proving the first comparison.

For integers $b \geq c \geq 1$ and $0 < a < 1$,

$$\frac{d}{da} \log \frac{\phi_b(a)}{\phi_c(a)} = \frac{a(b-c)}{(b+a)(c+a)} \geq 0.$$

The ratio is constant above one. Applying the same conditional-sum argument to products of these ratios proves monotonicity in the neighboring dimensions. Taking every $c_i = 1$ gives $\phi_1(a) = 1 + \min(a, 1)$ and recovers the star ratio, proving the second comparison. $\square$

Corollary 5.3. *In the Fano simple-graph mixed-blowup construction, a saturated vertex of degree at least six forces anticanonical tangent instability. In particular, if every vertex is saturated and the tangent bundle is stable, then the maximum graph degree is at most five. The star Y_5 attains degree five.*

Proof. Let v be saturated of degree $m \geq 6$. Theorems 5.1 and 5.2 give $A_v \geq A_m > 1$. Its unassigned ray is the negative sum of m basis vectors and is not collinear with another original ray in that factor. Original rays at other vertices lie in different direct summands, and exceptional rays have nonzero components in two summands. Thus its line has normalized

ray weight exactly $A_v > 1$ and defines a strict rank-one destabilizer. The last assertion follows from Theorem 5.1. $\square$

Corollary 5.4. *If a saturated degree-five vertex has at least four neighbors of factor dimension at least two, its unassigned divisor degree satisfies*

$$A_v \geq \frac{6244655501}{6233518181} > 1,$$

and its unassigned ray defines a strict rank-one destabilizer. Consequently, in a saturated simple graph with stable tangent bundle, each degree-five vertex has at least two degree-one neighbors.

Proof. Theorem 5.2 reduces the assertion to the neighboring dimensions $(2, 2, 2, 2, 1)$. Their simplex and facet volumes at capacity six are

$$Z(6) = \frac{6233518181}{362880}, \quad Z'(6) = \frac{892093643}{51840}.$$

To specify these rational calculations explicitly, let $c_{j,k}$ be the coefficient of $q^j u^k$ in

$$(2u^2 + 2u + 1 - q(3u + 1))^4(u + 1 - q).$$

The Laplace transforms of ϕ_1, ϕ_2 are respectively $(s + 1 - e^{-s})/s^2$ and $(2s^2 + 2s + 1 - e^{-s}(3s + 1))/s^3$. Thus

$$Z(6) = \sum_{j,k} c_{j,k} \frac{(6-j)^{14-k}}{(14-k)!}, \quad Z'(6) = \sum_{j,k} c_{j,k} \frac{(6-j)^{13-k}}{(13-k)!},$$

where $0 \leq j \leq 5$ and $0 \leq k \leq 9$. These finite sums give the displayed values. The unassigned ray is the only ray on its line, by the same direct-summand argument as in Corollary 5.3. Its weight exceeds one, so the line strictly destabilizes. In the saturated case neighboring dimensions equal neighboring degrees; at most three of the five neighbors can therefore have degree at least two. $\square$

Corollary 5.5. *A saturated five-arm tree is unstable whenever at least four of its arms have length at least two. In particular, subdivision of four arms of the stable five-arm star produces an unstable mixed blowup, and further subdivision of those arms preserves this obstruction.*

Proof. The central vertex has degree five and at least four degree-two neighbors, so Corollary 5.4 applies. $\square$

Here subdivision refers to the indexing graph: its factor dimensions and the dimension of the variety change. It is not a fan subdivision in fixed dimension. The obstruction rules out a general stability theorem obtained merely by making every arm of a mixed-blowup tree sufficiently long. It concerns saturated vertices and places no corresponding degree bound on vertices whose dimensions exceed their degrees.

6. A SHARP UNIVERSAL DEGREE THRESHOLD FOR BLOWUP TREES

We now prove that every tree of maximum degree three gives a stable blowup when the nonleaf dimensions equal their valences; the leaf dimensions can be arbitrary. The argument controls the incoming volume profiles uniformly under both the one-child and two-child integration operations. Four neighbours with sufficiently large profiles instead force the central radial foliation to destabilize. This yields a sharp universal degree threshold, while allowing the stable degree-four and degree-five arrangements of Section 5. A family with two branching vertices gives further stable arrangements.

Theorem 6.1. *Let G be a finite tree with at least one edge and maximum degree at most three. Set $b_v = d_v$ at every nonleaf vertex, and choose any positive integer b_v at each leaf. Then the mixed blowup $Y(G, b)$ has anticanonically stable tangent bundle. If $M = |E(G)|$, then*

$$n = 2M + \sum_{v \text{ leaf}} (b_v - 1), \quad \rho = 2M + 1.$$

It is smooth Fano and admits no torus-equivariantly locally trivial bundle presentation with positive-dimensional complete fibre and base. The normalized degrees of unassigned original divisors satisfy

$$(39) \quad A_v \leq \frac{25950132}{26414113} < 1 \quad (d_v = 3), \quad A_v \leq \frac{24}{25} < 1 \quad (d_v = 2).$$

In particular, every saturated tree of maximum degree three gives such a stable variety with $n = 2M$ and $\rho = n + 1$.

There is no restriction on the distances between branching vertices or on the relative sizes of the components of $G - v$. The numerical bounds in (39) concern the indicated individual divisor degrees; they do not give a uniform slope gap for all proper subspaces or for leaves of arbitrary dimension.

In the notation of Section 4, the two internal-vertex bounds imply

$$\sum_{e \ni v} J_{v,e} \leq \frac{24558189}{26414113} < 1 \quad (d_v = 3), \quad \sum_{e \ni v} J_{v,e} \leq \frac{22}{25} < 1 \quad (d_v = 2).$$

The proof therefore bounds the sum of the ordered exceptional intersections, including their dependence on all farther branches. Theorem 3.4 is the additional step that promotes these local inequalities to stability of the entire tangent bundle.

6.1. Comparison of simplex integrals. We use two elementary integral constructions together with Efron's conditional-sum theorem. A function in this subsection is positive and continuous on $[0, \infty)$ and constant on $[1, \infty)$. For $k \geq 1$ set

$$V_{p_1, \dots, p_k}(t) = \int_{\substack{x_i \geq 0 \\ \sum_i x_i \leq t}} \prod_i p_i(x_i) dx, \quad F_{p_1, \dots, p_k}(t) = V'_{p_1, \dots, p_k}(t),$$

and

$$B_{p_1, \dots, p_k}(t) = \int_{\substack{x_i \geq 0 \\ \sum_i x_i \leq t}} (t - \sum_i x_i) \prod_i p_i(x_i) dx.$$

The derivative F is the integral on the sum slice in lattice measure, obtained by eliminating one coordinate with coefficient one.

Lemma 6.2. *Let p_i, q_i be functions as above, with q_i log-concave. If every p_i/q_i is nonincreasing, then*

$$\frac{B_{p_1, \dots, p_k}(t)}{B_{q_1, \dots, q_k}(t)} \quad \text{and} \quad \frac{F_{p_1, \dots, p_k}(t)}{F_{q_1, \dots, q_k}(t)}$$

are nonincreasing for $t > 0$, and

$$(40) \quad \frac{F_{p_1, \dots, p_k}(t)}{V_{p_1, \dots, p_k}(t)} \leq \frac{F_{q_1, \dots, q_k}(t)}{V_{q_1, \dots, q_k}(t)}.$$

If every p_i/q_i is nondecreasing, both monotonicity assertions and inequality (40) reverse.

Proof. Take independent variables X_i with densities proportional to $e^{-x}q_i(x)$ on $[0, \infty)$, and two further independent exponential variables of rate one. These densities are log-concave. The bounded positive function $R(x) = \prod_i p_i(x_i)/q_i(x_i)$ has the asserted coordinatewise monotonicity and is independent of the two extra coordinates. Extend it to real arguments by their nonnegative parts. Efron's conditional-sum theorem [SW18, Theorems 3.2–3.3] makes its conditional expectation monotone in the total sum. The nonincreasing version follows by applying the theorem to $-R$.

On a slice of total t , the common exponential factor cancels. Integrating the two additional coordinates gives the factor $t - \sum_i x_i$, so the conditional expectation is exactly $B_p(t)/B_q(t)$. Omitting these two variables instead gives $h(t) = F_p(t)/F_q(t)$; for $k = 1$ this is just the assumed ratio order. Rescaling to a fixed compact simplex proves continuity of all slice ratios for $t > 0$. Their positive denominators ensure that the monotone versions supplied by the theorem agree with these ratios everywhere. In the nonincreasing case,

$$V_p(t) = \int_0^t F_q(s)h(s) ds \geq h(t)V_q(t), \quad F_p(t) = F_q(t)h(t).$$

Division proves (40). The same calculation with the reversed integral inequality proves the other case. $\square$

The following coefficient identity specifies the finite rational calculations used below. For a function p as above define the formal power series

$$(41) \quad Q_p(z) = p(1) + z \sum_{j \geq 0} \frac{z^j}{j!} \int_0^1 (1-a)^j p(a) da.$$

Only finitely many coefficients are needed in each application.

Lemma 6.3. *For $k \geq 1$, $r \geq 0$, and an integer $s \geq 0$,*

$$(42) \quad \int_{\substack{a_i \geq 0 \\ \sum_i a_i \leq k+r}} \frac{(k+r - \sum_i a_i)^s}{s!} \prod_i p_i(a_i) da = [z^{k+s}] e^{rz} \prod_i Q_{p_i}(z).$$

For $r > 0$, the facet volume is

$$F_{p_1, \dots, p_k}(k+r) = [z^{k-1}] e^{rz} \prod_i Q_{p_i}(z).$$

Proof. Partition the domain according to the set S of coordinates in $[0, 1]$. Subtract one from each of the remaining $k - |S|$ coordinates. Their integration leaves

$$\frac{(r + \sum_{i \in S} (1 - a_i))^{k-|S|+s}}{(k - |S| + s)!} \prod_{i \notin S} p_i(1)$$

times the product of the short-coordinate weights. Express this power as an exponential coefficient, integrate the short coordinates over $[0, 1]^S$, and sum over S . The result is (42). Differentiate the case $s = 0$ with respect to r to obtain the facet formula. $\square$

6.2. The proof for maximum degree three. Fix an oriented edge towards a parent vertex. With the parent's incidence coordinate equal to a , integrate over the component on the other side of the edge and denote its volume by $p(a)$. These incoming functions are positive, continuous, and constant for $a \geq 1$. Indeed, the only inequality involving the fixed coordinate is $a + x \geq 1$; above one it is redundant. Positivity follows by choosing all assigned coordinates slightly greater than one and all remaining coordinates small and

positive. The inequalities $b_v \geq d_v$ leave room in every vertex simplex. Continuity follows by dominated convergence inside the fixed compact product of these simplexes. Since removing a vertex separates the tree into disjoint components, the incoming functions multiply when its coordinates are fixed.

For a leaf of dimension b , the incoming function is

$$\phi_b(a) = \frac{(b + \min(a, 1))^b}{b!}.$$

At a nonleaf of degree two or three, integration of the coordinate assigned to the parent gives, for $0 \leq a \leq 1$,

$$(43) \quad T(p)(a) = B_p(2 + a),$$

$$(44) \quad \mathcal{G}(p, q)(a) = B_{p,q}(3 + a),$$

followed by constant continuation. There are respectively one and two child coordinates; the remaining interval has length $2 + a - x$ or $3 + a - x - y$.

Proof of Theorem 6.1. Put $P(a) = (2 + \min(a, 1))^3$, and write $p \preceq P$ if p/P is nonincreasing. The reference P is an auxiliary function, not the volume of an additional projective-space factor. It is log-concave: its logarithmic derivative is $3/(2 + a)$ below one and drops to zero above one. Every leaf satisfies $\phi_b \preceq P$, since

$$\frac{d}{da} \log \frac{\phi_b(a)}{P(a)} = \frac{b}{b+a} - \frac{3}{2+a} \leq 1 - \frac{3}{2+a} \leq 0 \quad (0 < a < 1).$$

This includes every positive leaf dimension.

We next prove closure under (43)–(44). Direct integration, or Lemma 6.3, gives

$$\begin{aligned} T(P)(a) &= \frac{363}{10} + \frac{173}{4}a + \frac{27}{2}a^2, \\ \mathcal{G}(P, P)(a) &= \frac{119659}{80} + \frac{148781}{80}a + \frac{3213}{4}a^2 + \frac{243}{2}a^3. \end{aligned}$$

For a polynomial g of degree at most three,

$$\left(\frac{g(a)}{(2+a)^3} \right)' = \frac{(2+a)g'(a) - 3g(a)}{(2+a)^4}.$$

The numerator coefficients are as follows; the cubic coefficient is zero.

g	1	a	a^2
$T(P)$	$-112/5$	$-65/2$	$-27/2$
$\mathcal{G}(P, P)$	$-12283/16$	$-20261/40$	$-297/4$

All are negative. Hence $T(P) \preceq P$ and $\mathcal{G}(P, P) \preceq P$. By Lemma 6.2, $T(p)/T(P)$ and $\mathcal{G}(p, q)/\mathcal{G}(P, P)$ are nonincreasing whenever $p, q \preceq P$. Multiplying positive nonincreasing ratios proves $T(p), \mathcal{G}(p, q) \preceq P$. Induction on the number of vertices in each oriented component now gives $p \preceq P$ for every incoming function. The two child inputs need not be equal.

At a degree-three vertex, the simplex capacity is four. For three reference inputs P , the volume and facet volume are

$$V_{P,P,P}(4) = \frac{26414113}{320}, \quad F_{P,P,P}(4) = \frac{6487533}{80}.$$

All arithmetic in these values and the two preceding reference polynomials follows from

$$Q_P(z) = 27 + \frac{65}{4}z + \frac{131}{20}z^2 + \frac{233}{120}z^3 + O(z^4)$$

and Lemma 6.3: the root values are the coefficients of z^3 and z^2 in $e^z Q_P(z)^3$, while the outgoing polynomials are $[z^2]e^{(1+a)z} Q_P(z)$ and $[z^3]e^{(1+a)z} Q_P(z)^2$. Inequality (40) therefore gives $A_v \leq 25950132/26414113$, whose deficit from one is $463981/26414113 > 0$.

At a degree-two vertex, each incoming function satisfies

$$\frac{p(a)}{p(1)} \geq \frac{P(a)}{P(1)} \geq a \quad (0 \leq a \leq 1), \quad (2+a)^3 - 27a = (1-a)^2(a+8) \geq 0.$$

Thus both inputs satisfy precisely (20). The two-input calculation (22)–(23), which requires only those inequalities and the constant continuation, gives $24V - 25F \geq 0$. Hence $A_v \leq 24/25$. At every leaf, Lemma 3.2 gives $A_v < 1$, also when its dimension exceeds one. All vertices satisfy Theorem 3.4, which proves stability against all proper subspaces and hence against all subsheaves. The geometric assertions and counts follow from Proposition 3.1 and $\sum_v d_v = 2M$. $\square$

6.3. A local obstruction at degree four.

Theorem 6.4. *Let $Y(G, b)$ be a Fano mixed blowup indexed by a finite tree, with $b_u \geq d_u$ at every vertex. Suppose that a vertex v and its four neighbors all have degree four and factor dimension four. Then*

$$(45) \quad A_v \geq \frac{40090063176909393024}{39498346443818421703} > 1.$$

The unassigned ray at v defines a strict rank-one destabilizer. No additional condition is imposed on the branches beyond these neighbors.

Proof. Let $\ell(a) = 1 + \min(a, 1)$. Every incoming branch function p in a Fano tree satisfies that p/ℓ is nondecreasing. To see this, let b be the dimension of its first vertex, x the coordinate assigned to the parent, and z its other $b-1$ coordinates. After integration over more distant vertices, their volume $\Psi(z)$ is coordinatewise nondecreasing, since increasing a coordinate only relaxes mixed inequalities. It is independent of x . For $0 \leq a \leq 1$, integration of x from $1-a$ to $b+1 - \sum z_j$ and rescaling give

$$p(a) = (b+a)^b \int_{\substack{y \geq 0 \\ \sum_j y_j \leq 1}} (1 - \sum_j y_j) \Psi((b+a)y) dy.$$

The integral factor is nondecreasing; the same formula includes $b=1$ as a zero-dimensional integral. Moreover,

$$\frac{d}{da} \log \frac{(b+a)^b}{1+a} = \frac{a(b-1)}{(b+a)(1+a)} \geq 0.$$

Both functions are constant above one, which proves the claim.

At a degree-four neighbor w of v , the incoming function is $g_w(a) = B_{p_1, p_2, p_3}(4+a)$ on $[0, 1]$, where p_i are its three child inputs. Apply the increasing case of Lemma 6.2 with reference inputs ℓ . This function is log-concave, and each p_i/ℓ is nondecreasing. It follows that g_w/g_4 is nondecreasing, where

$$(46) \quad g_4(a) = B_{\ell, \ell, \ell}(4+a) = \frac{5217 + 5965a + 2610a^2 + 520a^3 + 40a^4}{120} \quad (0 \leq a \leq 1),$$

with constant continuation.

For a further application of the lemma, we check log-concavity of g_4 . Writing $H = 120g_4$ on $[0, 1]$, direct differentiation gives

$$(H')^2 - HH'' = 8348485 + 14860260a + 11120040a^2 \\ + 4474400a^3 + 1020000a^4 + 124800a^5 + 6400a^6.$$

Its coefficients are positive. Also $H' > 0$, so the logarithmic derivative drops to zero at the constant continuation. Thus g_4 is log-concave on the whole half-line.

The four components at v supply four functions whose ratios to g_4 are nondecreasing. Their product is the outside volume at fixed central coordinates. The increasing case of (40) compares A_v with the ratio for four copies of g_4 at capacity five. The reference values are

$$V_0(5) = \frac{39498346443818421703}{23224320000}, \quad F_0(5) = \frac{104401206189868211}{60480000}.$$

For their exact evaluation, formula (41) gives

$$Q_{g_4}(z) = \frac{598}{5} + \frac{3683}{48}z + \frac{513}{16}z^2 + \frac{195899}{20160}z^3 + \frac{11549}{5040}z^4 + O(z^5).$$

Lemma 6.3 identifies $V_0(5), F_0(5)$ with $[z^4]e^z Q_{g_4}(z)^4$ and $[z^3]e^z Q_{g_4}(z)^4$. Their ratio proves (45); its excess over one is $591716733090971321/39498346443818421703$. The unassigned ray is the negative sum of four independent basis vectors and is the only ray on its line: other original rays lie on different lines or in different vertex summands, and exceptional rays have nonzero components in two summands. That line has normalized weight exactly $A_v > 1$, proving strict instability. $\square$

Corollary 6.5. *Three is the largest integer D for which every saturated mixed-blowup tree of maximum degree at most D has anticanonically stable tangent bundle. A counterexample at degree four has dimension 32 and Picard number 33.*

Proof. Theorem 6.1 proves the assertion for $D = 3$. For $D = 4$, take one degree-four vertex joined to four degree-four vertices, each with three additional pendant leaves, and saturate every vertex. There are sixteen edges, five dimension-four factors, and twelve unit leaves, so $n = 32$ and $\rho = 33$. All comparisons in Theorem 6.4 are equalities in this case. Thus (45) gives an unstable example of maximum degree four, which also excludes every $D \geq 4$. $\square$

6.4. Two branching vertices with arbitrary separation. The sharp universal threshold does not preclude stable trees containing degree-four or degree-five vertices. The following smaller family allows an arbitrary distance between two such vertices, while keeping their pendant edges of length one.

Proposition 6.6. *Let $d, e \in \{3, 4, 5\}$. Join vertices of degrees d, e by a path of $L \geq 1$ edges, and attach respectively $d - 1, e - 1$ unit-length pendant edges. Saturate every vertex. The resulting mixed blowup is smooth Fano, has anticanonically stable tangent bundle, and admits no nontrivial torus-equivariantly locally trivial bundle presentation. Writing $M = d + e + L - 2$, its dimension and Picard number are $n = 2M$ and $\rho = 2M + 1$.*

The proof is in Appendix A. It bounds the degree-two vertices uniformly and proves positivity at the two branching vertices by a linear functional on their incoming slice polynomials.

The fixed pendant lengths in Proposition 6.6 are essential to its statement. Corollary 5.4 prevents an extension allowing all pendant arms at a degree-five centre to have arbitrary lengths. The degree-four obstruction, in turn, concerns trees; its factorization argument does not establish the same assertion for simple graphs with cycles.

7. A SHARP PICARD BOUND FOR TORIC BUNDLES

We now consider bundles with fibre a product of degree-six del Pezzo surfaces and base a product of projective spaces. In this class the maximal Picard number of a stable example is $\lfloor 5n/4 \rfloor + 1$ for every $n \geq 4$. The upper bound allows arbitrary integral twists. Its proof combines three requirements: Fano positivity bounds the twisting at each base factor, stability requires each surface factor to meet at least two base factors, and stability excludes a product decomposition. The constructions in the following sections show that these requirements can be attained simultaneously.

7.1. Bundle fans and a relative tangent obstruction. Let F be a smooth projective toric variety with lattice N_F , and let $B = \prod_{c=1}^k \mathbb{P}^{b_c}$, where $k, b_c \geq 1$. Throughout this section a *toric bundle with fibre F over B* has the following explicit fan presentation. Its lattice is $N_F \oplus \bigoplus_c \mathbb{Z}^{b_c}$; its fibre rays are those of F in N_F ; and its other primitive rays are

$$(47) \quad f_a^c = (0, e_a^c) \quad (1 \leq a \leq b_c), \quad f_0^c = \left(t_c, -\sum_{a=1}^{b_c} e_a^c \right), \quad t_c \in N_F.$$

For every maximal fibre cone and every choice of one omitted ray from each set $\{f_0^c, \dots, f_{b_c}^c\}$, their remaining generators form a maximal cone of X , and these are all the maximal cones. This is the fan description of a locally trivial toric bundle [CLS11, Theorem 3.3.19]; it is part of our hypotheses, and no assertion about an arbitrary Mori fibre space is intended. The vectors t_c specify the twisting. The displayed normalization is obtained by an integral shear from the lifts of the standard base rays.

Write $P_F = \{x : \langle x, v \rangle \geq -1 \text{ for every fibre ray } v\}$ when F is Fano. All volume measures below are lattice normalized, without a factorial. The first obstruction only requires that P_F have barycentre zero. For the special case $F = \mathbb{P}^1$ and base dimension at least two, the strict inequality below recovers [BDGP21, Theorem 8.1].

Proposition 7.1 (Obstruction over projective space). *Let F be a smooth toric Fano variety of positive dimension f , and let $\pi : X \rightarrow \mathbb{P}^b$ be a toric bundle with the fan presentation (47) and fibre F , where $b \geq 1$. Assume that X is Fano and that the barycentre of the anticanonical polytope P_F is zero. Then*

$$\mu_{-K_X}(T_{X/\mathbb{P}^b}) \geq \mu_{-K_X}(T_X).$$

In particular, T_X is not anticanonically stable. The inequality is strict if $b \geq 2$ and the twisting vector t_1 in (47) is nonzero. It is an equality if $b = 1$ or $t_1 = 0$.

Proof. Write $n = f + b$ and specialize (47) to a single base factor. The lifted base rays are

$$(0, e_1), \dots, (0, e_b), \quad (t, -e_1 - \dots - e_b), \quad t \in N_F.$$

Put $s(x) = \langle x, t \rangle$ and $L(x) = b + 1 + s(x)$ for $x \in P_F$. The anticanonical polytope of X is

$$P_X = \{(x, y) : x \in P_F, y_i \geq -1, \sum_{i=1}^b (y_i + 1) \leq L(x)\}.$$

Ampleness of $-K_X$ implies $L(x) > 0$ on P_F . Indeed, the vertex of P_X corresponding to any maximal fibre cone together with the base cone generated by $e_1, \dots, e_b$ must satisfy the remaining base inequality strictly. This gives $L(x) > 0$ at every vertex of P_F , hence throughout P_F by affinity.

Let $A_q = \int_{P_F} L(x)^q dx$, with lattice-normalized Lebesgue measure. The slices are b -simplices, so

$$D := (-K_X)^n = \frac{n!}{b!} A_b.$$

Each of the $b+1$ base-ray divisors has the same degree

$$d = \frac{(n-1)!}{(b-1)!} A_{b-1}.$$

For the facets $y_i = -1$ this follows by integrating the corresponding $(b-1)$ -simplex volumes. All $b+1$ degrees are equal by the principal divisor relations from the characters dual to e_i . The relative tangent bundle has rank f and first Chern class equal to the sum of the fibre-ray divisors. Hence

$$\begin{aligned} \mu(T_{X/\mathbb{P}^b}) - \mu(T_X) &= \frac{D - (b+1)d}{f} - \frac{D}{n} \\ &= \frac{(n-1)!}{f(b-1)!} (A_b - (b+1)A_{b-1}) \\ &= \frac{(n-1)!}{f(b-1)!} \int_{P_F} s(x)(b+1+s(x))^{b-1} dx. \end{aligned}$$

The barycentre assumption implies $\int_{P_F} s(x) dx = 0$. Subtracting $(b+1)^{b-1} \int s$ from the last integral rewrites it as

$$\int_{P_F} s(x) \left((b+1+s(x))^{b-1} - (b+1)^{b-1} \right) dx.$$

This integrand is nonnegative, because the function $z \mapsto z^{b-1}$ is nondecreasing for $z > 0$. If $b \geq 2$ and $t \neq 0$, it is strictly positive whenever $s(x) \neq 0$. Since P_F is full-dimensional, that set has positive measure. If $b = 1$ or $t = 0$, the integrand vanishes. $\square$

Corollary 7.2 (The Assarf–Nill extremal bundles). *Let X_m be the toric variety associated with the smooth Fano polytope B_m of Assarf–Nill [AN16, Definition 41 and Proposition 43]. Thus $\dim X_m = 3m$, $\rho(X_m) = 4m+1$, and X_m is a toric S_6^m -bundle over $\mathbb{P}^m$, with nonzero twist having one ray component in each hexagon factor. Then T_{X_m} is not anticanonically stable for every $m \geq 1$, and is unstable for every $m \geq 2$.*

Proof. The anticanonical polytope of S_6^m is centrally symmetric and thus has barycentre zero. Apply the proposition with $b = m$ and $f = 2m$. $\square$

7.2. Incidences of hexagon fibre factors. Let S_6 be the blow-up of $\mathbb{P}^2$ at its three torus-fixed points. Its fan rays, in a suitable lattice basis r, q , are $\pm r, \pm q, \pm(r+q)$. Its anticanonical polytope is centrally symmetric. For fibre S_6^j , decompose $N_F = \bigoplus_{i=1}^j N_i$ and write $t_c = (t_{1c}, \dots, t_{jc})$. The *incidence graph of the twisting* is the bipartite graph on the fibre factors and base factors, with an edge ic precisely when $t_{ic} \neq 0$. Thus its edges record which base factors twist a given surface factor.

Theorem 7.3 (Picard bound for hexagon bundles). *Let $j, k \geq 1$, let $b_1, \dots, b_k \geq 1$, and let X be a locally trivial toric S_6^j -bundle over $\prod_{c=1}^k \mathbb{P}^{b_c}$, with the fan presentation (47). Assume that X is smooth Fano and that T_X is anticanonically stable. Then*

$$\rho(X) \leq n + \lfloor n/4 \rfloor + 1, \quad n = \dim X.$$

Here the twists in each hexagon lattice are arbitrary integral vectors; they are not required to have the graph form of Section 8. More precisely, if $s = \sum_c b_c$ and ℓ is the number of edges of the incidence graph, then

$$(48) \quad n = 2j + s, \quad \rho(X) = 4j + k, \quad 2j \leq \ell \leq s, \quad j + k - 1 \leq \ell.$$

Proof. Use the lattice decomposition and incidence graph defined above. Let ℓ be the number of incidences and put $s = \sum_c b_c$. Counting rays gives

$$n = 2j + s, \quad \rho(X) = 4j + k.$$

For each hexagon let ψ_i be the piecewise linear function equal to one on its primitive rays and linear on its fan cones. For an integral nonzero vector t , the positive integer $\psi_i(t)$ is the sum of its coefficients in the minimal containing unimodular cone. The rays of the c th base factor form a primitive collection: any proper subset lies in a lifted base cone, while the whole set does not. Their sum is t_c in the fibre lattice. Expressing it in the minimal containing product cone gives a primitive relation of degree

$$b_c + 1 - \sum_i \psi_i(t_{ic}).$$

The Fano condition makes this degree a positive integer [CLS11, Theorem 6.4.9(b)]. Therefore

$$\#\{i : t_{ic} \neq 0\} \leq \sum_i \psi_i(t_{ic}) \leq b_c, \quad \ell \leq s.$$

We next show that each hexagon vertex of the incidence graph has degree at least two. Let Q_i be its centrally symmetric anticanonical hexagon, and define

$$L_c(x) = b_c + 1 + \sum_i \langle x_i, t_{ic} \rangle, \quad G(x) = \prod_{c=1}^k \frac{L_c(x)^{b_c}}{b_c!}, \quad V = \int_{\prod_i Q_i} G(x) dx.$$

The Fano assumption gives $L_c > 0$ on the product of hexagons, by the same strict vertex-inequality argument as in the preceding proposition. The anticanonical polytope is a product of simplex slices over $\prod_i Q_i$, so V is its lattice volume. Since the base has positive dimension, $N_{i,\mathbb{C}}$ is a proper subspace, and no base ray lies in it. Its normalized ray-weight sum is

$$w(N_{i,\mathbb{C}}) = \frac{1}{V} \sum_{H \text{ facet of } Q_i} \int_{H \times \prod_{h \neq i} Q_h} G(x) d\sigma dx_{-i} = 2 + \frac{1}{V} \int_{\prod_i Q_i} x_i \cdot \nabla_i G(x) dx.$$

The second equality is the divergence theorem in the x_i variables. This is the slice-volume identity used in subspace concentration [HL14, Eq. (4.4)]; here we use the symmetry of the fibre factor, without assuming that the total polytope is centred. For a facet with primitive outward normal u and equation $\langle u, x_i \rangle = 1$, the normal component of x_i is $1/\|u\|$, exactly converting Euclidean facet measure to lattice facet measure. Thus the identity has no missing normalization factor.

If all t_{ic} vanish, then $w(N_{i,\mathbb{C}}) = 2$ and stability fails. If only t_{ic} is nonzero, write $z = \langle x_i, t_{ic} \rangle$ and $A = b_c + 1 + \sum_{h \neq i} \langle x_h, t_{hc} \rangle > 0$. Apart from a positive factor independent of x_i , the inner integral contributing to $\int x_i \cdot \nabla_i G$ is

$$b_c \int_{Q_i} z(A + z)^{b_c - 1} dx_i = b_c \int_{Q_i} z((A + z)^{b_c - 1} - A^{b_c - 1}) dx_i \geq 0.$$

Here $\int_{Q_i} z = 0$ by central symmetry, and positivity follows from monotonicity as before. Consequently $w(N_{i,\mathbb{C}}) \geq 2$, again contrary to stability. Thus $\ell \geq 2j$.

Finally, the incidence graph is connected. Otherwise, grouping the lattice summands according to its components gives a direct sum of proper subspaces containing all ray generators, so the ray matroid is disconnected, contrary to Lemma 2.3. A connected graph on $j + k$ vertices has at least $j + k - 1$ edges. Combining the inequalities gives

$$s \geq \ell \geq 2j, \quad s \geq \ell \geq j + k - 1.$$

Hence $n \geq 4j$ and

$$\rho(X) = 4j + k \leq 4j + s - j + 1 = n + j + 1 \leq n + \lfloor n/4 \rfloor + 1,$$

as claimed. $\square$

Corollary 7.4 (Necessary conditions for equality). *Under the hypotheses of Theorem 7.3, suppose that $n = 4j + r$ with $0 \leq r \leq 3$ and that $\rho(X) = \lfloor 5n/4 \rfloor + 1 = 5j + r + 1$. Then the incidence graph is a tree,*

$$k = j + r + 1, \quad \ell = \sum_c b_c = 2j + r, \quad \sum_{i=1}^j (\deg(i) - 2) = r.$$

Every nonzero t_{ic} is a primitive ray of its hexagon fan, and the number of incidences at each base vertex c is b_c .

Proof. Equality in the proof of Theorem 7.3 forces $s = \ell = j + k - 1$ and $j = \lfloor n/4 \rfloor$. Connectedness therefore makes the incidence graph a tree. Since the sum of all base capacities equals the number of incidences, every base inequality is an equality, and $\psi_i(t_{ic}) = 1$ for each nonzero twist. In a unimodular hexagon cone, a nonzero integral vector with coefficient sum one is a primitive ray. Finally $\sum_i \deg(i) = \ell$ gives the displayed excess-degree formula. $\square$

The coefficient $5/4$ follows from the separate geometric restrictions in (48). Each surface factor contributes two dimensions and four Picard classes, but stability forces at least two nonzero twists at that factor. Positivity of the base primitive relations requires a total base dimension at least equal to the number of such incidences. Thus the j surface factors require at least $2j$ base dimensions. Connectedness gives the remaining inequality needed to bound the number of base factors.

The equality conditions are necessary, not sufficient: they do not control the degrees of subspaces containing rays from several factors. Section 8 constructs the fans, and Section 10 proves stability for independently subdivided subcubic trees. Paths attain the bound when $4 \mid n$; the end attachments of Corollary 9.3 give attainment in every dimension $n \geq 4$. Hence Theorem 7.3 is a sharp bound in its stated bundle class. It is not an upper bound for arbitrary smooth toric Fano varieties.

8. HEXAGON BUNDLES OVER GRAPHS

We associate a toric fibre bundle to a graph whose vertices carry positive integers. Each edge contributes a degree-six del Pezzo surface, and each vertex contributes a projective-space factor to the base. The graph records which two base factors twist a given surface factor.

Definition 8.1. Let Γ be a finite nonempty graph without loops, with vertex set C and edge set I . Multiple edges are allowed. Write $k = |C|$ and $j = |I|$, choose an orientation $c_i^- \rightarrow c_i^+$ for every edge i , and assign an integer $b_c \geq 1$ to each vertex c . In the lattice

$$N = \bigoplus_{i \in I} \mathbb{Z}_i^2 \oplus \bigoplus_{c \in C} \mathbb{Z}^{b_c}$$

write $r_i = (1, 0)_i$, $q_i = (0, 1)_i$, and let $f_1^c, \dots, f_{b_c}^c$ be the standard basis of the last summand. Set

$$(49) \quad f_0^c = - \sum_{t=1}^{b_c} f_t^c + \sum_{c_i^+ = c} r_i - \sum_{c_i^- = c} r_i.$$

For each edge take the six rays generated by $\pm r_i, \pm q_i, \pm(r_i + q_i)$. Together with the rays generated by all f_t^c , these are the rays of Σ_Γ . Its maximal cones contain two cyclically adjacent hexagon rays for each i , and all but one of $f_0^c, \dots, f_{b_c}^c$ for each c ; all their faces are included. We write $X(\Gamma, b)$, or $X(\Gamma)$ when the integers are understood, for the associated toric variety.

The construction indeed gives a smooth projective fan, as proved below. The surface with the indicated hexagon fan is the blow-up of $\mathbb{P}^2$ at its three torus-fixed points; we denote it by S_6 , with the subscript indicating its anticanonical degree. Projection onto the last summands will identify $X(\Gamma)$ as a toric fibre bundle with fibre S_6^j over $\prod_c \mathbb{P}^{b_c}$. Reversing the orientation of an edge gives an isomorphic fan: multiplication by -1 on its summand $\mathbb{Z}_i^2$, and the identity on every other summand, gives the required lattice isomorphism. Thus no choice of edge orientation affects the variety.

We first record the primitive collections needed to test ampleness. A *primitive collection* of a simplicial fan is a set of rays which does not span a cone, whereas every proper subset does. For a complete simplicial fan, its *primitive relation* expresses the sum of these primitive ray generators as a positive linear combination of the ray generators of the unique cone whose relative interior contains that sum. The zero sum has the empty cone on the right.

Lemma 8.2. *Suppose a complete simplicial fan Σ is split by a base fan Σ_B and a fibre fan Σ_F : there is a lattice projection with kernel N_F and a subfan $\widehat{\Sigma}_B$ projecting cone by cone isomorphically onto Σ_B , and the cones of Σ are exactly $\widehat{\sigma} + \tau$, with $\widehat{\sigma} \in \widehat{\Sigma}_B$ and $\tau \in \Sigma_F$. The primitive collections of Σ are precisely the fibre primitive collections and the lifts of the base primitive collections. Fibre primitive relations are unchanged. In a lifted base primitive relation, the coefficients of the lifted base rays on the right are those of the base relation; the other terms are a nonnegative combination of fibre rays.*

Proof. The two summands of $\widehat{\sigma} + \tau$ have independent linear spans: projection is injective on the first and vanishes on the second. Consequently a set of rays spans a cone exactly when its fibre part spans a fibre cone and its base part projects to a base cone. The simplicial complex of Σ is therefore the join of the complexes of Σ_F and Σ_B . A minimal nonface of a join is a minimal nonface of one factor, proving the assertion about collections.

Let x be the sum of the generators of a lifted base collection, and write its minimal containing cone as $\widehat{\sigma} + \tau$. Projection maps the relative interior of this cone onto the relative interior of σ . Thus σ is the minimal containing cone of the projected sum. Comparing coefficients in the linearly independent generators of σ identifies all base coefficients; the remaining coefficients are the nonnegative fibre coefficients in x . If instead $x \in N_{F, \mathbb{R}}$, its minimal containing cone has $\widehat{\sigma} = 0$, so the fibre relation is unchanged. $\square$

Proposition 8.3. *The variety $X(\Gamma)$ is smooth and projective, and is a Zariski locally trivial toric fibre bundle*

$$X(\Gamma) \longrightarrow \prod_{c \in C} \mathbb{P}^{b_c} \quad \text{with fibre } S_6^j.$$

Its dimension and Picard number are

$$(50) \quad n = 2j + \sum_c b_c, \quad \rho = 4j + k.$$

Its ray matroid is connected if and only if Γ is connected. Furthermore,

$$(51) \quad X(\Gamma) \text{ is Fano} \iff \deg_\Gamma(c) \leq b_c \text{ for every } c \in C,$$

where the degree counts edges with multiplicity.

Proof. We verify completeness and projectivity directly. Let

$$(52) \quad Q = \{(a, z) \in \mathbb{R}^2 : |a| \leq 1, |z| \leq 1, |a + z| \leq 1\},$$

the anticanonical polygon of S_6 . In coordinates (m_i, y_t^c) on $M_{\mathbb{R}}$, with $m_i = (a_i, z_i)$, put $s_c = \sum_{c_i^+ = c} a_i - \sum_{c_i^- = c} a_i$. Choose integers $A_c > \deg_\Gamma(c)$ and consider the polytope

$$m_i \in Q, \quad y_t^c \geq -1, \quad \sum_{t=1}^{b_c} (y_t^c + 1) \leq A_c + s_c.$$

Every slice over Q^j is a product of full-dimensional simplices, since $A_c + s_c > 0$. At any point at most b_c inequalities of the c th simplex are active. A vertex therefore requires $2j$ independent active hexagon inequalities, so its projection is a vertex of Q^j . Conversely, each such vertex and each choice of a vertex in every simplex give a vertex of the full polytope. Exactly n inequalities are active, and their inward normals are precisely the generators of a maximal cone in Definition 8.1. The fan of inward normals is consequently Σ_Γ ; in particular it is complete and projective. Each maximal cone is unimodular: its hexagon generators give a basis of $\bigoplus_i \mathbb{Z}_i^2$, and modulo that sublattice its remaining generators form a standard maximal cone of $\prod_c \mathbb{P}^{b_c}$. This also shows that the polytope just constructed has integral vertices.

Projection to the base splits the fan in the sense of Lemma 8.2, with fibre the product hexagon fan. The toric local triviality theorem gives the asserted bundle [CLS11, Theorem 3.3.19]. There are $6j + \sum_c (b_c + 1)$ rays. Subtracting n gives (50).

The fibre primitive collections are the nonadjacent pairs in a hexagon. Their relations are either $v + (-v) = 0$, of degree 2, or $v + v'' = v'$, of degree 1. Here the degree of a primitive relation is the number of terms on the left minus the sum of coefficients on the right. The other primitive collections, by Lemma 8.2, are $\{f_0^c, \dots, f_{b_c}^c\}$, with relations

$$(53) \quad f_0^c + \dots + f_{b_c}^c = \sum_{c_i^+ = c} r_i + \sum_{c_i^- = c} (-r_i).$$

The right-hand side consists of one ray from each incident hexagon, so these rays span a cone and all coefficients are 1. Its degree is $b_c + 1 - \deg_\Gamma(c)$. Batyrev's primitive-collection criterion for ampleness, in the form of [CLS11, Theorem 6.4.9(b)], says that $-K_X$ is ample exactly when all these degrees are positive. Since the degrees are integers, this is (51).

For the matroid assertion, the six rays of each hexagon belong to one matroid component: the antipodal pairs and the relation $r_i + q_i - (r_i + q_i) = 0$ connect all six. For each c , the set

$$\{f_0^c, \dots, f_{b_c}^c\} \cup \{r_i : i \text{ is incident to } c\}$$

is a circuit. Indeed, projection to $\mathbb{Z}^{b_c}$ forces the coefficients of all the f_t^c in any relation to agree; the independent r_i then determine all the other coefficients, and a nonzero relation has no zero coefficient. These circuits join exactly the vertex and edge factors incident in Γ . Thus a connected graph gives a connected matroid. Distinct connected components of Γ use

disjoint lattice summands, giving a direct sum of nonempty ray matroids and proving the converse. $\square$

Lemma 8.4 (Recovery of the graph from the fan). *Let Γ be a connected loopless graph with at least one edge. The fan Σ_Γ of Definition 8.1 determines Γ and the integers b_c up to isomorphism. In particular, isomorphic toric fans give isomorphic graphs with the same vertex dimensions.*

Proof. We first recover the fibre rays without using the given presentation. They are precisely the primitive ray generators whose negatives also occur in the fan. Every fibre ray has this property. A base ray cannot be opposite a fibre ray, since its projection to the base lattice is nonzero. Rays belonging to different base factors cannot be opposite, since their projections lie in independent summands. Within a base factor of dimension at least two, the standard projective-space rays contain no opposite pair. If $b_c = 1$, its two lifted rays have sum

$$f_0^c + f_1^c = \sum_{c_i^+ = c} r_i - \sum_{c_i^- = c} r_i.$$

This sum is nonzero: connectedness and the existence of an edge exclude an isolated vertex, and the vectors from different incident edges belong to independent hexagon summands. The loopless hypothesis ensures that no edge contributes both signs at c .

Restrict the ray matroid to the recovered fibre rays. Its connected components are exactly the six-ray sets of the individual hexagons. Indeed, the opposite pairs and the circuit on the three ray lines connect all rays of one hexagon, whereas independent hexagon planes admit no circuit meeting two factors. We therefore recover each plane W_i and their direct sum $W = \bigoplus_i W_i$.

The primitive collections consisting entirely of the remaining rays are precisely the groups $C_c = \{f_0^c, \dots, f_{b_c}^c\}$, by Lemma 8.2. Thus these groups, and the integers $b_c = |C_c| - 1$, are determined by the fan. Their vector sums lie in W :

$$s_c := \sum_{u \in C_c} u = \sum_{c_i^+ = c} r_i - \sum_{c_i^- = c} r_i.$$

The component of s_c in the recovered direct summand W_i is nonzero exactly when c is an endpoint of i . This recovers the incidence graph, including parallel edges, and hence Γ . Primitive ray generators, opposite pairs, matroid components, primitive collections, and these vector sums are all preserved by a lattice isomorphism of fans, proving the assertion. $\square$

Corollary 8.5. *If Γ is connected and $X(\Gamma)$ is Fano, then*

$$(54) \quad \rho \leq 5 \left\lfloor \frac{n}{4} \right\rfloor + 1, \quad \rho - n \leq \frac{n}{4} + 1.$$

Equality in the first bound holds exactly when Γ is a tree with $j = \lfloor n/4 \rfloor$ edges. In that case

$$\sum_c (b_c - \deg_\Gamma(c)) = n - 4 \lfloor n/4 \rfloor.$$

In particular, for $j \geq 1$ and $n = 4j$, equality is equivalent to a tree with $b_c = \deg_\Gamma(c)$ at every vertex.

Proof. Connectedness gives $k \leq j + 1$, with equality exactly for a tree. The Fano condition gives $\sum_c b_c \geq \sum_c \deg_\Gamma(c) = 2j$. Hence $n \geq 4j$ and

$$\rho = 4j + k \leq 5j + 1 \leq 5 \lfloor n/4 \rfloor + 1.$$

Equality in this chain holds precisely when $k = j + 1$ and $j = \lfloor n/4 \rfloor$, giving the stated characterization. The formula for the sum follows from $n - 4j = \sum_c (b_c - \deg_\Gamma(c))$. $\square$

The corollary includes the graph with one vertex and no edges: $X(\Gamma) = \mathbb{P}^{b_c}$, $\rho = 1$, and equality in its first bound holds exactly for $1 \leq b_c \leq 3$. The empty graph is excluded throughout. If Γ is disconnected, the lattice and fan decompose over its connected components, so $X(\Gamma)$ is a product of positive-dimensional varieties. Its tangent bundle is not stable for any polarization: it splits into nonzero direct summands, and at least one summand has slope at least the slope of their direct sum. The bound (54) concerns this graph construction; it makes no claim for arbitrary smooth toric Fano varieties.

Example 8.6. For $j \geq 1$, let Γ_j be the path with vertices $0, \dots, j$, edges $i : i \rightarrow i + 1$ for $0 \leq i < j$, and weights $b_0 = b_j = 1$, $b_c = 2$ for $0 < c < j$. We denote $X(\Gamma_j)$ by X_j . It is Fano, has connected ray matroid, and satisfies

$$\dim X_j = 4j, \quad \rho(X_j) = 5j + 1.$$

Thus these paths attain the bound in Corollary 8.5. Connectedness alone does not establish stability; the slope inequalities for this family will require the degree formulas below.

8.1. Anticanonical degrees. Assume henceforth that $b_c \geq \deg_\Gamma(c)$. For $a = (a_i) \in [-1, 1]^j$ define

$$(55) \quad s_c(a) = \sum_{c_i^+ = c} a_i - \sum_{c_i^- = c} a_i, \quad L_c(a) = b_c + 1 + s_c(a), \quad \Phi(a) = \prod_{c \in C} \frac{L_c(a)^{b_c}}{b_c!}.$$

In particular $L_c \geq 1$. All lattice volume measures below assign volume 1 to a fundamental parallelepiped, without a factorial. Put

$$d\nu(a) = \prod_{i \in I} (2 - |a_i|) da_i.$$

For a hexagon ray v in the i th factor, let $d\nu^{(i,v)}$ be obtained by replacing the i th factor of $d\nu$ with the measure dE_v on $[-1, 1]$ specified by

$$(56) \quad \begin{aligned} dE_{r_i} &= \delta_{-1}, & dE_{-r_i} &= \delta_1, \\ dE_{r_i+q_i} &= dE_{-q_i} = \mathbf{1}_{[-1,0]}(a_i) da_i, & dE_{q_i} &= dE_{-r_i-q_i} = \mathbf{1}_{[0,1]}(a_i) da_i. \end{aligned}$$

Here $\delta_{\pm 1}$ denote unit point masses, not normalized divisor degrees. Each measure in (56) has total mass 1.

Proposition 8.7. Write $D = (-K_X)^n$ and $d_v = D_v \cdot (-K_X)^{n-1}$ for $X = X(\Gamma)$. Then

$$(57) \quad D = n! \int_{[-1,1]^j} \Phi d\nu,$$

$$(58) \quad d_{f_t^c} = (n-1)! \int_{[-1,1]^j} \frac{b_c}{L_c} \Phi d\nu \quad (0 \leq t \leq b_c),$$

$$(59) \quad d_v = (n-1)! \int_{[-1,1]^j} \Phi d\nu^{(i,v)} \quad (v \text{ in hexagon } i).$$

These formulas also hold for $j = 0$, with the integral over a point.

Proof. The moment polytope of $-K_X$ is given by

$$m_i = (a_i, z_i) \in Q, \quad y_t^c \geq -1, \quad -\sum_{t=1}^{b_c} y_t^c + s_c(a) \geq -1.$$

After translating y_t^c by 1, its slice over $(m_i) \in Q^j$ is

$$\prod_c \{(u_t^c) : u_t^c \geq 0, \sum_t u_t^c \leq L_c(a)\},$$

whose volume is $\Phi(a)$. For fixed $a \in [-1, 1]$, the allowable z in (52) form an interval of length $2 - |a|$. Fubini's theorem and $D = n! \operatorname{vol}(P_{-K_X})$ therefore give (57).

For $t \geq 1$, the facet corresponding to f_t^c sets $u_t^c = 0$. It replaces one simplex volume by $L_c^{b_c-1}/(b_c-1)!$, multiplying the integrand by b_c/L_c . The facet corresponding to f_0^c sets $\sum_t u_t^c = L_c(a)$; eliminating one coordinate gives the same volume. This elimination is an isomorphism of the affine facet lattice with the coordinate lattice, since the coefficient of the eliminated coordinate is 1 and s_c is integral affine. Thus no additional factor occurs in its lattice measure. The standard facet-volume identity $d_v = (n-1)! \operatorname{vol}(F_v)$ proves (58), including $b_c = 1$.

A facet for a hexagon ray lies over the corresponding edge of Q . Each such edge has lattice length 1. The edges $a = -1$ and $a = 1$ project to the unit point masses in (56). On each other edge, projection to a is a lattice isomorphism onto an interval of length 1. Specifically, $a + z = -1$ and $z = 1$ give $[-1, 0]$, whereas $z = -1$ and $a + z = 1$ give $[0, 1]$. These are exactly the four interval measures in (56). Fubini on the facet proves (59). $\square$

Remark 8.8. If $Z = \int \Phi d\nu$, the normalized degrees $\delta_v = nd_v/D$ are the respective integrals in (58) and (59), divided by Z . Thus the factorials cancel before applying the slope criterion. In particular the $b_c + 1$ rays associated to a vertex have equal degrees.

9. EXTREMAL FAMILIES AND THE EFFECT OF SEPARATION

We first state the stable path, end-attachment and three-arm families needed for the extremum and the general subdivision theorem. Their complete degree calculations and uniform estimates appear in Appendices B–D. The distinction between the two uses of these results is useful: paths and attachments attain the numerical bound, whereas the quantitative three-arm estimate is the initial case for arbitrarily many branching vertices.

9.1. Paths. Take the graph of Section 8 to be a path with vertices $0, \dots, j$ and edges $i: i \rightarrow i+1$, $0 \leq i < j$. Put $b_c = 2$ at the interior vertices and $b_0, b_j \in \{1, 2, 3\}$ at the ends. Denote the resulting variety by $X_j^{(b_0, b_j)}$. Its dimension and Picard number are

$$(60) \quad n = 4j - 2 + b_0 + b_j, \quad \rho = 5j + 1.$$

The Fano condition holds at every vertex. We write $X_j = X_j^{(1,1)}$.

Theorem 9.1. *For every $j \geq 1$ and every $b_0, b_j \in \{1, 2, 3\}$, the tangent bundle of $X_j^{(b_0, b_j)}$ is slope stable with respect to its anticanonical polarization.*

The proof is in Appendix B.

9.2. Attainment in every dimension. Put $r = (1, 0)$ and $q = (0, 1)$ in the hexagon lattice and define

$$A_0 = (-r), \quad A_1 = (q, -r - q), \quad A_2 = (-r, q, -q), \quad A_3 = (r, -r, q, -r - q).$$

For $j \geq 1$, take j hexagon factors. Between hexagons $i - 1$ and i place a $\mathbb{P}^2$ base factor with twisting vector $r_{i-1} - r_i$. At the first hexagon attach one $\mathbb{P}^1$ base factor for each root in A_a ; at the last attach one for each root in $-A_b$. Use the locally trivial toric bundle fan over the product of these projective spaces, with fibre S_6^j , and denote the resulting variety by $X_j(a, b)$.

Theorem 9.2. *For every $j \geq 1$ and*

$$(a, b) \in \{(1, 0), (2, 0), (1, 1), (3, 0), (1, 2)\},$$

$X_j(a, b)$ is smooth Fano and its tangent bundle is anticanonically stable. Moreover,

$$\dim X_j(a, b) = 4j + a + b, \quad \rho(X_j(a, b)) = 5j + a + b + 1.$$

The proof is in Appendix C.

Corollary 9.3 (Sharpness in the bundle class). *For every $n \geq 4$, the largest Picard number of a smooth toric Fano n -fold with stable tangent bundle among the locally trivial toric S_6^j -bundles over products of positive-dimensional projective spaces, with the fan presentation (47), is*

$$\left\lfloor \frac{5n}{4} \right\rfloor + 1.$$

Proof. Theorem 7.3 gives the upper bound. Write $n = 4j + r$, $0 \leq r \leq 3$. For $r = 0$ use the ordinary path X_j from Theorem 9.1. For $r = 1, 2, 3$ use respectively the pairs $(1, 0), (2, 0), (3, 0)$ in Theorem 9.2. Their Picard numbers are $5j + r + 1 = \lfloor 5n/4 \rfloor + 1$. $\square$

9.3. The first branching case. Stability at the Picard bound is compatible with branching, even when the branches have different lengths. Let $\Gamma_{a,b,c}$ be the tree with one vertex of degree three and arms containing a, b, c edges. Put $b_v = \deg(v)$ and let $Y_{a,b,c} = X(\Gamma_{a,b,c})$ be the graph bundle of Section 8. Its base has one factor $\mathbb{P}^3$, three factors $\mathbb{P}^1$, and $a + b + c - 3$ factors $\mathbb{P}^2$.

Theorem 9.4. *For all integers $a, b, c \geq 2$, the tangent bundle of $Y_{a,b,c}$ is stable with respect to $-K_{Y_{a,b,c}}$. Moreover,*

$$\dim Y_{a,b,c} = 4(a + b + c), \quad \rho(Y_{a,b,c}) = 5(a + b + c) + 1.$$

Thus branching examples attain the bundle bound in every dimension divisible by four and at least twenty-four. Their fans are not lattice-isomorphic to the ordinary path fans in the same dimension: the central $\mathbb{P}^3$ rays form a primitive collection of size four, whereas the path primitive collections have sizes two or three by Lemma 8.2.

The Fano property and connected ray matroid follow from Proposition 8.3. The proof keeps independent subspace choices on the three arms. A cubic integral gives their interaction at the center. Contraction then bounds the effect of changing one arm along the others, uniformly even when their lengths are very different.

The proof is in Appendix D.

The contraction estimate behind the proof explains why separation is effective. At a degree-two vertex, integration over the base simplex acts on a positive function by the operator

$$(\mathcal{T}f)(a') = \int_{-1}^1 f(a) \frac{(3 + a - a')^2}{2} (2 - |a|) da.$$

Thus a path applies the same positive integral operator repeatedly. For positive continuous functions define

$$d_H(f, g) = \log \left(\sup_a \frac{f(a)}{g(a)} \sup_a \frac{g(a)}{f(a)} \right).$$

Lemma B.2 proves that for arbitrary positive continuous initial functions and every $i \geq 1$,

$$(61) \quad d_H(\mathcal{T}^i f, \mathcal{T}^i g) \leq \log(81/25)(2/7)^{i-1}.$$

The corresponding ray degrees are ratios of positive integrals. For each hexagon or interior-base ray type, with left and right chain indices $i, k \geq 1$, its normalized degree $\delta(i, k)$ consequently satisfies

$$|\log \delta(i, k) - \log \delta_\infty| \leq \log(81/25)((2/7)^{i-1} + (2/7)^{k-1}).$$

The limiting degree is independent of the distant end data. An end-ray degree has only a one-sided limit and can retain dependence on its own end factor. These are convergence estimates; they do not assert monotonicity in the path lengths.

Stability also requires a uniform bound for the deficits of all relevant subspaces. Section 10 supplies the rank formula and the gluing argument that preserve those bounds when smaller trees are joined.

10. SUBDIVIDED TREES AND ABUNDANT EXTREMIZERS

The sharp bundle bound admits extremizers with arbitrary fixed subcubic branching patterns. Sufficiently long paths between branching vertices make the divisor degrees nearly independent of distant branches. An inequality for joining subtrees then controls subspaces meeting both sides of a path. Together these arguments prove stability, with no requirement that the subdivision lengths agree.

For a finite tree T of maximum degree three, suppress its degree-two vertices; the resulting tree will be called its skeleton. Let s be the number of trivalent vertices. If $s \geq 1$, the skeleton has $s + 2$ leaves and $2s + 1$ edges. Replacing its edges by paths of positive integral lengths gives a subdivision Γ . Throughout this section put $b_v = \deg_\Gamma(v)$ in Definition 8.1.

Theorem 10.1. *There are explicit integers L_s , depending only on the number $s \geq 1$ of trivalent vertices, such that $T_{X(\Gamma)}$ is anticanonically stable whenever every skeleton edge is replaced by a path of length at least L_s . The lengths may be chosen independently. One can take $L_1 = 2$ and $L_s = O(\log s)$ as $s \rightarrow \infty$. If j is the total number of edges of Γ , then*

$$\dim X(\Gamma) = 4j, \quad \rho(X(\Gamma)) = 5j + 1.$$

In particular every fixed finite tree of maximum degree three and at least one edge gives stable tangent bundles after all sufficiently long independent edge subdivisions.

The case with no trivalent vertex is Theorem 9.1. The examples attain the Picard bound in Theorem 7.3. The graph can also be recovered from the fan, so varying its branching pattern produces distinct extremizers in a common dimension.

Corollary 10.2 (Abundance of extremal fans). *For every integer $q \geq 1$, there is an integer J_q such that for every $j \geq J_q$ there are at least q pairwise nonisomorphic toric fans of smooth toric Fano varieties with stable tangent bundle, dimension $4j$, and Picard number $5j + 1$. Each belongs to the bundle class of Theorem 7.3 and attains its maximum in that dimension.*

Proof. Choose q finite subcubic trees with an edge and with pairwise distinct numbers of trivalent vertices, all positive. Such trees exist: take a path of s vertices and attach leaves so that each path vertex has degree three, with the three-arm tree for $s = 1$. Their branching patterns remain distinct under subdivision.

For each chosen tree, first suppress degree-two vertices and give every skeleton edge a length satisfying Theorem 10.1. Let j_t be the resulting total length for the t th tree, and put $J_q = \max_t j_t$. For any $j \geq J_q$, increase the length of one skeleton edge in the t th tree by $j - j_t$. The stability theorem allows independent lengths, so every resulting graph bundle is stable. Each subdivided tree has j edges and $j + 1$ vertices, and $\sum_v b_v = \sum_v \deg(v) = 2j$. Proposition 8.3 therefore gives dimension $4j$ and Picard number $5j + 1$.

Subdivisions of the chosen trees are nonisomorphic because their numbers of trivalent vertices differ. Lemma 8.4 then excludes isomorphisms of their toric fans. Finally, Theorem 7.3 identifies $5j + 1$ as the maximum within the bundle class. $\square$

This corollary concerns isomorphisms of toric fans. It neither classifies all extremizers nor asserts a growth rate for their number.

10.1. Subdivision and dependence on distant branches. The estimate (61) makes dependence on distant branches exponentially small along a subdivided edge.

Convergence of individual ray degrees does not itself imply stability. The proof below combines it with the rank formula at a cut and an induction on the number of trivalent vertices. The initial three-arm case is the quantitative form of Theorem 9.4. Its finite rational comparisons, and those for paths, are given in Appendices D and B; their uniform error bounds apply to every allowed length. The resulting sufficient lengths are explicit, though not claimed optimal:

s	1	2	3	4	5	10	25	50	100	1000
L_s	2	27	31	31	33	35	41	43	47	57.

No enumeration of tree shapes is required. This subdivision theorem concerns the hexagon bundles of Definition 8.1. It is distinct from subdivision in the blowup construction, which can produce instability as in Corollary 5.5.

For clarity, we fix the slice functions used in the proof. Put $d\nu(t) = (2 - |t|) dt$ on $[-1, 1]$ and $\mathcal{K}(s, t) = (3 + s - t)^2/2$. The right operator is

$$(\mathcal{T}_R h)(s) = \int_{-1}^1 \mathcal{K}(s, t) h(t) d\nu(t).$$

Write h_∞ for its positive projective limit, obtained by iterating from $h_0(t) = 2 + t$ and normalizing at zero. The contraction estimate makes the limit independent of the positive initial function. At a trivalent vertex, integrating the other two branches with incoming functions u, v gives the outgoing slice function

$$J_{u,v}(s) = \iint_{[-1,1]^2} \frac{(4 - s - t - z)^3}{6} u(t)v(z) d\nu(t)d\nu(z).$$

This is the cubic integral of (89); it is a function of a boundary coordinate, distinct from the scalar exceptional contributions $J_{v,e}$ in the blowup construction.

10.2. Rank at a cut. Use the independent hexagon reflections and untwisted-base permutations from Appendix D, without permuting branches. An invariant ray-spanned subspace selects each hexagon as $H_e = 0, 1, 2$ (empty, distinguished line, or plane), and at each vertex selects the untwisted rays and twisted ray with indicators C_v, T_v . Delete the edges

with $H_e > 0$. A component of the remaining forest is called complete for this selection if $C_v = T_v = 1$ at every vertex in it. If c_0 counts these components, the rank is

$$(62) \quad r = \sum_e H_e + \sum_v b_v C_v + \sum_v T_v - c_0.$$

Indeed, first quotient by the selected hexagon directions and untwisted base coordinates. A selected twisted ray with $C_v = 0$ has a private base coordinate and contributes one dimension. The remaining twisted rays are rows of the incidence matrix of the zero-edge forest. A dependence among selected rows extends by zero to a function constant on each component, and exists precisely on a complete component.

For a subtree rooted at a cut, postpone subtracting the relation of its root component. Write $a = 1$ if this component is complete, and $a = 0$ otherwise. Let e record a nonempty selection. Let $f = 1$ mean that every hexagon is a plane, every vertex with $b_v \geq 2$ has $C_v = 1$, and every leaf has $C_v \vee T_v = 1$. These are precisely the local requirements for full rank when the tree is completed. Necessity follows from the independent nontrivial representations at hexagons and bases, and from the private coordinate of each leaf; they also suffice. The possible triples are

$$(a, e, f) = (0, 0, 0), (0, 1, 0), (1, 1, 0), (0, 1, 1), (1, 1, 1).$$

Let c_* count the complete components other than the root component. If $w_e(H)$ is the selected hexagon weight and δ_v one base-ray weight, define the open deficit

$$A = \sum_e (H_e - w_e(H_e)) + \sum_v (b_v C_v + T_v)(1 - \delta_v) - c_*.$$

This is the rank deficit with the single boundary relation left uncounted. Joining two rooted pieces with open deficits A, B and indicators a, b across a hexagon H gives final deficit

$$(63) \quad \begin{cases} A + B - ab, & H = 0, \\ A + B + H - w(H) - a - b, & H > 0. \end{cases}$$

For $H = 0$ the two root components merge and are complete exactly when $a = b = 1$; for $H > 0$ they close separately. The combined selection is nonempty exactly when $e \vee e' \vee (H > 0)$, and has full rank exactly when $f \wedge f' \wedge (H = 2)$.

10.3. An inequality inherited from a completed tree. Consider a finite subtree ending at a degree-two vertex adjacent to the cut. Give it the limiting degrees obtained by continuing the cut to an ordinary leaf whose distance tends to infinity. Equivalently, its incoming function at the cut is h_∞ . Let α be its all-selected open deficit.

Lemma 10.3. *Suppose every sufficiently long completion to that leaf has deficit at least $\mu > 0$ for every invariant selection of rank strictly between zero and the ambient dimension. For the retained subtree,*

boundary condition	inequality
$a = 0, e = 1$	$A \geq \mu$
$a = 1, f = 0$	$A - \alpha \geq \mu$
$a = 1, f = 1$	$A - \alpha \geq 0.$

The empty selection has $A = 0$.

Proof. For $a = 0, e = 1$, select no ray in the continuation or its joining hexagon. The completed selection is proper and nonempty, including when the retained subtree has $f = 1$.

Formula (63) makes its deficit exactly A . Pass to the limit in the finitely many retained degrees.

For $a = 1, f = 0$, select every ray in the continuation and take $H = 2$ at the cut. The full-rank failure in the retained subtree persists. Subtract the all-selected deficit, which is zero. The continuation and joining terms cancel exactly, leaving $A - \alpha$. Only then pass to the limit, so no unbounded sum is interchanged with a limit. For $a = f = 1$, adding omitted rays preserves the open rank and boundary indicator and decreases the deficit, since their weights are positive. Thus the all-selected configuration minimizes this state. $\square$

10.4. Nonnegative limiting transitions. Let w_1, w_2 be the limiting weights of a distinguished hexagon line and full plane, and let δ be one limiting degree-two-base weight. Appendix B gives $w_2 + 3\delta = 4$. The rational moment matrices at index 40, with Lemma B.2, give

$$(64) \quad \frac{187}{100} < w_2 < \frac{15}{8}, \quad 0 < w_1 < \frac{14}{25}.$$

The rational approximations are

$$w_1 \simeq 0.5579289459, \quad w_2 \simeq 1.8740976113.$$

For the exact index-40 rational values, with $D = 11757/10000$, $B = 81$ and $y = 2D(2/7)^{39}$, the errors are at most $2B(y + y^2)$ and $6B(y + y^2)$, respectively; these bounds are smaller than the distances to the endpoints in (64).

Put $\beta = w_2/2$ and replace the open deficit by $A - \beta a$. Extending across one hexagon H and one new degree-two vertex C, T changes the boundary indicator to $a' = CTa$ for $H = 0$, and $a' = CT$ for $H > 0$. The adjusted increment is

$$(65) \quad (2C + T)(1 - \delta) + H - w(H) - a\mathbf{1}_{\{H>0\}} + \beta(a - a').$$

For $C = T = 0$ its three values, indexed by $H = 0, 1, 2$, are

$$\begin{array}{l|ll} a = 0 & 0 & 1 - w_1 & 2 - w_2 \\ a = 1 & \beta & \beta - w_1 & 1 - \beta. \end{array}$$

When $CT = 0$ and $(C, T) \neq (0, 0)$, add $(2C + T)(1 - \delta)$. For $C = T = 1$ the values are

$$\begin{array}{l|lll} a = 0 & w_2 - 1 & \beta - w_1 & 1 - \beta \\ a = 1 & w_2 - 1 & w_2 - w_1 - 1 & 0. \end{array}$$

All nonzero entries exceed $1/16$ by (64); also $1 - \delta = (w_2 - 1)/3 > 0.29$. The only zero transitions are $(a, C, T, H) = (0, 0, 0, 0)$ and $(1, 1, 1, 2)$, the empty and all-selected periods. In the same adjusted deficits the joining correction is

$$(66) \quad \begin{cases} \beta(a + b) - ab, & H = 0, \\ H - w(H) + (\beta - 1)(a + b), & H > 0. \end{cases}$$

Its only zeros are $(a, b, H) = (0, 0, 0)$ and $(1, 1, 2)$; all other values exceed $1/16$. Thus arbitrarily many limiting periods may be inserted without accumulating a negative increment.

10.5. Uniform control of the geometric error. Split an internal skeleton edge of length $2q + 1 + m$, $m \geq 0$, leaving q edges and q degree-two vertices at each end. Call these retained paths the two necks. The middle consists of $m + 1$ hexagons and m degree-two bases. The two components, capped by a leaf on the chosen edge, have $s_1, s_2 \geq 1$ trivalent vertices, with $s_1 + s_2 = s$. Retain all their other edge lengths.

Define comparison weights by putting the incoming function h_∞ at each cut and using the limiting weights on the middle. Every individual ray weight is at most $B = 81$, provided all skeleton lengths are at least two. Indeed a junction slice function $J_{u,v}$ in (89) has oscillation ratio at most 27. Each hexagon has at least one endpoint of degree two or one; that side's function has oscillation at most nine. The facet measure has mass one and ν has mass three, giving $27 \cdot 9/3 = 81$. The base-ray bounds are at most 1, 2, 3 at leaves, degree-two vertices and junctions.

Changing one incoming function by Hilbert distance η changes $J_{u,v}$ by at most η : normalize its pointwise ratio to lie in $[1, e^\eta]$ and integrate the positive cubic. At each degree-two vertex the distance contracts by $\tau = 2/7$. There is a unique path from the cut to each vertex. Consequently only one incoming argument changes at every later junction. The disturbance can enter both outgoing branches, but does not add to itself at a junction.

Set

$$(67) \quad x_q = D\tau^{q-1}, \quad c_s = \frac{156s + 272}{5}, \quad E_s(q) = Bc_s x_q(1 + 2x_q), \quad D = \frac{11757}{10000}.$$

Lemma 10.4. *If all skeleton lengths are at least two and $2x_q \leq 1$, the actual and comparison deficits of every selection differ by at most $E_s(q)$, independently of all subdivision lengths.*

Proof. The incoming function at either cut has traversed at least q degree-two vertices from the opposite junction. Its distance from the limit is at most $x = x_q$. Along the retained neck it contracts geometrically toward the junction. Six hexagon rays and three base rays per step give a sum of logarithmic degree errors at most $9x/(1 - \tau)$. The base closest to the cut has error at most x ; the adjacent retained hexagon has an additional contraction, which this estimate discards.

A capped component with s_i junctions has $2s_i + 1$ skeleton edges. One is the neck just counted; each other edge contributes at most $9x/(1 - \tau)$, by contraction along its interior. We discard any attenuation before reaching that edge. Its s_i junction bases add $4s_i x$, and its $s_i + 1$ genuine leaf bases add $2(s_i + 1)x$. The endpoint total is at most

$$\left\{ \frac{9(2s_i + 1)}{1 - \tau} + 6s_i + 2 \right\} x.$$

Number the chosen bridge edges $i = 0, \dots, L - 1$, where $L = 2q + 1 + m$, and its internal bases $c = 1, \dots, L - 1$. On the middle edges $q \leq i \leq L - q - 1$, the omitted left and right distances are at most $D\tau^{i-1}$ and $D\tau^{L-2-i}$. On middle bases $q + 1 \leq c \leq L - q - 1$, they are at most $D\tau^{c-2}$ and $D\tau^{L-2-c}$. Each pair is at most $2x$; summing either set over its positions gives at most $2x/(1 - \tau)$. The six hexagon and three base rays therefore add at most $18x/(1 - \tau)$.

The sum of all logarithmic ray errors is consequently at most

$$\left\{ \frac{9(2s + 4)}{1 - \tau} + 6s + 4 \right\} x = c_s x.$$

Each individual error η is at most $2x \leq 1$, and $e^\eta - 1 \leq \eta(1 + 2x)$. Multiplying by B bounds the sum of absolute ray-weight changes by $E_s(q)$. The selected rank is unchanged, so the same bound holds for every deficit. $\square$

10.6. Joining and induction. Let α_i be the all-selected open deficit of endpoint i . Glue two identical copies of this endpoint with a bridge of length $2q + 1$. Its actual all-selected

deficit is zero. The comparison deficit is $2\alpha_i - w_2$. Lemma 10.4, applied with $2s_i$ junctions and the same q , gives

$$|\alpha_i - \beta| \leq \frac{1}{2}E_{2s_i}(q).$$

This uses the degree identity for the doubled tree, without a stability assumption. Since c_s is affine,

$$(68) \quad \frac{1}{2}\{E_{2s_1}(q) + E_{2s_2}(q)\} = E_s(q).$$

Suppose the two completed endpoint families have invariant margins $\mu_1, \mu_2 \leq \mu_0 := 2698/100000 < 1/16$. Lemma 10.3 and (68) show that a positive-margin endpoint contributes its inherited margin minus its potential error in the adjusted deficit. An endpoint with $a = f = 1$ contributes at least minus that error, and an empty endpoint contributes zero. All intervening increments and the final joining correction are nonnegative.

If neither endpoint has a positive-margin state, a proper nonempty global selection must contain a positive increment or joining correction. Otherwise the zero transitions propagate only empty to empty or full to full, giving an excluded global selection. Thus the comparison deficit is at least $\min(\mu_1, \mu_2) - E_s(q)$. Subtracting the geometric error proves the joining inequality

$$(69) \quad \text{actual deficit} \geq \min(\mu_1, \mu_2) - 2E_s(q).$$

Proof of Theorem 10.1. For $s \geq 2$, let q_s be the least positive integer such that

$$(70) \quad 2x_{q_s} \leq 1, \quad E_s(q_s) \leq \frac{\mu_0}{4s(s-1)}, \quad L_s = 2q_s + 1.$$

Put $L_1 = 2$ and $\mu_s = \frac{\mu_0}{2}(1 + 1/s)$. The minimum exists by geometric decay. These thresholds are nondecreasing: c_s increases and the allowed error decreases. We prove the stronger assertion that every invariant selection of rank strictly between zero and $4j$ has deficit at least μ_s . The case $s = 1$ is the quantitative conclusion of Theorem 9.4.

For $s \geq 2$, remove any internal skeleton edge and cap each exposed junction by a leaf. The two smaller trees have $s_1, s_2 \geq 1$ junctions, with $s_1 + s_2 = s$. Their unchanged lengths satisfy the inductive thresholds since $L_s \geq L_{s_i}$. Their new leaf edges may be continued to arbitrary sufficiently large lengths, giving the families required by Lemma 10.3. Retain q_s neck edges at either end of the chosen original edge; its length is at least $2q_s + 1$. Formula (69) now gives

$$\text{deficit} \geq \mu_{s-1} - 2E_s(q_s) \geq \mu_{s-1} - \frac{\mu_0}{2s(s-1)} = \mu_s > 0.$$

The independent local symmetries and connected ray matroid therefore give stability by Proposition 2.4.

For the growth bound, it suffices to choose

$$q \geq 1 + \left\lceil \log_{7/2} \left(\frac{8BDc_s s(s-1)}{\mu_0} \right) \right\rceil.$$

Then $2x_q \leq 1$ and $E_s(q) \leq 2Bc_s x_q \leq \mu_0/[4s(s-1)]$. Thus $L_s = O(\log s)$. The dimension and Picard number follow from Proposition 8.3, since $\sum_v b_v = 2j$ and a tree has $j + 1$ vertices. For a tree with original degree-two vertices, the skeleton lengths are sums of the original subdivision lengths. The case $s = 0$ is Theorem 9.1. $\square$

The quantitative margin $\mu_s > 0.01349$ concerns these locally invariant selections only. The theorem asserts stability for all subspaces through the symmetry criterion, without a uniform quantitative claim for them.

Corollary 10.5. *Suppose a tree has two trivalent vertices joined by a path of length at least 27, with two outer arms at each vertex. If all four outer arms have length at least two, the graph bundle with $b_v = \deg(v)$ has anticanonically stable tangent bundle.*

Proof. The two completed endpoint families are three-arm trees, so both inherited margins are μ_0 even when the outer lengths are less than L_2 . Lemma 10.4 needs only outer lengths at least two. With $q = 13$, formula (69) gives

$$\mu_0 - 2E_2(13) = \frac{122113777399878205459586741}{5986913480642700450031250000} > 0.02039.$$

□

11. FURTHER QUESTIONS

The two constructions give complete extremal statements within their classes, but do not classify all stable members. For blowups, the vertex criterion separates this classification problem from the subspace-rank calculation. For bundles, the subdivision theorem gives many extremizers without a characterization of every stable twist.

Question 11.1. *Which saturated blowup trees of maximum degree four or five have stable tangent bundle? What additional subspace inequalities are needed for the same construction on graphs with cycles?*

The degree-four obstruction is sufficient, whereas Proposition 6.6 gives stable higher-degree arrangements. Larger factor dimensions at nonleaf vertices also lie beyond the subcubic stability theorem. The exceptional mixed-intersection formula in Section 4 remains available without a tree, but controlling its associated foliation alone does not control every higher-rank subsheaf.

Question 11.2. *Is there an absolute integer L_* such that every finite subcubic tree with an edge gives a stable graph bundle whenever all skeleton edges are subdivided to length at least L_* ?*

Theorem 10.1 gives a sufficient length growing logarithmically with the number of trivalent vertices. It does not show that this dependence is necessary. More generally, the necessary incidence conditions in Corollary 7.4 do not decide which primitive twisting vectors produce stable extremizers.

Finally, let $R(n)$ be the largest Picard number of a smooth toric Fano n -fold with anticanonically stable tangent bundle. The bundle theorem gives the lower bound below. For comparison, a published splitting theorem supplies a general upper bound.

Corollary 11.3 (Consequence of Assarf–Nill). *For $n \geq 4$,*

$$\left\lfloor \frac{5n}{4} \right\rfloor + 1 \leq R(n) \leq 2n - \left\lfloor \frac{\sqrt{60n + 1309} - 37}{30} \right\rfloor.$$

Proof. The lower bound is Theorem 1.3. For a stable X , put $k = 2n - \rho(X) \geq 0$, using Casagrande’s bound [Cas06]. Assarf–Nill [AN16, Theorem 8] show that $n > 15k^2 + 37k + 1$ forces a factor S_6 in a product decomposition. For $n \geq 3$ this is a nontrivial product, whose tangent bundle is not stable. Thus $n \leq 15k^2 + 37k + 1$; solving for the nonnegative integer k gives the upper bound. □

Assarf–Nill [AN16, Conjecture 13] conjecture the stronger bound $\rho \leq (4n+3)/3$ for smooth toric Fano varieties of dimension at least three that are not nontrivial products. Tangent stability implies this nonproduct condition.

Question 11.4. *Does every smooth toric Fano n -fold with anticanonically stable tangent bundle satisfy $\rho \leq (4n + 3)/3$ for $n \geq 3$? What is $\limsup_{n \rightarrow \infty} R(n)/n$?*

The results here give an asymptotic lower ratio of $5/4$. The proposed upper ratio $4/3$ remains conjectural. A general proof would have to show that excessive Picard number forces excessive degree in some proper tangent subspace. Neither the exceptional-intersection identity nor the restricted bundle incidence argument supplies that implication.

APPENDIX A. STAR TRANSITIONS AND TWO-CENTRE COMPARISONS

We give the complete calculations for Theorem 5.1 and Proposition 6.6, retaining the notation of Sections 5 and 6.

A.1. Proof of the star transition.

Proof. The geometric assertions follow from Proposition 3.1. The unassigned divisor degree at every leaf is strictly less than one by the local identity in Lemma 3.2. The tree criterion, Theorem 3.4, therefore reduces stability to $A_m < 1$. This reduction includes $m = 1$, when both vertices are leaves.

Use affine coordinates $a_i = 1 + \langle x, e_i \rangle$ and coordinates $y_i = \langle x, u_i \rangle$. The anticanonical polytope is

$$a_i \geq 0, \quad \sum_{i=1}^m a_i \leq m + 1, \quad -\min(a_i, 1) \leq y_i \leq 1.$$

Define, for $a \geq 0$ and $t > 0$,

$$(71) \quad \ell(a) = 1 + \min(a, 1), \quad Z_m(t) = \int_{\substack{a_i \geq 0 \\ \sum_i a_i \leq t}} \prod_{i=1}^m \ell(a_i) da_1 \cdots da_m.$$

The lattice volume of the polytope is $Z_m(m + 1)$. Eliminating one coordinate from $\sum_i a_i = t$ gives the lattice measure on its central facet, whose volume is $Z'_m(t)$. Consequently

$$(72) \quad A_m = \frac{Z'_m(m + 1)}{Z_m(m + 1)}.$$

There is no Euclidean normal-length factor in this formula.

For $t > m$, the integral has the finite expression

$$(73) \quad Z_m(t) = \sum_{j=0}^m (-1)^j \binom{m}{j} \sum_{h=0}^{m-j} \binom{m-j}{h} \frac{(t-j)^{2m-h}}{(2m-h)!}.$$

Indeed, the Laplace transform of ℓ is $(\lambda + 1 - e^{-\lambda})/\lambda^2$. The simplex integral in (71) has transform $(\lambda + 1 - e^{-\lambda})^m/\lambda^{2m+1}$; expanding the numerator and inverting each term proves (73). Differentiating that finite sum gives the exact values of $1 - A_m$ in Table 1. They prove stability for $1 \leq m \leq 5$ and $A_m > 1$ for $6 \leq m \leq 14$. The entries are comparisons of the individual divisor degree A_m with one. In particular, the entry at $m = 1$ is not the weight deficit of the line $\mathbb{C}e_0$, which then also contains e_1 .

It remains to prove $A_m > 1$ for all $m \geq 15$. Set $t = m + 1$ and

$$J_m(t) = \int_{\substack{a_i \geq 0, \sum_i a_i \leq t \\ 0 \leq a_1 \leq 1}} a_1 \prod_{i=2}^m \ell(a_i) da_1 \cdots da_m.$$

TABLE 1. Exact values of $1 - A_m$, obtained from (73) and its derivative at $t = m + 1$.

m	$1 - A_m$
1	3/7
2	7/31
3	149/1193
4	3039/47119
5	2509/103289
6	-13135/2987177
7	-41300337/1594796251
8	-8432407681/197900582879
9	-92165038835/1646595334057
10	-1447101945437/21629833922043
11	-289731964699063/3811832962784653
12	-244482437035005131/2920544373257729893
13	-11134987212181695069/123294500114195565143
14	-901920150115619173247/9391892098065855984793

Rescale the simplex in (71) to a fixed simplex and differentiate. Since $\ell'(a) = 1$ on $(0, 1)$ and is zero on $(1, \infty)$, symmetry among the coordinates gives

$$(74) \quad tZ'_m(t) = mZ_m(t) + mJ_m(t).$$

The kink of ℓ occurs on a set of measure zero and does not contribute an additional term.

For every integer $p \geq 1$, $s > 0$, and $0 < \theta \leq 1$, the inequalities $\theta\ell(a) \leq \ell(\theta a) \leq \ell(a)$ imply

$$\theta^{2p}Z_p(s) \leq Z_p(\theta s) \leq \theta^pZ_p(s).$$

Using the upper bound for Z_m and the lower bound for J_m yields

$$\begin{aligned} Z_m(t) &= \int_0^t \ell(a)Z_{m-1}(t-a) da \\ &\leq 2Z_{m-1}(t) \int_0^t (1-a/t)^{m-1} da = \frac{2t}{m}Z_{m-1}(t), \\ J_m(t) &= \int_0^1 aZ_{m-1}(t-a) da \\ &\geq Z_{m-1}(t) \int_0^1 a(1-a/t)^{2(m-1)} da. \end{aligned}$$

For $0 \leq a \leq 1$, the inequality $\log(1-x) \geq -x/(1-x)$ gives

$$2(m-1)\log(1-a/t) \geq -\frac{2(m-1)a}{t-a} \geq -2a.$$

Hence

$$\int_0^1 a(1-a/t)^{2(m-1)} da \geq \int_0^1 ae^{-2a} da = \frac{1-3e^{-2}}{4} > \frac{1}{7}.$$

The last strict inequality follows from $e^2 > 7$: the terms of its power series through degree four sum to seven and the remaining terms are positive. It follows that $J_m(t)/Z_m(t) > m/(14t)$.

Substituting in (74), with $t = m + 1$, proves

$$(75) \quad A_m - 1 = \frac{mJ_m(t)/Z_m(t) - 1}{m + 1} > \frac{m^2 - 14m - 14}{14(m + 1)^2} > 0 \quad (m \geq 15).$$

The numerator on the right equals one at $m = 15$ and is increasing thereafter.

For $m \geq 6$, the ray e_0 is the only ray on its line. The other central rays are not collinear with the negative sum of at least two basis vectors, and every other ray has a nonzero leaf component. Thus $\mathbb{C}e_0$ has normalized ray weight $A_m > 1$ and gives the stated strict destabilizer. Together with the finite table and the tree criterion, this proves the theorem. $\square$

Corollary A.1. *For $m \geq 6$, the first term of the Harder–Narasimhan filtration of T_{Y_m} is the line bundle $\mathcal{O}_{Y_m}(D_{e_0})$.*

Proof. Put $A = A_m$, and let B, C be the normalized degrees of $-u_i, s_i$, respectively. The degrees of u_i, e_i are $B - C, A - C$, so $(m + 1)A + 2mB - mC = 2m$. On the simplex $\Delta = \{a_i \geq 0 : \sum_i a_i \leq m + 1\}$, put $F(a) = \prod_{j \neq i} \ell(a_j)$ and $Z = Z_m(m + 1)$. The leaf facets give

$$B = \frac{\int_{\Delta} F da}{Z}, \quad C = \frac{\int_{\Delta \cap \{a_i \leq 1\}} F da}{Z}.$$

Since $\ell \leq 2$, strictly on an open set, $B > 1/2$. Also $Z(1 - 2B + C) = \int_{\Delta \cap \{a_i \leq 1\}} a_i F da > 0$. Thus $2B - C < 1$, and positivity of $B - C$ gives $C < 2B - C < 1$.

The first Harder–Narasimhan term is unique, hence preserved by the torus and by arm permutations. Its generic subspace is ray-spanned: otherwise taking the span of its contained rays preserves its positive degree and decreases its rank, contradicting maximality of slope. Define

$$L = \mathbb{C}e_0, \quad \mathcal{E} = \text{span}(e_i), \quad \mathcal{U} = \text{span}(u_i), \quad \mathcal{S} = \text{span}(s_i).$$

An invariant ray-spanned subspace is generated by complete ray orbits. The two leaf-sign orbits have the same span; any two of $\mathcal{E}, \mathcal{U}, \mathcal{S}$ span the ambient space, and $L \subset \mathcal{E}$. Thus the nonzero proper possibilities are $L, \mathcal{E}, \mathcal{U}, \mathcal{S}, L + \mathcal{U}, L + \mathcal{S}$. Their normalized slopes at the three rank- m spaces are

$$\hat{\mu}(\mathcal{E}) = 2 - 2B < 1, \quad \hat{\mu}(\mathcal{U}) = 2B - C < 1, \quad \hat{\mu}(\mathcal{S}) = C < 1.$$

The last two spaces have slopes $(A + m(2B - C))/(m + 1)$ and $(A + mC)/(m + 1)$, both strictly below $A > 1$. Hence this first term has generic subspace L . It is saturated and rank one, hence reflexive and a line bundle on the smooth variety. The induced filtration jumps only at e_0 , with multiplicity one, so this line bundle is $\mathcal{O}_{Y_m}(D_{e_0})$. $\square$

Rank-one maximal destabilizing subsheaves also occur in the four-dimensional examples of Reynolds [Rey23, Appendix A, p. 68].

A.2. Proof for two branching vertices.

Proof. For the unit leaf input ℓ , the incoming function supplied by a degree- d branching vertex with its $d - 1$ pendant leaves is

$$g_d(a) = [z^d] e^{(1+a)z} Q_\ell(z)^{d-1} \quad (0 \leq a \leq 1),$$

by Lemma 6.3. It is constant above one. The three polynomials are

$$\begin{aligned} g_3(a) &= \frac{137 + 155a + 60a^2 + 8a^3}{12}, \\ g_4(a) &= \frac{5217 + 5965a + 2610a^2 + 520a^3 + 40a^4}{120}, \\ g_5(a) &= \frac{122889 + 141357a + 66120a^2 + 15760a^3 + 1920a^4 + 96a^5}{720}. \end{aligned}$$

The degree- d centre receives $T^{L-1}g_e$ from the connecting path. For any positive input p , write $c = p(1)$, $i = \int_0^1 p(a) da$, and $j = \int_0^1 ap(a) da$. Then

$$(76) \quad T(p)(a) = (2i - j + c/2) + (i + c)a + (c/2)a^2.$$

All coefficients are positive, since $2i - j > 0$.

First consider the degree-two vertices of the connecting path. The polynomials g_d are convex on $[0, 1]$, and their endpoint differences are

$$g_3(1) - g'_3(1) = \frac{61}{12}, \quad g_4(1) - g'_4(1) = \frac{1447}{120}, \quad g_5(1) - g'_5(1) = \frac{3821}{144}.$$

The tangent-line inequality at one gives $g_d(a) \geq ag_d(1)$. Equation (76) also gives

$$T(p)(a) - aT(p)(1) = (1 - a)(2i - j + c/2 - ac/2) \geq 0.$$

Every input at a connecting degree-two vertex therefore satisfies (20). The two-input calculation in the proof of Theorem 3.6 again gives $A_v \leq 24/25 < 1$ at each of these vertices, regardless of L .

It remains to treat the two branching vertices. For $d \in \{3, 4, 5\}$ define the linear functional

$$(77) \quad D_d(p) = [z^d]e^z Q_\ell(z)^{d-1} Q_p(z) - [z^{d-1}]e^z Q_\ell(z)^{d-1} Q_p(z).$$

It equals $V - F$ at the degree- d centre, so positivity is equivalent to $A_v < 1$. When a monomial a^k is used as an input, it is understood to be continued constantly with value one above one. Then

$$T(a^k) = \left(\frac{2}{k+1} - \frac{1}{k+2} + \frac{1}{2} \right) + \left(\frac{1}{k+1} + 1 \right) a + \frac{1}{2} a^2.$$

The required values of $D_d(T(a^k))$ are printed in Table 2. They are exact evaluations of (41) and (77).

TABLE 2. Positive values of $D_d(T(a^k))$ for the capped monomial inputs.

d	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
3	4477/360	611/80	1483/240	7909/1440	367/72	24403/5040
4	362/15	19847/1440	15361/1440	44209/4800	15077/1800	79019/10080
5	102241/3360	725819/60480	398891/60480	55289/13440	410239/151200	15739/8640

For $L = 2$, the input before the final application of T is g_e , of degree at most five with positive coefficients. For $L \geq 3$, it is $T^{L-2}g_e$, a quadratic with positive coefficients by (76). In either case it is a nonnegative linear combination of the capped monomials $1, a, \dots, a^5$, with at least one positive coefficient. Linearity and Table 2 prove $D_d(T^{L-1}g_e) > 0$ for every $L \geq 2$. Interchanging d, e proves the inequality at the other centre.

For $L = 1$, the inputs are g_e, g_d directly. The six possibilities up to interchange are listed in Table 3. Their strictly positive values follow from the displayed polynomials and coefficient formula (77), divided by the corresponding positive volumes.

TABLE 3. The smaller normalized deficit at the two adjacent branching vertices.

(d, e)	$\min(1 - A_{\text{left}}, 1 - A_{\text{right}})$
(3, 3)	104355/994027
(3, 4)	2643419/54754807
(3, 5)	14018811/1317077851
(4, 4)	97814081/2152964421
(4, 5)	1281377927/155298531575
(5, 5)	8349076421/1244343460105

Every pendant leaf satisfies $A_v < 1$ by Lemma 3.2. Theorem 3.4 now proves stability. The remaining assertions follow from Proposition 3.1. $\square$

APPENDIX B. PATH DEGREES, RANKS AND UNIFORM ESTIMATES

We prove Theorem 9.1. Retain the notation $X_j^{(b_0, b_j)}$ from Section 9. The proof is computer assisted. We first express the divisor degrees by products of fixed rational matrices and the relevant subspace ranks by a recurrence with three states. A contraction estimate then reduces all path lengths to a specified finite calculation. The matrices, recurrence, rational bounds, and error estimate are given below.

B.1. Divisor degrees along the path. Write $a_i = \langle m_i, r_i \rangle \in [-1, 1]$ for the coordinate on the i th dual hexagon and $d\nu(a) = (2 - |a|) da$. The factors in the volume formula are

$$L_0 = b_0 + 1 - a_0, \quad L_c = 3 + a_{c-1} - a_c \quad (1 \leq c < j), \quad L_j = b_j + 1 + a_{j-1}.$$

Set

$$(78) \quad \mathcal{K}(a, a') = \frac{(3 + a - a')^2}{2}, \quad \ell_0(a) = \frac{(b_0 + 1 - a)^{b_0}}{b_0!}, \quad h_0(a) = \frac{(b_j + 1 + a)^{b_j}}{b_j!}.$$

Define positive functions recursively by

$$(79) \quad \ell_{i+1}(a') = \int \ell_i(a) \mathcal{K}(a, a') d\nu(a), \quad h_{k+1}(a) = \int \mathcal{K}(a, a') h_k(a') d\nu(a').$$

All integrals without limits in this section are over $[-1, 1]$. If Z_j is the volume of the moment polytope, then

$$Z_j = \int \ell_i(a) h_{j-1-i}(a) d\nu(a), \quad D = n! Z_j.$$

For a hexagon ray v at position i , and an interior base ray at position c , respectively, the normalized degrees are

$$(80) \quad \delta_{i,v} = \frac{\int \ell_i h_{j-1-i} dE_v}{\int \ell_i h_{j-1-i} d\nu},$$

$$(81) \quad \delta_c = \frac{\iint \ell_{c-1}(a) (3 + a - a') h_{j-1-c}(a') d\nu(a) d\nu(a')}{\iint \ell_{c-1}(a) \mathcal{K}(a, a') h_{j-1-c}(a') d\nu(a) d\nu(a')}.$$

Here E_v is the push-forward of lattice length on the corresponding hexagon edge. For $v = r, -r$ it is unit mass at $-1, 1$, respectively; for the other four rays it is Lebesgue measure on $[-1, 0]$ or $[0, 1]$, twice each. At the left end, the numerator for a single ray is obtained by replacing ℓ_0 with $(b_0 + 1 - a)^{b_0 - 1} / (b_0 - 1)!$; the right end is analogous. These formulas follow by integrating the product of simplex volumes in Section 8.

For completeness, the rational matrices needed to evaluate the integrals are as follows. On $1, a, a^2$, the moment matrix and the kernel coefficient matrix are

$$(82) \quad W = \begin{pmatrix} 3 & 0 & 5/6 \\ 0 & 5/6 & 0 \\ 5/6 & 0 & 7/15 \end{pmatrix}, \quad \kappa = \begin{pmatrix} 9/2 & -3 & 1/2 \\ 3 & -1 & 0 \\ 1/2 & 0 & 0 \end{pmatrix}.$$

For cubic end factors, append a fourth row and column, using

$$W_{pq} = I_{p+q}, \quad I_t = \begin{cases} 0 & t \text{ odd}, \\ 2(2/(t+1) - 1/(t+2)) & t \text{ even}, \end{cases} \quad 0 \leq p, q \leq 3,$$

and extend κ by zero. If λ, η are the coefficient vectors of ℓ_0, h_0 , then

$$(83) \quad Z_j = \lambda^\top (W\kappa)^{j-1} W\eta.$$

Replace the relevant W by $W_v = (\int a^{p+q} dE_v)_{pq}$ to compute a hexagon degree. For an interior base degree replace the relevant κ by the coefficient matrix of $3+a-a'$. The four edge moment sequences are $(-1)^t, 1, (-1)^t/(t+1)$, and $1/(t+1)$. Thus every finite-path degree used below is rational and is determined by the displayed data.

B.2. Invariant subspaces and ranks. Let A be generated by the independent hexagon reflections $(x, y) \mapsto (x - y, -y)$, the permutations of the two untwisted rays at each interior base factor, and the permutations of the b untwisted rays at each end $\mathbb{P}^b$. These preserve the fan and the degrees. By Proposition 2.4, it suffices to test the A -invariant ray-spanned subspaces, since the path has connected ray matroid.

Taking all rays in the span of a ray subset preserves its rank and increases its weight. The two nontrivial reflection orbits in a hexagon each span its plane. Hence it suffices to consider the following *configurations*: choose, at each hexagon, nothing, the pair $\{r_i, -r_i\}$, or all six rays; at each base factor, choose all its untwisted rays or none, and choose its twisted ray independently. Configurations need not be closed under taking the rays in their span, but include every invariant closed set. Positivity for all nonempty configurations of proper rank is therefore equivalent to stability.

Write $H_i \in \{0, 1, 2\}$ for the hexagon choice, and $(C_c, T_c) \in \{0, 1\}^2$ for the choices of the untwisted rays and twisted ray at base vertex c . Let $g_c = \sum_{t=1}^{b_c} f_t^c$. The nontrivial representation contributed by a selected hexagon plane has dimension one; that contributed by selected untwisted base rays has dimension $b_c - 1$. For $b_c = 1$ this latter contribution is zero. The remaining vectors lie in the space with basis $r_0, \dots, r_{j-1}, g_0, \dots, g_j$, where

$$(84) \quad f_0^0 = -g_0 - r_0, \quad f_0^c = -g_c + r_{c-1} - r_c \quad (1 \leq c < j), \quad f_0^j = -g_j + r_{j-1}.$$

Lemma B.1. *The rank of a configuration is computed as follows. A selected hexagon line or plane contributes H_i ; selected untwisted rays at vertex c contribute $b_c C_c$. Each selected twisted ray with $C_c = 0$ contributes one further dimension. For the remaining twisted rays, form a graph with vertices $r_0, \dots, r_{j-1}$ and one additional vertex $*$. Identify every selected r_i with $*$. Include $r_{c-1} r_c$ when $C_c T_c = 1$, and include the edges $*r_0$ and $r_{j-1}*$ when $C_0 T_0 = 1$ and $C_j T_j = 1$. Add the rank of this graphic matroid to the preceding contributions.*

A configuration spans the whole lattice space if and only if every hexagon is a plane, every interior base has $C_c = 1$, and each end has $C_c = 1$ or has $b_c = 1$ and $T_c = 1$.

Proof. The nontrivial representations of the independent group factors are distinct and occur with multiplicity one. After accounting for them and the selected basis vectors r_i, g_c , a twisted ray whose g_c is not selected has a coordinate occurring in no other remaining vector. It adds one dimension and may be removed. The other vectors in (84), modulo the selected basis vectors, are precisely the incidence vectors of the stated graph. Their rank is its graphic-matroid rank. Full rank requires each nontrivial representation, hence every hexagon plane and every interior untwisted pair. All r_i are then selected, and the only remaining condition is to obtain each end's base space, as stated. $\square$

Here is an explicit recurrence for the minimum. After processing hexagon i , the state of r_i is G if it is selected, g if it is unselected but joined to $*$ through the processed edges, and u otherwise. A period consists of an interior base factor followed by its right-hand hexagon. Let w_h be the weight of hexagon choice h , where $w_0 = 0$, and let b be the weight of one base ray. For a choice (C, T, h) put

$$L = CT, \quad a(C, T) = 2C + T(1 - C).$$

The rank increment, new state, and weight increment are given by

	condition	rank increment	new state	weight increment
(85)	$h \geq 1$	$a(C, T) + h + L \mathbf{1}_{s=u}$	G	$(2C + T)b + w_h$
	$h = 0, L = 1$	$a(C, T) + 1$	g if $s \neq u$, u otherwise	$(2C + T)b$
	$h = 0, L = 0$	$a(C, T)$	u	$(2C + T)b$

The left end contributes $b_0 C_0 + T_0(1 - C_0)$ dimensions before the first hexagon; its possible remaining edge starts at $*$. The right end contributes $b_j C_j + T_j(1 - C_j) + C_j T_j \mathbf{1}_{s=u}$ dimensions. Its weight is $(b_j C_j + T_j) \delta_j$, and similarly at the left. To process the initial hexagon, use (85) with incoming state G , $L = C_0 T_0$, and $a(C, T) = 0$, since the left base contribution has already been included; only the hexagon weight is subtracted at this step.

Keep Boolean variables e, f recording whether any ray has been selected and whether the full-rank conditions on the processed factors still hold. After choosing the left end, initialize

$$e = C_0 \vee T_0, \quad f = C_0 \vee (b_0 = 1 \wedge T_0).$$

After the initial hexagon, replace these by $e \vee (H_0 > 0)$ and $f \wedge (H_0 = 2)$. Each interior period updates them by

$$e \mapsto e \vee C \vee T \vee (h > 0), \quad f \mapsto f \wedge C \wedge (h = 2).$$

At the right end, the final values are $e \vee C_j \vee T_j$ and $f \wedge (C_j \vee (b_j = 1 \wedge T_j))$. At each step retain the least cost, defined as rank minus weight, for each state and these two variables. At the end discard final $e = 0$ and final $f = 1$. This gives the exact *minimum invariant rank deficit*, denoted $m_A(j; b_0, b_j)$. Only its positivity is needed; no equality with the minimum over arbitrary subspaces is asserted.

B.3. Uniform convergence. For positive functions on $[-1, 1]$ define Hilbert's projective distance by

$$d_H(f, g) = \log \left(\sup_a \frac{f(a)}{g(a)} \sup_a \frac{g(a)}{f(a)} \right), \quad \Delta = \log(81/25), \quad \tau = 2/7.$$

Lemma B.2. *For $i, i' \geq 1$ one has $d_H(\ell_i, \ell_{i'}) \leq \Delta \tau^{\min(i, i')-1}$, and the same holds for the right-hand functions. The bound also compares chains starting from different end factors. For a hexagon or interior-base degree ratio, viewed as a function $\delta(i, k)$ of its left and right chain indices,*

$$|\log \delta(i, k) - \log \delta(i', k')| \leq d_H(\ell_i, \ell_{i'}) + d_H(h_k, h_{k'}).$$

The limits $\delta(i, \infty)$, $\delta(\infty, k)$, and $\delta_\infty = \delta(\infty, \infty)$ exist. In particular,

$$\begin{aligned} |\log \delta(i, k) - \log \delta_\infty| &\leq \Delta(\tau^{i-1} + \tau^{k-1}) & (i, k \geq 1), \\ |\log \delta(i, k) - \log \delta(i, \infty)| &\leq \Delta \tau^{k-1} & (i \geq 0, k \geq 1). \end{aligned}$$

These bulk limits are independent of the end factors. An end-ray degree has a one-sided limit, obtained by sending the opposite chain index to infinity; its logarithmic error satisfies the corresponding one-index bound above, with the end factor fixed. This limit may depend on that end factor. Every individual ray weight, including all these limiting weights, is at most 27.

Proof. The Birkhoff–Hopf theorem for a strictly positive continuous kernel gives projective diameter equal to the logarithm of its maximal cross-ratio and contraction factor $\tanh(\Delta/4)$ [EN95, Theorems 3.5 and 6.3]. The measure ν has full support. For $a > b$, the cross-ratio of $3 + a - a'$ increases with a' and decreases with b' ; at $a' = 1, b' = -1$ it is

$$\frac{(2+a)(4+b)}{(4+a)(2+b)} \leq \frac{9}{5}.$$

This increases with a and decreases with b , and equality holds at $a = 1, b = -1$. Exchanging the pairs treats $a < b$. Squaring gives $81/25$ for $\mathcal{K}$, and $\tanh(\Delta/4) = 2/7$. One application puts any two positive initial functions at distance at most Δ ; each subsequent application contracts this distance by τ .

Each degree ratio is the ratio of two integrals of the chain functions against positive measures. If $m \leq f/\tilde{f} \leq M$, replacing f by $\tilde{f}$ changes such a ratio by a factor between m/M and M/m . Applying this to the two chain functions proves the logarithmic estimate. It makes their degree ratios Cauchy and proves the stated limits without any spectral assumption.

For the last assertion, each initial end polynomial has maximum divided by minimum at most 9. For fixed a , the ratio of the largest to the smallest value of $\mathcal{K}(a, a')$ as a' varies is at most 9; the transposed kernel has the same property. All chain functions therefore have oscillation ratio at most 9. A hexagon edge measure has total mass 1 and ν has mass 3, so (80) is at most $9^2/3 = 27$. An interior base weight is an average of $2/(3 + a - a') \leq 2$, and an end weight is an average of $b/(b + 1 \pm a) \leq 1$. The bounds persist under limits. $\square$

B.4. The finite calculation and its uniform error bound. At limiting weights an interior period has total weight exactly 4. Indeed, Lemma B.2 bounds the difference between the total weight of a path and $(j-1)$ times its limiting period weight by a constant independent of j : the geometric tails are summable and the end factors have bounded weight. The total weight is $n = 4j - 2 + b_0 + b_j$, so division by j gives the assertion.

Consequently the all-selected period at state G has cost zero, as does the empty period at state u . For the limiting period recurrence, every other simple directed cycle has cost greater than 0.1259. This is a finite rational verification with controlled error: substitute the chain indices (80, 80) for (∞, ∞) in (80)–(81), enumerate the 36 transitions of (85), and then the simple cycles of length at most three. Table 4 lists the least transition costs after removing the two zero loops. Each positive entry lies within 10^{-6} of the displayed number; the limiting

entries marked 0 are exactly zero. There is no transition from u to g . Thus those remaining zero transitions lie on no zero-cost cycle.

TABLE 4. Least limiting transition costs, with the all-selected $G \rightarrow G$ and empty $u \rightarrow u$ loops removed.

	G	g	u
G	0.125902	0.874098	0
g	0	0.874098	0
u	0.125902	—	0.291366

We now specify the finite comparison for a path of length $j = 2q + m$, where $q = 40$ and $m \geq 0$. A period has index p when it consists of the base factor p and hexagon p . Define comparison weights by using $\delta(i, \infty)$ at the left end and periods $1, \dots, q - 1$, δ_∞ at periods $q, \dots, q + m - 1$, and $\delta(\infty, k)$ at periods $q + m, \dots, j - 1$ and the right end. The initial hexagon belongs to the left end. Let $M(m)$ denote the costs of configurations with these comparison weights.

There are at most $9q + 4$ rays in each end zone. At a hexagon p the chain indices are $(p, j - 1 - p)$; at an interior base factor p they are $(p - 1, j - 1 - p)$. Put

$$\bar{\Delta} = \frac{11757}{10000} > \Delta, \quad x = \bar{\Delta}\tau^{q-2}, \quad y = 2\bar{\Delta}\tau^{79}, \quad s = \frac{2x}{1 - \tau}.$$

The strict upper bound on Δ follows, using rational arithmetic, from $\sum_{r=0}^7 \bar{\Delta}^r / r! > 81/25$. Since $e^t - 1 \leq t + t^2$ for $0 \leq t \leq 1$, Lemma B.2 bounds the sum of all weight changes between a true path and $M(m)$ by

$$(86) \quad E_{\text{path}} = 27\{9s(1 + 2x) + 2(9q + 4)(x + x^2)\}.$$

The first term sums the two geometric tails in the central periods. The second treats both end zones, using the weaker index $q - 2$ that also applies to the first base factor of the right zone when $m = 0$. This bound is uniform in m and in the configuration selected.

For the four comparison paths $M(0), \dots, M(3)$, replacing every infinite chain index by 80 changes the cost of any configuration by at most

$$(87) \quad E_{\text{rat}} = 27\{9(2q + 4) + 4\}(y + y^2) < 6 \cdot 10^{-39}.$$

The same bound more than covers a cycle of length at most three. All such substituted computations use rational numbers only. Their all-selected costs have absolute value below 10^{-36} . Thus the true all-selected cost of $M(m)$ has absolute value below $2 \cdot 10^{-36}$ for every m , because adding a limiting full period changes that cost by zero.

Table 5 gives rational lower bounds from the recurrence, first for every true path $1 \leq j \leq 84$, and then for each of the four rational comparison paths. The bounds in the table are terminating decimals, not rounded approximations to the minima. Reversing the path exchanges its ends, so the six rows cover all nine ordered end types.

The calculation is completely specified by (82), the moment sequences following it, the end polynomials (78), and the recurrence (85), with the two rank flags described there. The accompanying implementation reproduces these tables using exact rational arithmetic; Appendix F describes additional independent checks.

Proof of Theorem 9.1. The finite calculation proves positivity for $j \leq 84$. For the remaining lengths use $M(m)$ as above. Every configuration of proper nonzero rank in $M(0), \dots, M(3)$ costs more than $0.0295 - E_{\text{rat}}$. The empty configuration costs zero. A full-rank configuration

TABLE 5. Lower bounds for the minimum invariant rank deficit of nonempty configurations of proper rank. The comparison column uses chain index 80 in place of infinity.

(b_0, b_j)	every true path $1 \leq j \leq 84$	each comparison path $M(0), \dots, M(3)$
(1, 1)	0.029505	0.029505
(2, 1)	0.029505	0.029505
(2, 2)	0.058900	0.058900
(3, 1)	0.003851	0.029505
(3, 2)	0.035003	0.039341
(3, 3)	0.039341	0.039341

costs at least the all-selected configuration, because all weights are positive, and hence at least $-2 \cdot 10^{-36}$.

For $m \geq 4$, two boundaries of the central periods have the same state. Delete the periods between them. The remaining configuration belongs to $M(m')$ with $m' < m$ and has the same rank increments outside the deleted walk. If the walk consists only of the all-selected zero loop or only of the empty zero loop, deletion preserves both its cost and whether the whole configuration is nonempty of proper rank. In the first case all deleted hexagons and base factors satisfy the full-rank conditions; the remaining state G prevents the new configuration from being empty. In the second case the remaining state u prevents it from having full rank. If the walk is not of either kind, its decomposition into simple cycles contains a positive-cost cycle and its cost exceeds 0.1259.

Induction therefore gives a lower bound $0.0295 - E_{\text{rat}} - 2 \cdot 10^{-36}$ for every nonempty proper-rank configuration of $M(m)$. Subtracting the true-path error, every relevant rank deficit is therefore greater than

$$0.0295 - (E_{\text{path}} + 2E_{\text{rat}} + 2 \cdot 10^{-36}) > 0.0295 - 6 \cdot 10^{-17} > 0.$$

Apply Proposition 2.4 to conclude stability. $\square$

Corollary B.3. *For every $n \geq 4$ there is a smooth toric Fano n -fold with anticanonically stable tangent bundle and Picard number $5\lfloor n/4 \rfloor + 1$. This is the largest Picard number of a stable example among the connected Fano hexagon bundles over graphs defined in Section 8.*

Proof. Write $n = 4j + t$, $0 \leq t \leq 3$. Choose the end types $(1, 1), (2, 1), (2, 2), (3, 2)$ for $t = 0, 1, 2, 3$, respectively. Equation (60) gives the claimed dimension and Picard number, and Theorem 9.1 gives stability. The upper bound within this class is the graph bound of Section 8. $\square$

APPENDIX C. PROOF FOR THE END ATTACHMENTS

We prove Theorem 9.2, using the varieties and root lists defined in Section 9.

C.1. Geometry and divisor degrees. The lattice is the direct sum of the hexagon lattices and the base lattices. In a base factor of dimension d , the lifted rays are $e_1, \dots, e_d, t - \sum e_i$, where t is its stated twisting vector. Maximal cones contain adjacent ray pairs in each hexagon and omit one ray in each base factor. The simplex lengths in the anticanonical polytope are at least one, so its normal fan is precisely the displayed complete bundle fan; the block triangular maximal-cone matrices are unimodular. The primitive collections are

the hexagon primitive collections and the lifted primitive collections of the base. Each leaf relation has degree $2 - 1 = 1$, and each internal base relation has degree $3 - 2 = 1$. The fibre primitive relations have degrees one and two. Thus the variety is Fano. The incidence graph is connected; within each hexagon the rays form one matroid component, and each lifted base relation connects its base coordinates to all the incident hexagon components. Hence the ray matroid is connected.

The number of base factors is $j + a + b + 1$ and their total dimension is $2j + a + b$. The dimension and Picard formulas follow. We prove stability using the following explicit degree and rank calculations. The dual anticanonical hexagon is

$$Q = \{(x, z) : |x|, |z|, |x + z| \leq 1\}.$$

Its vertices, cyclically ordered, are $(1, 0), (1, -1), (0, -1), (-1, 0), (-1, 1), (0, 1)$. For $A_a = (t_1, \dots, t_{a+1})$ set

$$P_a(x, z) = \prod_{\ell=1}^{a+1} (2 + \langle (x, z), t_\ell \rangle), \quad m_a(p) = \int_Q x^p P_a(x, z) dx dz \quad (0 \leq p \leq 2).$$

The three-entry moment vectors are

a	$m_a(0)$	$m_a(1)$	$m_a(2)$
0	6	$-5/6$	$5/3$
1	$139/12$	$-5/3$	$10/3$
2	$67/3$	$-31/10$	$31/5$
3	43	$-86/15$	$1373/120$

For a hexagon ray h , define $m_{a,h}$ by the same formula integrated over its facet $\langle (x, z), h \rangle = -1$ with lattice measure. For either ray of leaf ℓ , define $m_{a,\ell}$ by omitting the factor $2 + \langle (x, z), t_\ell \rangle$ from P_a . These prescriptions specify every modified moment vector without rounding. Reflection changes a right-hand moment vector v_p to $(-1)^p v_p$. The reflected local hexagon ray h is the global ray $-h$.

Write

$$W = \begin{pmatrix} 3 & 0 & 5/6 \\ 0 & 5/6 & 0 \\ 5/6 & 0 & 7/15 \end{pmatrix}, \quad \kappa = \begin{pmatrix} 9/2 & -3 & 1/2 \\ 3 & -1 & 0 \\ 1/2 & 0 & 0 \end{pmatrix}, \quad \kappa_1 = \begin{pmatrix} 3 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

These are respectively the moments of $(2 - |x|)dx$, the coefficients of $(3 + x - x')^2/2$, and the coefficients of $3 + x - x'$. For $j \geq 2$, let $L_0 = m_a$ be a row vector and let R_0 be the reflected m_b as a column vector. Set

$$L_{i+1} = L_i \kappa W, \quad R_{i+1} = W \kappa R_i.$$

The anticanonical polytope volume is $Z = L_0 \kappa R_{j-2}$, and $(-K_X)^n = n!Z$. Normalized weights are facet volumes divided by Z . For the left end replace L_0 by its modified vector; for the right replace R_0 by its reflected modified vector. For the internal base factor c , $1 \leq c \leq j-1$, each of its three rays has weight

$$\beta_c = \frac{L_{c-1} \kappa_1 R_{j-1-c}}{Z}.$$

For an internal hexagon i , $1 \leq i \leq j-2$, replace W in $L_{i-1} \kappa W \kappa R_{j-2-i}$ by its facet moment matrix W_h . Its denominator is Z . These formulas follow by integrating the simplex slices and applying Fubini. The case $j = 1$ is simply the integral of $P_a(x, z) P_b(-x, -z)$ over Q .

C.2. End subspaces and the rank recurrence. An end of type a has lattice $E_a = \mathbb{Z}^2 \oplus \mathbb{Z}^{a+1}$ and rays

$$(h, 0) \ (h \in \{\pm r, \pm q, \pm(r+q)\}), \quad (0, e_\ell), \ (t_\ell, -e_\ell) \quad (1 \leq \ell \leq a+1).$$

Enumerate the ray-spanned subspaces by starting with zero, adjoining each ray in turn, and closing under inclusion of all rays in the span. This finite, exact linear-algebra procedure gives respectively 13, 34, 92, 241 subspaces for $a = 0, 1, 2, 3$, including zero and the whole space. All their ray subsets and ranks are supplied with the computation. For such an end subspace S , record its rank, selected ray weight, whether $r \in S$, and whether $S + \mathbb{C}r = E_{a,\mathbb{C}}$. No symmetry of either end is assumed.

Use only the independent reflections of the internal hexagons fixing r_i , and the transpositions of the two untwisted rays of each internal $\mathbb{P}^2$. These preserve the entire fan, including the end pieces. Every invariant ray-spanned subspace is represented by an end choice on each side, choices $H_i \in \{0, 1, 2\}$ of empty set, distinguished line, or whole plane in each internal hexagon, and binary choices C_c, T_c for the two untwisted base rays and the twisted ray. Passing to closure can only increase weight at fixed rank, so minimizing over all these configurations suffices.

The outgoing boundary state is G if its distinguished ray was selected, g if that line is spanned through the preceding relations, and u if it is not spanned. Initialize at the left end with G when $r \in S$ and u otherwise, with rank $\dim S$. Adjoining an internal base and the following hexagon adds the rank

$$d = 2C + T(1 - C) + \begin{cases} H + \mathbf{1}_{CT=1, s=u}, & H \geq 1, \\ CT, & H = 0, \end{cases}$$

and gives state G when $H \geq 1$, state g when $H = 0$, $CT = 1$, and $s \neq u$, and state u otherwise. Its selected weight is $(2C + T)\beta + \omega(H)$, where $\omega(H)$ is the empty, line, or plane weight. This is the usual path relation $-g_c + r_{c-1} - r_c$ modulo previously selected subspaces.

For the selected subspace V in the processed ambient space E , define binary indicators $\eta = \mathbf{1}_{V \neq 0}$ and $F = \mathbf{1}_{E/(V+\mathbb{C}r)=0}$, where r is the outgoing distinguished ray. Initially $F = \mathbf{1}_{S+\mathbb{C}r=E_{a,\mathbb{C}}}$. At an internal step, $\eta' = \mathbf{1}_{\eta=1 \text{ or } d>0}$ and

$$F' = F \text{ and } (\mathbf{1}_{s=u} + 4 - d = \mathbf{1}_{s'=u}).$$

Indeed, if F holds, the old rank deficiency is exactly $\mathbf{1}_{s=u}$. Otherwise, a missing direction independent of the boundary can never be supplied by the remaining chain. This also proves that $F = 0$ implies $F' = 0$.

To close with right end subspace S_R , let $e = \dim S_R$ and let g_R mean $r \in S_R$. The last base and right end add rank

$$d_R = 2C + T(1 - C) + e + \mathbf{1}_{CT=1} \mathbf{1}_{s=u \text{ or } \neg g_R}.$$

The whole configuration has full rank precisely when

$$F \text{ and } (\mathbf{1}_{s=u} + 2 + (b+3) - d_R = 0).$$

The formula for d_R follows because the remaining relation $r_{j-2} - r_{j-1}$ is dependent exactly when both boundary lines are already spanned. Minimize rank minus selected weight, retaining only the least cost for each state and pair of binary indicators. Discard the empty and full-rank final configurations. This proves the finite recurrence; it is not inferred from comparisons with numerical ranks.

C.3. Uniform comparison and stability. The internal kernel is positive and has Hilbert projective contraction factor $\tau = 2/7$ and diameter $\Delta = \log(81/25)$, as in the path theorem. The end polynomials are positive on Q . Equivalently, their moment vectors are those of the positive density

$$g_a(x) = \frac{1}{2 - |x|} \int_{\{z: (x,z) \in Q\}} P_a(x, z) dz$$

against $(2 - |x|)dx$. After one application of the kernel the outgoing functions have oscillation ratio at most nine. Thus they converge to the same limiting functions as the original path ends. Every internal hexagon weight is at most 27, and every base weight is at most two. At an end, $1 \leq P_a \leq 3^4 = 81$ and the incoming function has oscillation ratio at most nine. Since a hexagon facet has lattice length one and $\text{vol}(Q) = 3$, every end hexagon weight is at most $81 \cdot 9/3 = 243$. End leaf weights are at most one. These bounds also hold for limiting and finite comparison weights.

Take $q_0 = 20$, comparison index $K = 40$, and $j = 2q_0 + m$. The left end and periods $1, \dots, q_0 - 1$ retain their left chain indices and use the limiting right function. Periods $q_0, \dots, q_0 + m - 1$ use both limiting functions. The remaining periods and the right end retain their right indices and use the limiting left function. There are at most $9q_0 + 10$ rays in either end zone. Write

$$\bar{\Delta} = 11757/10000, \quad x = \bar{\Delta}\tau^{q_0-3}, \quad y = 2\bar{\Delta}\tau^{K-2}, \quad B = 243.$$

The deliberately weaker exponents cover the first kernel application from an end piece, including its adjacent base factor. The contraction bound and $e^t - 1 \leq t + t^2$ imply the uniform true-path error

$$E_{\text{path}} = B \left\{ 9 \frac{2x}{1 - \tau} (1 + 2x) + 2(9q_0 + 10)(x + x^2) \right\}.$$

For each of the four comparison paths $m = 0, 1, 2, 3$, replacing infinite indices by K changes every configuration cost by at most

$$E_{\text{rat}} = B \{ 9(2q_0 + 4) + 10 \} (y + y^2).$$

This also covers every simple internal transition cycle.

The limiting all-selected period has weight four: summable convergence errors make the difference between total weight and the number of periods times the limiting weight bounded independently of j , whereas the total weight is $4j + a + b$. Hence the all-selected G loop and empty u loop have exactly zero cost. All other simple cycles are the same as in the original path theorem and have cost exceeding 0.1259. The exact rational comparison calculation gives the following strict lower bounds; these are terminating decimal lower bounds, not rounded minima: Their all-selected comparison costs have absolute value less than $6 \cdot 10^{-25}$.

TABLE 6. Strict lower bounds for nonempty proper-rank configurations.

(a, b)	every true $2 \leq j \leq 44$	each comparison $m = 0, 1, 2, 3$
(1, 0)	0.029505	0.029505
(2, 0)	0.029486	0.029486
(1, 1)	0.030639	0.030639
(3, 0)	0.028404	0.028404
(1, 2)	0.029486	0.029486

The exact rational bounds give

$$E_{\text{path}} + 2E_{\text{rat}} + 6 \cdot 10^{-25} < 0.00006523.$$

The finite inputs are fully specified by the displayed rays, moment integrals, matrices, recurrence, and constants. The supplied code uses rational arithmetic to evaluate them. The $j = 1$ cases are checked by positive rational decompositions into ray bases with strongly connected exchange graphs, as in Lemma F.3. Their normalized degrees independently agree with direct integration of the product of the stated linear factors over Q and its facets. The decompositions are supplied in the ancillary data.

Completion of the proof of Theorem 9.2. Apply the cycle-deletion argument of Appendix B to the central periods. A walk between repeated states either contains a positive-cost simple cycle, or consists solely of one of the two zero loops. Deleting all-selected G loops preserves nonemptiness and proper rank. Deleting empty u loops does likewise: the boundary at which deletion occurs is an internal hexagon, so state u implies that this hexagon's independent nondistinguished direction is missing; the shortened configuration cannot become full rank. Thus the end pieces cause no new exception in the zero-loop argument. A full-rank reduced configuration has cost at least that of the all-selected configuration, by positivity of weights. Deleting cycles therefore reduces every comparison configuration to $m \leq 3$, or supplies the positive cycle bound. Subtracting all errors, every nonempty proper-rank configuration at arbitrary length has deficit

$$> 0.028404 - 0.00006523 = 0.02833877 > 0.$$

Together with the finite cases and connectedness of the ray matroid, the symmetry criterion proves the theorem. $\square$

APPENDIX D. PROOF FOR THREE INDEPENDENT ARMS

We prove Theorem 9.4. The notation $Y_{a,b,c}$ and the construction are fixed in Section 9.

D.1. Independent arm subspaces and the central relation. Use the independent local reflections $r \mapsto r$, $q \mapsto -r - q$, the untwisted-ray transpositions at internal $\mathbb{P}^2$ factors, and the permutations of the three central untwisted rays. They preserve the distinguished vectors and the sums of untwisted base coordinates, hence every twist. They also preserve adjacent hexagon pairs and omitted base-ray sets. They therefore generate a group A of fan automorphisms for arbitrary arm lengths.

Every A -invariant closed ray set selects each hexagon as empty, its distinguished line, or its whole plane, denoted $H = 0, 1, 2$. At an internal base the two untwisted rays are selected together, and the twisted ray independently, with indicators C, T . At a leaf its two rays are chosen independently. At the center C_*, T_* select all three untwisted rays and the twisted ray. These configurations include every invariant flat; the arm choices need not agree.

Read each arm from the leaf to the center using the path recurrence (85). Its state $s = G, g, u$ records whether the outgoing distinguished line is selected, is spanned through earlier relations, or is unspanned. Let e indicate a nonempty arm selection. Let $f = 1$ when every hexagon is a plane, every internal base has $C = 1$, and the leaf has at least one ray selected. The reachable terminal triples are

$$(G, 1, 0), \quad (G, 1, 1), \quad (g, 1, 0), \quad (u, 0, 0), \quad (u, 1, 0).$$

Lemma D.1. *Suppose the three arm selections have ranks r_i , deficits v_i , and terminal triples (s_i, e_i, f_i) . Adding the central choice gives rank*

$$\sum_{i=1}^3 r_i + 3C_* + T_*(1 - C_*) + C_*T_*\mathbf{1}_{\{\exists i:s_i=u\}}$$

and deficit

$$(88) \quad \sum_{i=1}^3 v_i + 3C_* + T_*(1 - C_*) + C_*T_*\mathbf{1}_{\{\exists i:s_i=u\}} - (3C_* + T_*)\delta_*.$$

Here δ_* is the normalized degree of one central ray. The selection is empty exactly when all e_i, C_*, T_* vanish, and has full rank exactly when $C_* = f_1 = f_2 = f_3 = 1$.

Proof. The arm spans occupy disjoint coordinate blocks. The central untwisted rays add $3C_*$ dimensions. If $C_* = 0$ and $T_* = 1$, the twisted ray has a new central coordinate and adds one dimension. If $C_* = T_* = 1$, its remaining image is minus the sum of the terminal distinguished vectors in the direct sum of the three arm quotients. This sum is nonzero precisely when at least one summand is nonzero, giving the displayed rank. Subtracting selected weights proves (88).

Full rank requires each hexagon's nontrivial reflection representation, each internal base's nontrivial transposition representation, and the central base's two-dimensional standard representation. Thus every hexagon must be a plane, every internal untwisted pair must be selected, and $C_* = 1$. The remaining leaf coordinate requires at least one leaf ray. These conditions also suffice directly. Emptiness is immediate. $\square$

Minimize each arm deficit for its five terminal triples, and combine the results for the four central choices. This gives $5^3 \cdot 4 = 500$ entries before the empty and full-rank exclusions. Their positivity implies stability by Proposition 2.4. All quantitative deficits below concern these A -invariant configurations.

D.2. The mixed cubic integral. Write $L_1 = a, L_2 = b, L_3 = c$. On each arm number the hexagons from the center by $p = 0, \dots, L_i - 1$. With $d\nu(t) = (2 - |t|) dt$ on $[-1, 1]$ and the kernel $\mathcal{K}$ of (78), put

$$h_0(t) = 2 + t, \quad h_{k+1}(s) = \int \mathcal{K}(s, t) h_k(t) d\nu(t).$$

For positive functions u, v define

$$(89) \quad J_{u,v}(s) = \iint \frac{(4 - s - t - z)^3}{6} u(t)v(z) d\nu(t)d\nu(z),$$

$$(90) \quad J_{u,v}^-(s) = \iint \frac{(4 - s - t - z)^2}{2} u(t)v(z) d\nu(t)d\nu(z).$$

Set $h^{(i)} = h_{L_i-1}$. The left initial function on arm i is $\ell_{0,i} = J_{h^{(j)}, h^{(k)}}$, where $\{i, j, k\} = \{1, 2, 3\}$. Propagate it by the left operator in (79). The central ray degree is

$$(91) \quad \delta_* = \frac{\int J_{h^{(j)}, h^{(k)}}^-(s) h^{(i)}(s) d\nu(s)}{\int J_{h^{(j)}, h^{(k)}}(s) h^{(i)}(s) d\nu(s)}.$$

The value is independent of i . The other degrees are the path ratios (80)–(81), using left index p and right index $L_i - 1 - p$ for hexagon p , and left index $c - 1$ and right index $L_i - 1 - c$ for

internal base c . The leaf numerator replaces $2 + t$ by 1. These formulas follow by integrating the other two arms in Proposition 8.7.

To specify the new polynomials, put $M_r(u) = \int t^r u(t) d\nu(t)$. The coefficient of s^d in $J_{u,v}$ is

$$(92) \quad \sum_{\substack{r,t \geq 0 \\ d+r+t \leq 3}} \frac{(-1)^{d+r+t} 4^{3-d-r-t} M_r(u) M_t(v)}{d! r! t! (3-d-r-t)!}.$$

Replacing 3 by 2 gives $J_{u,v}^-$. Thus the four-by-four moment matrices already printed in Appendix B determine every finite degree exactly.

D.3. Uniform estimates with different arm lengths. Keep $\Delta = \log(81/25)$ and $\tau = 2/7$. Lemma B.2 applies to any positive initial function. The left and right chains have positive projective limits ℓ_∞, h_∞ , independent of their initial functions, with index- k error at most $\Delta \tau^{k-1}$ for $k \geq 1$. Each degree ratio has logarithmic error at most the sum of the projective errors of its factors.

Every individual ray weight in the following actual and comparison configurations is at most $B = 81$. Indeed, for fixed t, z the oscillation ratio of $(4 - s - t - z)^3$ is at most 27, and this persists under positive integration. Right functions have oscillation ratio at most 9. Hexagon facet measures have mass one and ν has mass three, giving $27 \cdot 9/3 = 81$. Internal, leaf and central base weights are positive averages bounded by 2, 1 and 3 respectively.

Fix $q = 16$. Call an arm short if $2 \leq L_i < 2q$, and otherwise write $L_i = 2q + m_i$, $m_i \geq 0$. First replace the incoming function $h^{(i)}$ of every long arm by h_∞ , keeping short incoming functions unchanged. Denote these functions by $\hat{h}^{(i)}$. Use them in the central degree and in each initial left function; for the moment retain every actual right chain on the arms.

Put $\bar{\Delta} = 11757/10000 > \Delta$, using the rational exponential estimate of Appendix B. For a long arm set $\epsilon_i = \bar{\Delta} \tau^{L_i-2}$, and for a short arm set $\epsilon_i = 0$. Positivity of the mixed integral gives

$$d_H(J_{h^{(j)}, h^{(k)}}, J_{\hat{h}^{(j)}, \hat{h}^{(k)}}) \leq \epsilon_j + \epsilon_k.$$

Let $z = 3\bar{\Delta} \tau^{2q-2}$. Each sum on the right is at most z . After propagation along arm i , the logarithmic weight change is at most $z \tau^p$ for hexagon p , $z \tau^{c-1}$ for internal base c , and $z \tau^{L_i-1}$ for the leaf. The central logarithmic change is at most $\sum_i \epsilon_i \leq z$.

Using $e^t - 1 \leq t + t^2$ for $0 \leq t \leq 1$, sum the six hexagon rays, three internal-base rays and two leaf rays on each arm. Their geometric sum is bounded by $9/(1 - \tau) + 2$. The four central rays give the uniform junction error

$$(93) \quad E_J = B \left(10 + \frac{27}{1 - \tau} \right) (z + z^2).$$

In particular, increasing the length of another arm does not multiply this perturbation by its number of rays.

Next replace each long arm by three zones. Read an internal period as base c followed by hexagon $c - 1$ and give both the index $p = c - 1$; hexagon p has the same zone index. For $p < q$, keep the finite left function and use h_∞ on the right. For $q \leq p < q + m_i$, use both limiting functions. For $p \geq q + m_i$, keep the right function and use ℓ_∞ on the left. The leaf uses the limiting left function. Short arms retain the preceding mixed-junction weights. Write $M(m_1, \dots, m_k)$ for this comparison, where k is the number of long arms. Its initial junction functions depend on the short lengths and the set of long arms, but not on any m_i .

Put $x = \bar{\Delta}\tau^{q-2}$ and $S = 2x/(1 - \tau)$. Each boundary zone has at most $9q + 4$ rays and omitted-chain logarithmic error at most x . In a middle period p the error is at most

$$t_p = \bar{\Delta}(\tau^{p-1} + \tau^{L_i-3-p}), \quad t_p \leq 2x, \quad \sum_p t_p \leq S.$$

There are at most nine rays per period. Summing over at most three long arms gives

$$(94) \quad E_Z = 3B\{9S(1 + 2x) + 2(9q + 4)(x + x^2)\}.$$

Hence every selected-ray deficit changes by at most $E_J + E_Z$ from the actual geometry to its comparison, uniformly in all arm lengths.

D.4. Finite comparisons and independent arm shortening. Consider the thirty short lengths $2, \dots, 31$ and the four long-arm middle lengths $m = 0, 1, 2, 3$. Their unordered triples give 7140 finite comparisons. Ordering removes duplicate geometries only; every case still includes independent selections on its arms.

For rational evaluation put $P = 40$. Replace h_∞ by h_P and ℓ_∞ by the index- P left chain starting at $2 - s$. The common limit is independent of this initial choice. With $y = 3\bar{\Delta}\tau^{P-1}$, each ray degree has logarithmic error at most y : a finite mixed left function changes by at most $2\bar{\Delta}\tau^{P-1}$, and an omitted right or left limit by at most $\bar{\Delta}\tau^{P-1}$. A finite comparison has at most $27(2q + 3) + 1$ rays, so its deficit error is bounded by

$$(95) \quad E_R = B\{27(2q + 3) + 1\}(y + y^2).$$

Let μ be the smallest proper nonempty rational deficit in these comparisons, and let F_0 be the largest absolute all-selected cost. Table 7 records the finite minima by number of long arms; each displayed lower bound is rounded down. The arm recurrence, the five terminal states, (88), and the mixed moment formula specify the finite calculation completely.

TABLE 7. Strict rational lower bounds for proper nonempty configurations in the finite mixed-arm comparisons. All short lengths are $2, \dots, 31$; each long arm has middle length $0, 1, 2, 3$.

Long arms	Comparisons	Lower bound
0	4960	0.029198008
1	1860	0.029198008
2	300	0.029198008
3	20	0.029505518

The exact rational calculations give

$$\mu > 0.029198, \quad \gamma > 0.1259, \quad F_0 < 10^{-15},$$

and

$$E_J < 6.506 \cdot 10^{-13}, \quad E_Z < 0.002216796, \quad E_R < 1.634 \cdot 10^{-16}.$$

At true limiting weights, the empty u -loop and all-selected G -loop have cost zero. The latter assertion follows because a limiting period has total weight four, as proved in Appendix B. The same period calculation gives a positive rational simple-cycle gap γ , after the two zero loops are removed. A simple cycle uses at most three nine-ray periods, so its true gap is at least $\gamma - E_R$.

Set $F = F_0 + E_R$. Inserting a full middle period changes neither the all-selected cost nor the junction data of a true comparison. Therefore the all-selected cost is at least $-F$ for

every choice of middle lengths. Every full-rank selection has cost at least this value: adding its omitted positive-weight rays changes no rank. Empty selections have cost zero.

Proof of Theorem 9.4. We first prove that every proper nonempty true comparison has deficit at least

$$\min\{\mu - E_R, \gamma - E_R - F\}.$$

Induct on the sum of its middle lengths. If each is at most three, the finite comparison applies. Otherwise choose an arm with at least four middle periods. Two of its boundary states agree among G, g, u . Delete the closed walk between them on this arm. The retained indices from the center and leaf are unchanged, as are the junction functions, so the resulting weights are those of the shorter comparison.

If the deleted walk contains only the empty u -loop, it removes no selected ray, and the retained u boundary prevents full rank. If it contains only the full G -loop, the retained G boundary prevents emptiness, and deleting selected planes and untwisted pairs removes no failure of a full-rank condition. Proper nonemptiness is therefore preserved in both zero-cost cases, and induction applies.

Every other closed walk contains a positive simple cycle, with cost at least $\gamma - E_R$. The remaining configuration is proper nonempty, empty, or full rank. Apply induction, the zero empty cost, or the bound $-F$, respectively. This proves the comparison inequality; it uses the gap of one arm, without an equal-length or simultaneous-deletion assumption.

Subtracting the actual-to-comparison error gives every proper nonempty A -invariant configuration deficit at least

$$\min\{\mu - E_R, \gamma - E_R - F\} - E_J - E_Z > 0.02698.$$

The symmetry criterion proves anticanonical stability. The dimension and Picard number follow from the graph construction. $\square$

APPENDIX E. STABILITY FOR CYCLES

The cycle construction gives another infinite family. Its additional symmetry allows a short induction in place of the finite reduction needed for paths.

Theorem E.1. *Let C_j be a cycle with $j \geq 3$ vertices and edges, or the graph with two vertices and two parallel edges when $j = 2$. Put $b_c = 2$ at every vertex. Then $X(C_j)$ is a smooth toric Fano variety with stable tangent bundle, dimension $4j$, and Picard number $5j$.*

Proof. The geometric assertions and connectedness of the ray matroid follow from Proposition 8.3. Number vertices and edges modulo j , orienting edge i from i to $i + 1$, and write $g_c = f_1^c + f_2^c$. The twisted base ray at c is

$$f_0^c = -g_c + r_{c-1} - r_c.$$

Rotation sends vertex c to $c + 1$ and hexagon i to $i + 1$, preserving the coordinates in each transported summand. Reflection sends c to $-c$ and i to $-i - 1$, and negates both coordinates in each transported hexagon. Both maps send the displayed twisted ray to the twisted ray at the image vertex. They preserve adjacent pairs of hexagon rays and the choices of an omitted base ray, hence preserve the fan. The same reflection works for $j = 2$, where it exchanges the parallel edges and fixes the vertices.

Together with the independent hexagon reflections $r_i \mapsto r_i$, $q_i \mapsto -r_i - q_i$ and the transpositions of f_1^c, f_2^c , these maps give four ray orbits: the $2j$ distinguished hexagon rays $\pm r_i$, the other $4j$ hexagon rays, the $2j$ untwisted base rays, and the j twisted base rays. A

ray-spanned invariant subspace containing any nondistinguished hexagon ray must contain the whole plane of every hexagon. Thus it suffices to consider a common hexagon choice $H \in \{0, 1, 2\}$, for empty, line, or plane, and common binary choices C, T for untwisted and twisted base rays.

Let a, b, c be the normalized degrees of a distinguished hexagon ray, a nondistinguished hexagon ray, and a base ray, respectively. Symmetry gives the first two equalities across factors. Principal divisor relations give equal degrees for all three base rays at a vertex, so there is only one base value c . Their sum satisfies

$$(96) \quad 2a + 4b + 3c = 4.$$

The nine proper nonempty configurations are listed in Table 8. For their ranks, selected untwisted base rays contribute $2jC$, and hexagon choices contribute jH . If $C = 0$, each selected twisted ray has an independent base coordinate. If $C = T = 1$, its residual vector is $r_{c-1} - r_c$: these vectors span a space of dimension $j - 1$ when $H = 0$, and vanish modulo the selected hexagon lines when $H > 0$. This gives every rank in the table. The other three configurations are empty or have full rank.

TABLE 8. Proper cycle configurations and their weights.

H	C	T	Rank	Weight divided by j
0	0	1	j	c
0	1	0	$2j$	$2c$
0	1	1	$3j - 1$	$3c$
1	0	0	j	$2a$
1	0	1	$2j$	$2a + c$
1	1	0	$3j$	$2a + 2c$
1	1	1	$3j$	$2a + 3c$
2	0	0	$2j$	$2a + 4b$
2	0	1	$3j$	$2a + 4b + c$

We will prove the bounds

$$(97) \quad a < \frac{1}{2}, \quad b > \frac{1}{4}, \quad \frac{2}{3} < c < \frac{5}{6}.$$

They imply every strict weight inequality in the table. For example, $3jc < (5/2)j \leq 3j - 1$ for $j \geq 2$; the row $(1, 1, 1)$ follows from $2a + 3c = 4 - 4b < 3$; and the rows with $H = 2$ follow from $2a + 4b = 4 - 3c$. The remaining inequalities follow directly from (97). Proposition 2.4 will then prove stability.

Let W, κ be the three-dimensional matrices in (82), and put $M = W\kappa$. For $0 \leq p, q \leq 2$, set

$$(W_r)_{pq} = (-1)^{p+q}, \quad (W_-)_{pq} = \frac{(-1)^{p+q}}{p+q+1}, \quad \kappa_1 = \begin{pmatrix} 3 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Expanding the cyclic product of kernels and integrating each variable contracts their coefficients cyclically. Consequently the volume and facet integrals of Section 8 are

$$\begin{aligned} Z_j &= \text{tr}(M^j), & A_j &= \text{tr}(W_r \kappa M^{j-1}), \\ B_j &= \text{tr}(W_- \kappa M^{j-1}), & C'_j &= \text{tr}(W \kappa_1 M^{j-1}). \end{aligned}$$

Here $a = A_j/Z_j$, $b = B_j/Z_j$, $c = C'_j/Z_j$. The denominator is the integral of the strictly positive cyclic kernel, so $Z_j > 0$. Define four rational sequences by

$$s_j^{(1)} = \frac{1}{2}Z_j - A_j, \quad s_j^{(2)} = B_j - \frac{1}{4}Z_j, \quad s_j^{(3)} = C'_j - \frac{2}{3}Z_j, \quad s_j^{(4)} = \frac{5}{6}Z_j - C'_j.$$

The characteristic polynomial of M is

$$x^3 - \frac{27}{2}x^2 + \frac{2491}{240}x - \frac{127}{864}.$$

By Cayley–Hamilton, each sequence satisfies

$$(98) \quad s_{j+3} = \frac{27}{2}s_{j+2} - \frac{2491}{240}s_{j+1} + \frac{127}{864}s_j.$$

The starting values, obtained from the displayed matrices, are in Table 9. Index one denotes a formal trace value, not a looped graph construction.

TABLE 9. Exact starting values for the four cycle inequalities.

j	$s_j^{(1)}$	$s_j^{(2)}$	$s_j^{(3)}$	$s_j^{(4)}$
1	$\frac{9}{4}$	$\frac{9}{8}$	0	$\frac{9}{4}$
2	$\frac{8431}{240}$	$\frac{6149}{480}$	$\frac{1141}{180}$	$\frac{2963}{144}$
3	$\frac{324403}{720}$	$\frac{464599}{2880}$	$\frac{184207}{2160}$	$\frac{1100717}{4320}$

All first terms are nonnegative, all second and third terms are positive, and componentwise

$$s_3 - 12s_2 = \left(\frac{20887}{720}, \frac{21871}{2880}, \frac{19903}{2160}, \frac{34037}{4320} \right) > 0.$$

Suppose $s_k > 12s_{k-1} > 0$ and $s_{k-2} \geq 0$. Recurrence (98) gives

$$s_{k+1} > \left(\frac{27}{2} - \frac{2491}{2880} \right) s_k > 12s_k > 0,$$

since $27/2 - 2491/2880 - 12 = 1829/2880 > 0$. Induction proves positivity for each sequence at every $j \geq 2$. These four signs are exactly (97), completing the proof. $\square$

All data used by the induction are printed above. Exact checks reproduce the seeds and compare the trace degrees with the independently integrated graph formulas for $j = 2, 3$.¹ These checks verify the finite matrix calculations; the induction proves the assertion for every length.

APPENDIX F. THE FINITE CLASSIFICATIONS

The path family explains why the inequality $\rho \leq n + 2$ eventually fails for stable tangent bundles. We now determine where the first failure occurs in the classifications of smooth toric Fano varieties. Throughout this section stability refers to the polarization $-K_X$.

Theorem F.1. *Let X be a smooth toric Fano variety of dimension $n \leq 7$. If T_X is stable, then $\rho(X) \leq n + 2$. For each $n = 4, 5, 6, 7$ this bound is attained, by respectively 3, 1, 10, 15 isomorphism classes.*

¹The checks are implemented in `figures/code/compute06_cycle_tables.py`.

Theorem F.2. *The largest Picard number of a smooth toric Fano eightfold with stable tangent bundle is 11. Exactly one isomorphism class attains it: the path variety X_2 of Example 8.6.*

F.1. An independent test using bases. For $S \subseteq \mathcal{R}$, put $\delta(S) = \sum_{\rho \in S} \delta_\rho$. Lemma 2.1 is equivalent to

$$(99) \quad \delta(S) < r(S) \quad \text{for every nonempty proper } S \subset \mathcal{R},$$

with weak inequalities for semistability. Indeed, if $r(S) < n$, apply the subspace criterion to the span of S . If $r(S) = n$, positivity of the weights gives $\delta(S) < \delta(\mathcal{R}) = n$. Conversely, for a proper subspace V containing rays, use the set of all rays in V ; its rank is at most $\dim V$.

A basis of the ray matroid is a subset $B \subset \mathcal{R}$ for which $\{u_\rho : \rho \in B\}$ is a basis of $N_{\mathbb{C}}$. Write $\mathbf{1}_B$ for its incidence vector. The convex hull of these vectors is the *base polytope* of the ray matroid.

Lemma F.3. *Suppose the normalized degree vector admits a decomposition*

$$\delta = \sum_{k=1}^{\ell} \lambda_k \mathbf{1}_{B_k}, \quad \lambda_k > 0, \quad \sum_{k=1}^{\ell} \lambda_k = 1,$$

where each B_k is a basis of the ray matroid. Then T_X is semistable. Form a directed graph on $\mathcal{R}$ with an edge $e \rightarrow e'$ whenever, for some k , one has $e \notin B_k$, $e' \in B_k$, and $(B_k \setminus \{e'\}) \cup \{e\}$ is also a basis. If this graph is strongly connected, then T_X is stable.

Proof. For every $S \subseteq \mathcal{R}$,

$$\delta(S) = \sum_k \lambda_k |B_k \cap S| \leq r(S),$$

because $B_k \cap S$ is independent. Equality holds precisely when $|B_k \cap S| = r(S)$ for every k . If a nonempty proper S gives equality and the directed graph is strongly connected, an edge $e \rightarrow e'$ leaves S . For its corresponding basis B_k , replacing $e' \notin S$ by $e \in S$ produces a basis containing $r(S) + 1$ members of S , a contradiction. Thus (99) holds. This argument does not use Proposition 2.4. $\square$

F.2. Coverage and exact slope comparisons. We use the complete classification of smooth Fano polytopes obtained by Øbro's algorithm [Øbr07], with the low-dimensional data from polyDB and the seven- and eight-dimensional lists distributed by Paffenholz [Paf]. A smooth Fano polytope is the convex hull of the primitive ray generators, and its facets determine the maximal cones. The facet normals of the dual moment polytope thus give the same ray configuration. This distinction matters because the two distributions use different polytope conventions.

The computations through dimension six are those of [Wue26]. In dimension four, the classification by tangent bundle stability was already obtained by Reynolds [Rey23, Appendix A]; our four-dimensional calculation is an independent verification, rather than a new classification. The additional coverage used here is precisely

n	Picard numbers examined	number of varieties
7	$\rho \geq 9$	1277
8	$\rho \geq 11$	1582.

Thus all sevenfolds at or above $\rho = n + 2$ are covered, whereas in dimension eight the complete calculation starts at $\rho = n + 3$. The eight-dimensional stratum $\rho = 10$ has not been classified here by stability. No stability classification in dimension nine is used.

For each ray configuration, the intersection numbers d_v are computed as normalized facet volumes of the moment polytope. Candidate facets and vertices obtained numerically are checked by rational linear algebra before evaluating the volumes. The identities

$$\sum_v d_v = D, \quad \sum_v d_v v = 0$$

provide consistency checks, while each final slope comparison is a comparison of rational numbers. To disprove stability it suffices to exhibit a proper ray-spanned subspace W for which

$$(100) \quad \frac{\sum_{v \in W} d_v}{\dim W} \geq \frac{D}{n}.$$

The search first examines subspaces spanned by at most four rays. If it finds no such subspace, all ray-spanned subspaces are examined. Consequently failure to find a small-dimensional subspace is never used as evidence of stability.

There are two distinct kinds of non-stability record. A strict inequality in (100) proves instability. An equality excludes stability but does not exclude the existence of a different subspace with strictly larger slope. We therefore retain the term *equality witness* for the latter records, without classifying all of them as semistable.

Proof of Theorem F.1. The bound in dimensions at most six follows from the complete low-dimensional calculation cited above and, in dimensions one and two, from the classifications of smooth toric Fano curves and surfaces. The numbers of varieties with $\rho \geq n+3$ in dimensions 4, 5, 6 are 2, 7, 52; none has stable tangent bundle. Of the 1277 sevenfolds examined, 1013 have $\rho = 9$ and 264 have $\rho \geq 10$. Among the latter, 100 admit a strict witness and 164 have a recorded equality witness. Hence none is stable. Among the former, exactly 15 are stable. Together with the stable counts 3, 1, 10 at $\rho = n+2$ in dimensions 4, 5, 6, this proves both assertions. $\square$

ρ	total	stable	unstable	equality witness
11	1286	1	665	620
12	243	0	85	158
13	40	0	12	28
14	11	0	0	11
15	1	0	0	1
16	1	0	0	1

TABLE 10. All smooth toric Fano eightfolds with $\rho \geq 11$. The final column excludes stability but does not assert semistability.

Proof of Theorem F.2. Table 10 accounts for all 1582 varieties with $\rho \geq 11$: one is stable, 762 have a strict destabilizing witness, and 819 have an equality witness. The stable variety has $\rho = 11$ and identifier `P8D.smooth_fano_8d.gz.389596` in Paffenholz’s list. Its anticanonical degree is $D = 5753440$.

Here is an explicit identification with X_2 . Order the coordinates of its lattice as

$$(r_0, q_0, r_1, q_1, f_1^0, f_1^1, f_2^1, f_1^2),$$

with the orientations in Example 8.6. In the database coordinates the lattice map is

$$(101) \quad A = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Direct evaluation gives $\det A = 1$ and shows that A maps the 19 ray generators of X_2 bijectively onto those of the listed variety. Since both fans are the fans over the faces of their smooth Fano polytopes, this ray bijection identifies the fans as well.

There is also a finite verification of stability directly from the database rays, independent of the path-family proof. Its rational base-polytope expression consists of 19 bases with positive rational coefficients summing to 1. The weighted sum of their incidence vectors equals $(\delta_v)_v$, and their directed exchange graph is strongly connected. Lemma F.3 therefore proves stability. These finite checks and the absence of any other stable record prove uniqueness and maximality. $\square$

F.3. The varieties of Picard number $n + 2$. For clarity, define the even-dimensional del Pezzo variety V_{2m} as the smooth toric Fano variety whose fan is the fan over the faces of the polytope

$$(102) \quad \text{conv}\{\pm e_1, \dots, \pm e_{2m}, \pm(e_1 + \dots + e_{2m})\} \subset \mathbb{Z}_{\mathbb{R}}^{2m}.$$

This is the del Pezzo variety of Voskresenskii and Klyachko [VK85]. It has $4m + 2$ rays and Picard number $2m + 2$; in particular $V_2 = S_6$. A toric fibre bundle with this fibre means a Zariski locally trivial toric morphism with fibre V_{2m} , equivalently a fan split by the base fan and the fan of V_{2m} as in Lemma 8.2.

Theorem F.4. *Suppose $4 \leq n \leq 7$, X is a smooth toric Fano n -fold, $\rho(X) = n + 2$, and T_X is stable. Then either n is even and $X \cong V_n$, or X is a toric fibre bundle with fibre V_{2m} , $0 < 2m < n$, over a smooth projective toric variety Y satisfying $\rho(Y) = \dim Y$. The numbers of isomorphism classes are as follows:*

n	V_n	V_2 -bundles	V_4 -bundles	total
4	1	2	0	3
5	0	1	0	1
6	1	7	2	10
7	0	14	1	15.

Proof. For the 29 stable varieties counted in Theorem F.1, we examine rational subspaces spanned by rays in even dimension. The fibre configurations are identified with (102) in integral coordinates. In each of the 27 proper-fibre cases, the non-fibre rays project bijectively onto the rays of a smooth complete quotient fan. Moreover, the maximal cones of the original fan are exactly all sums of a fibre maximal cone and a chosen lift of a quotient maximal cone. Thus the full split-fan condition holds, and the toric local triviality theorem applies. The other two configurations are V_4 and V_6 themselves.

The base is projective: a base maximal cone lifts to a cone of the original fan, and the orbit closure associated to any fixed fibre maximal cone is a section isomorphic to the base.

Since X is projective, so is this closed subvariety. Finally, counting fibre and base rays in a split fan gives

$$\rho(X) = \rho(V_{2m}) + \rho(Y), \quad \dim X = 2m + \dim Y.$$

Substituting $\rho(X) = n + 2$ and $\rho(V_{2m}) = 2m + 2$ yields $\rho(Y) = \dim Y$. $\square$

The statement describes a finite part of the classification, not a criterion asserting stability for every del Pezzo bundle of these dimensions. Nor does it imply that the base is a product of projective lines. Already in dimension six there are stable hexagon bundles over other smooth toric bases of dimension and Picard number four.

F.4. Reproducibility. The ancillary package includes standalone rational-arithmetic programs for the uniform path, end-attachment, branching and cycle calculations. Their inputs are the rays, moment matrices, rank recurrences and comparison bounds stated above. In particular, the end-attachment and branching programs reproduce the finite minima, all-selected costs and error bounds. The unequal-arm calculation evaluates all 500 central choices for each of 7140 mixed comparisons, including every short-length case required by the contraction argument. The general-tree theorem then uses an induction rather than an enumeration of tree shapes. Its additional arithmetic consists of the limiting transition inequalities and the explicit subdivision thresholds. From the ancillary directory, the commands

```
python3 -S code/end_attachment_proof.py --uniform
python3 -S code/unequal_spider_proof.py --uniform
python3 -S code/two_junction_gluing.py
python3 -S code/subcubic_thresholds.py
python3 -S code/research_graph_targets.py
```

require only the Python standard library. The package retains the exact rational outputs and 93 positive basis decompositions for independently checking selected finite examples. Running `python3 -S code/verify_certificate.py certificates/*.json` checks all these decompositions. Numerical searches for decompositions are not dependencies of their rational verification or of the uniform stability proofs.

For the mixed-blowup construction, the tree criterion, the all-length path theorem and the saturated-vertex comparison are proved by identities and inequalities without a finite graph enumeration. The fourteen exact fractions used in the star-transition proof are reproduced directly from (73) and its derivative by

```
python3 -S code/mixed_star_table.py --check-json
data/mixed_star_table/signs.json
```

The second line continues the same command. This calculation covers only $m \leq 14$; the estimate (75) proves the assertion for every $m \geq 15$.

The finite rational inputs for the tree stability theorem and the local degree-four and degree-five obstructions are reproduced by

```
python3 -S code/mixed_tree_arithmetic.py --check-json
data/mixed_tree_arithmetic/signs.json
```

This also checks the eighteen positive linear-functional values and six adjacent pairs in Proposition 6.6. The conclusions for arbitrary trees and connecting lengths use the monotonicity and induction arguments in Section 6.

The ancillary data distinguish input polytopes from varieties whose stability has actually been decided. The seven-dimensional records cover all 1277 varieties with $\rho \geq 9$; the eight-dimensional records cover precisely the 1582 with $\rho \geq 11$. They retain identifiers, ray coordinates, divisor degrees and the subspaces proving non-stability. The stable-boundary

list is restricted to $\rho = n + 2$ in dimensions four through seven. In particular it does not include X_2 , whose Picard number is $n + 3$.

For the stable eightfold, the rational expression in 19 bases can be checked using only the ray matrix, the normalized degrees and rational arithmetic: verify each determinant is nonzero, each coefficient is positive, their sum is 1, the weighted incidence-vector identity holds, and the directed exchange graph is strongly connected. This check uses no fan symmetry. The identification matrix (101) gives a separate comparison with the graph construction.²

APPENDIX G. FURTHER EXAMPLES AND RELATIVE TANGENT SLOPES

We record several consequences of the methods used above. All slopes in this appendix are taken with respect to the anticanonical polarization of the total space.

G.1. Del Pezzo and pseudo del Pezzo varieties. Let $n = 2m \geq 2$, let $e_1, \dots, e_n$ be the standard basis of $\mathbb{Z}^n$, and put $s = e_1 + \dots + e_n$. The del Pezzo variety V_n and the pseudo del Pezzo variety $\tilde{V}_n$ have ray generators

$$V_n : \quad \pm e_1, \dots, \pm e_n, \pm s, \quad \tilde{V}_n : \quad \pm e_1, \dots, \pm e_n, s,$$

respectively; their fans consist of the cones over the faces of the convex hulls of these generators [VK85, Ewa88, Nil06]. Thus V_2 is the del Pezzo surface of degree six and $\tilde{V}_2$ the one of degree seven.

Proposition G.1. *For every $m \geq 1$, both V_{2m} and $\tilde{V}_{2m}$ are smooth toric Fano varieties with stable tangent bundles. Their Picard numbers are $2m + 2$ and $2m + 1$, respectively.*

Proof. Their anticanonical polytopes are

$$[-1, 1]^n \cap \left\{ \left| \sum_i x_i \right| \leq 1 \right\}, \quad [-1, 1]^n \cap \left\{ \sum_i x_i \geq -1 \right\}.$$

Here n is even, so no cube vertex lies on a cutting hyperplane. The remaining cube vertices have coordinate sum zero in the first case and nonnegative coordinate sum in the second. Every new vertex lies on a cube edge: one coordinate is zero and the remaining coordinates are ± 1 with sum 1 or -1 in the first case, and sum -1 in the second. The primitive normals of the active facets are a signed coordinate basis at an old vertex. At a new vertex they are $n - 1$ signed coordinate vectors together with s or $-s$; their determinant is ± 1 . Both polytopes are therefore integral and simple with unimodular normal cones. The origin is interior, and all displayed inequalities define facets. This proves the smooth Fano assertion.

For V_n , use the equivalent lattice presentation $\mathbb{Z}^{n+1}/\mathbb{Z}(e_0 + \dots + e_n)$, with rays $\pm e_i$. The symmetric group S_{n+1} acts through its irreducible standard representation on $N_{\mathbb{C}}$. Its ray matroid is connected: opposite rays are parallel and $\{e_0, \dots, e_n\}$ is a circuit meeting every parallel class. Proposition 2.4 proves stability.

For $\tilde{V}_n$, the group S_n permuting the coordinate vectors preserves the fan. Its only nonzero proper invariant subspaces are $\mathbb{C}s$ and $\{\sum_i x_i = 0\}$; this follows from the decomposition into the trivial and standard representations, which are non-isomorphic irreducibles, including when $n = 2$. The latter hyperplane contains no ray. The circuit $\{e_1, \dots, e_n, s\}$ and the

²The computations are implemented by `screen_high_rank.py`, `matroid_certificate.py` and `verify_certificate.py`. The stable-eightfold expression is retained as `P8D_389596.json`. Only completed calculations enter the ancillary tables.

opposite pairs show that the ray matroid is connected. Write $d_{e_i} = p$, $d_{-e_i} = q$, and $d_s = h$. Lemma 2.2 gives $q = p + h$, so

$$D = n(p + q) + h = n(2p + h) + h > nh.$$

Thus the only invariant subspace spanned by rays, $\mathbb{C}s$, satisfies the strict slope inequality. Proposition 2.4 applies. Finally, subtracting the dimension from the number of rays gives the two Picard numbers. $\square$

G.2. Extensions with base $\mathbb{P}^1$. Let $Z = X_{\Sigma_Z}$ be a positive-dimensional smooth projective toric variety with lattice N_Z , and let $t \in N_Z$. In $N_Z \oplus \mathbb{Z}$, take the cones generated by a cone of Σ_Z and either $(t, 1)$ or $(0, -1)$, together with all their faces. Denote the resulting variety by X_t ; projection to the last coordinate gives a toric bundle $X_t \rightarrow \mathbb{P}^1$ with fibre Z .

Proposition G.2. *The fan of X_t is smooth and complete. The variety X_t is Fano if and only if Z is Fano and*

$$t \in Q \cap N_Z, \quad Q = \text{conv}\{u_\rho : \rho \in \Sigma_Z(1)\}.$$

When Z is Fano, $Q \cap N_Z$ consists exactly of the origin and the vertices of Q .

Proof. On the upper half-space, the shear $(x, y) \mapsto (x - yt, y)$ identifies the cones with the product of Σ_Z and the nonnegative ray. On the lower half-space they already have product form. This proves completeness, and each maximal cone is unimodular. If X_t is Fano, restriction of $-K_{X_t}$ to a fibre gives $-K_Z$, so Z is Fano.

Assume henceforth that Z is Fano. For a facet F of Q , let $u_F \in M_Z$ satisfy $\langle u_F, v \rangle = 1$ on F and be strictly less than one at every other vertex. The linear forms supporting the two proposed facets over F are

$$(u_F, -1), \quad (u_F, 1 - \langle u_F, t \rangle).$$

Evaluating each on the opposite apex shows that the strict convexity condition is $\langle u_F, t \rangle < 2$. Since this quantity is integral, the condition is equivalent to $\langle u_F, t \rangle \leq 1$ for every facet, hence to $t \in Q$. These are exactly the strict support-function inequalities for ampleness of $-K_{X_t}$. Finally, the simplices $\text{conv}(0, F)$ cover Q and are unimodular, so they contain no lattice points other than their vertices. $\square$

G.3. Bundles over products of projective lines. Let F be a smooth toric Fano variety of dimension $f \geq 1$, and set

$$P_F = \{m \in M_{F, \mathbb{R}} : \langle m, v \rangle \geq -1 \text{ for every ray generator } v \text{ of } F\}.$$

For $k \geq 1$, choose each t_i to be zero or a primitive ray generator of F . The bundle $X = X_F(t_1, \dots, t_k)$ over $(\mathbb{P}^1)^k$ has rays $(v, 0)$, (t_i, e_i) , and $(0, -e_i)$, with maximal cones obtained by choosing a maximal fibre cone and one ray from each base pair. Proposition G.2, applied successively, shows that X is smooth Fano. Put $n = f + k$, $[k] = \{1, \dots, k\}$, and

$$I(S) = \int_{P_F} \prod_{i \in S} \langle m, t_i \rangle dm, \quad S \subseteq \{1, \dots, k\},$$

where dm gives the character lattice covolume one and an empty product is one.

Proposition G.3. *The anticanonical degree is*

$$D = n! \int_{P_F} \prod_{i=1}^k (2 + \langle m, t_i \rangle) dm = n! \sum_{S \subseteq [k]} 2^{k-|S|} I(S).$$

For each i , the two lifted base rays have equal divisor degree $(n-1)!J_i$, where

$$J_i = \int_{P_F} \prod_{j \neq i} (2 + \langle m, t_j \rangle) dm.$$

The relative tangent bundle satisfies

$$(103) \quad \mu(T_{X/(\mathbb{P}^1)^k}) \geq \mu(T_X) \iff \sum_{S \subseteq [k]} |S| 2^{k-|S|} I(S) \geq 0.$$

The same equivalence holds with strict inequalities.

Proof. The anticanonical polytope is

$$\{(m, y) : m \in P_F, -1 - \langle m, t_i \rangle \leq y_i \leq 1 \text{ for } 1 \leq i \leq k\}.$$

All interval lengths are positive. Integration gives $D = n! \text{vol}(P_X)$. Each base facet projects by a lattice isomorphism onto the region with the i -th interval omitted; this gives its degree $(n-1)!J_i$. The fibre subspace corresponds to the relative tangent bundle, which has rank f and degree $D - 2(n-1)! \sum_i J_i$. Comparing its slope with D/n is equivalent to $kD \geq 2n! \sum_i J_i$. A term $I(S)$ occurs in the last sum exactly $k - |S|$ times, with coefficient $2^{k-1-|S|}$ each time. Subtracting gives (103). $\square$

For $k = 1$, write $b_F = \text{vol}(P_F)^{-1} \int_{P_F} m dm$. The relative tangent slope is at least $\mu(T_X)$ exactly when $\langle b_F, t_1 \rangle \geq 0$. Thus a Fano toric bundle of this form over $\mathbb{P}^1$ cannot have stable tangent bundle if $b_F = 0$.

For $k = 2$ and centrally symmetric P_F , the condition is

$$B(t_1, t_2) := \int_{P_F} \langle m, t_1 \rangle \langle m, t_2 \rangle dm \geq 0.$$

For $F = V_{2m}$, use its quotient-lattice presentation with rays $\pm e_0, \dots, \pm e_{2m}$. The form B is positive definite and S_{2m+1} -invariant. Irreducibility of the standard representation implies that it is a positive multiple of the form with entries $\delta_{ij} - 1/(2m+1)$. Hence, for nonzero twists $t_1 = \varepsilon_1 e_i$, $t_2 = \varepsilon_2 e_j$, with $\varepsilon_1, \varepsilon_2 \in \{1, -1\}$, the relative tangent bundle has slope greater than $\mu(T_X)$ precisely when

$$t_2 = t_1 \quad \text{or} \quad i \neq j \quad \text{and} \quad \varepsilon_1 \varepsilon_2 = -1.$$

For opposite twists with the same index, and for equal signs with different indices, this particular subsheaf has smaller slope. These comparisons alone do not decide stability of T_X .

G.4. Projective-line bundles over a toric base. Let Y be a smooth projective toric variety of dimension $n-1 \geq 1$, and let $X = \mathbb{P}_Y(\mathcal{O} \oplus L)$ be a toric projective-line bundle. Choose its fan presentation with fibre rays $(0, \pm 1)$ and lifted base rays (u_ρ, a_ρ) . Set $L' = \sum_\rho a_\rho D_\rho$ on Y ; the line bundle L corresponds to L' or $-L'$, according to the projectivization convention. Define

$$g(y) = (-K_Y + yL')^{n-1}.$$

Proposition G.4. *Assume X is Fano. Then $-K_Y \pm L'$ are ample, and*

$$D = n \int_{-1}^1 g(y) dy, \quad d_{(0,1)} = g(-1), \quad d_{(0,-1)} = g(1).$$

Consequently

$$\begin{aligned} \mu(T_{X/Y}) \geq \mu(T_X) &\iff g(1) + g(-1) \geq \int_{-1}^1 g(y) dy \\ &\iff \sum_{\substack{2 \leq j \leq n-1 \\ j \text{ even}}} \frac{j}{j+1} \binom{n-1}{j} (-K_Y)^{n-1-j} \cdot (L')^j \geq 0. \end{aligned}$$

For $n = 3$ this is $(L')^2 \geq 0$, and for $n = 4$ it is $(-K_Y) \cdot (L')^2 \geq 0$.

Proof. The two sections have normal bundles L and L^{-1} ; adjunction therefore identifies the restrictions of $-K_X$ with $-K_Y + L'$ and $-K_Y - L'$. They are ample, hence in particular nef. Their convex combinations $-K_Y + yL'$ are ample for $-1 \leq y \leq 1$. At height y , the anticanonical polytope has slice

$$P_y = \{m : \langle m, u_\rho \rangle \geq -1 - a_\rho y \text{ for every } \rho\},$$

the polytope of $-K_Y + yL'$, of volume $g(y)/(n-1)!$. Integrating gives D ; the facets at $y = -1$ and $y = 1$ give the two divisor degrees. The relative tangent line bundle is $\mathcal{O}_X(D_{(0,1)} + D_{(0,-1)})$, so its degree is their sum. Finally, expand $g(y) = \sum_j c_j y^j$. The difference between the endpoint sum and the integral is $2 \sum_{j \geq 2, j \text{ even}} c_j j/(j+1)$, giving the assertion. $\square$

Example G.5. A smaller relative tangent slope does not imply stability. Let $Y = \mathbb{P}^2 \times \mathbb{P}^1$ and $L' = 2H - F$, where H and F are the hyperplane classes of the two factors. Both $-K_Y \pm L'$ are ample; the strict support-function inequalities on the two sets of maximal bundle cones then show that X is Fano. Its rays may be written

$$-e_1, -e_4, e_2 - e_4, -e_2, e_4, -e_3, e_1 + e_3 + 2e_4.$$

This is the variety F.4D.0019 in the smooth Fano classification [Paf]. Here $g(y) = 3(3 + 2y)^2(2 - y)$, so

$$D = 400, \quad \mu(T_X) = 100, \quad \deg T_{X/Y} = g(1) + g(-1) = 84.$$

The divisor degrees in the displayed ray order are $(64, 75, 62, 62, 9, 64, 64)$. For example, the facet for $-e_1$ has degree $6 \int_{-1}^1 (3 + 2y)(2 - y) dy = 64$, while the two facets for the $\mathbb{P}^1$ base directions have degree $3 \int_{-1}^1 (3 + 2y)^2 dy = 62$. The plane $\text{span}(e_2, e_4)$ contains four rays of total degree $75 + 62 + 62 + 9 = 208$. Its subsheaf has slope $104 > 100$, proving that T_X is unstable.

REFERENCES

- [AN16] B. Assarf, B. Nill, *A bound for the splitting of smooth Fano polytopes with many vertices*, J. Algebraic Combin. **43** (2016), 153–172, <https://doi.org/10.1007/s10801-015-0630-1>.
- [BDGP21] I. Biswas, A. Dey, O. Genc, M. Poddar, *On stability of tangent bundle of toric varieties*, Proc. Indian Acad. Sci. Math. Sci. **131** (2021), article 36, <https://doi.org/10.1007/s12044-021-00623-w>.
- [Cas06] C. Casagrande, *The number of vertices of a Fano polytope*, Ann. Inst. Fourier **56** (2006), 121–130.
- [CNT25] A. Clarke, A. Napame, C. Tipler, *Toric sheaves and flips*, Épijournal Géom. Algébrique **9** (2025), article 8, <https://doi.org/10.46298/epiga.2024.12468>.
- [CLS11] D. A. Cox, J. B. Little, H. K. Schenck, *Toric varieties*, Grad. Stud. Math. **124**, Amer. Math. Soc., 2011.
- [DDK20] J. Dasgupta, A. Dey, B. Khan, *Stability of equivariant vector bundles over toric varieties*, Doc. Math. **25** (2020), 1787–1833.
- [DDK21] J. Dasgupta, A. Dey, B. Khan, *Erratum for “Stability of equivariant vector bundles over toric varieties”*, Doc. Math. **26** (2021), 1271–1274.

- [Dev22] L. Devey, *A combinatorial description of stability for toric vector bundles*, arXiv:2212.11020, 2022 (version 3, 2023).
- [EN95] S. P. Eveson, R. D. Nussbaum, *An elementary proof of the Birkhoff–Hopf theorem*, Math. Proc. Cambridge Philos. Soc. **117** (1995), 31–55.
- [Ewa88] G. Ewald, *On the classification of toric Fano varieties*, Discrete Comput. Geom. **3** (1988), 49–54.
- [HL14] M. Henk, E. Linke, *Cone-volume measures of polytopes*, Adv. Math. **253** (2014), 50–62, <https://doi.org/10.1016/j.aim.2013.11.015>.
- [HNS22] M. Hering, B. Nill, H. Süß, *Stability of tangent bundles on smooth toric Picard-rank-2 varieties and surfaces*, in: Facets of Algebraic Geometry, Vol. II, LMS Lecture Note Ser. 473, Cambridge Univ. Press, 2022, 1–25.
- [Kly90] A. A. Klyachko, *Equivariant bundles on toral varieties*, Math. USSR Izv. **35** (1990), 337–375.
- [NT24] A. Napame, C. Tipler, *Toric sheaves, stability and fibrations*, J. Pure Appl. Algebra **228** (2024), no. 3, article 107494, <https://doi.org/10.1016/j.jpaa.2023.107494>.
- [Nil06] B. Nill, *Classification of pseudo-symmetric simplicial reflexive polytopes*, in: Algebraic and Geometric Combinatorics, Contemp. Math. **423**, Amer. Math. Soc., 2006, 269–282.
- [Øbr07] M. Øbro, *An algorithm for the classification of smooth Fano polytopes*, arXiv:0704.0049.
- [Paf] A. Paffenholz, *Smooth Fano polytopes*, classification data, <https://polymake.org/polytopes/paffenholz/www/fano.html>.
- [Rey23] W. Reynolds, *On slope stability of tangent sheaves on smooth toric Fano varieties*, Ph.D. thesis, University of Edinburgh, 2023, <https://hdl.handle.net/1842/40752>.
- [VK85] V. E. Voskresenskiĭ, A. A. Klyachko, *Toric Fano varieties and systems of roots*, Math. USSR Izv. **24** (1985), 221–244.
- [Wue26] B. J. Wuebben, *Slope stability of tangent bundles of smooth toric Fano varieties*, arXiv:2608.20411.
- [SW18] A. Saumard, J. A. Wellner, *Efron’s monotonicity property for measures on $\mathbb{R}^2$* , J. Multivariate Anal. **166** (2018), 212–224, <https://doi.org/10.1016/j.jmva.2018.03.005>.